Salesforce Competition in Residential Solar*

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Job Market Paper

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Abstract

I study how door-to-door salespeople affect market outcomes in the California residential solar industry. I combine data on installations with a novel dataset on registered salespeople, as well as utility billing records, to estimate a model of demand and supply where firms endogenously set prices and hire local salesforces. Salespeople increase adoption by 17 percent relative to a counterfactual where salespeople are banned. Yet much of the salesforce exists because of business stealing: salesforces that maximized joint industry profits would be 55 percent smaller with only 4 percent fewer systems sold. On average, agents cost less than the environmental benefit of the additional installations they bring, but the marginal agent costs nearly twice as much as its environmental benefit. Most households brought into the market by agents earn internal rates of return above typical borrowing rates, but 86 percent would lose money at higher rates that concern regulators. Because net metering subsidizes adoption at the expense of other ratepayers, the market expansion due to agents lowers social welfare.

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1 Introduction

Sellers do not just set prices and wait for buyers. Sellers sell, investing in salespeople and marketing as a competitive tool to raise demand for their products and increase profits.[1] Their increased demand comes in two forms, with potentially opposing welfare effects (Grossman and Shapiro, 1984). First, *market expansion*: sales activity brings marginal buyers into the market who would not otherwise purchase due to, for example, search or information frictions. Second, *business stealing*: one firm's sales effort diverts inframarginal buyers from its competitors, wasting additional resources for the same consumers.[2]

Additional features of real-world markets complicate any empirical analysis of selling. With heterogeneous consumers and firms, selling can change who buys and from whom, which can have distributional or efficiency consequences. Selling also changes how firms compete on price, with increased concentration that softens competition or expanded choice sets that intensify it. With externalities, the social value of selling differs from its private value to firms and consumers. A firm makes its selling decisions without internalizing its effects on rivals and society, putting sales competition potentially at odds with social welfare. In this paper, I study how much selling brings new consumers into the market rather than moving them between firms, whether it softens or intensifies price competition, and what it costs firms, consumers, and society. I answer these questions in the context of door-to-door salespeople hired by installing firms in the California residential solar industry. Many features of this market make it an ideal setting for studying the economics of selling and how these effects matter for market outcomes and policy. Surveys report that around half of installations occur through a salesperson (Sigrin et al., 2017). As a new, rapidly growing technology, residential solar has a large extensive margin of potential adopters. Many heterogeneous installing firms compete for these households, so selling dictates which firms serve which households and at what prices. With the goal of reducing greenhouse gas emissions, environmental policy has subsidized solar through federal tax credits and net metering, at a cost to taxpayers and other ratepayers (Borenstein, 2024; Borenstein and Davis, 2025). However, the recent growth of door-to-door sales has raised consumer protection concerns from regulators, who worry about high-pressure sales tactics

[1] As evidence of how widespread this activity is in the economy, combined wages paid to sales occupations are around 2.5 percent of GDP (U.S. Bureau of Labor Statistics, 2024), alongside economy-wide advertising expenditures of 1 to 2 percent of GDP (Misra, 2019; Baslandze et al., 2023).

[2] See Kaldor (1950) for an early example analyzing the economics of advertising. Even earlier, Pigou (1929, p. 200) reasons about the social costs of business stealing, "For, clearly, if each of two rivals makes equal efforts to attract the favour of the public away from the other, the total result is the same as it would have been if neither had made any effort at all."

and deceptive marketing that misleads households about savings and financing (Khan, 2024; Consumer Financial Protection Bureau, 2024). In August 2026, British Columbia banned unsolicited door-to-door solar sales entirely (Consumer Protection BC, 2026).[3]

Prior empirical work has studied selling in markets that lack one or more of these features. In many settings, like real estate, pension accounts, or retail electricity, the pool of participants is largely fixed, so sales agents primarily shift existing consumers between firms (Hsieh and Moretti, 2003; Barwick and Pathak, 2015; Hastings, Hortaçsu and Syverson, 2017; Kahn-Lang, 2024). In prescription drug markets, prices are negotiated between manufacturers, insurers, and providers, which limits how prices respond to selling (Shapiro, 2018; Sinkinson and Starc, 2019; Grennan et al., 2025). And in consumer product markets for "sin goods" like alcohol, junk food, or e-cigarettes, the additional consumption that advertising brings reinforces rather than offsets the losses from business stealing (Rojas and Peterson, 2008; Dubois, Griffith and O'Connell, 2018; Tuchman, 2019).

To understand the effect of solar salespeople, I combine three datasets on firms, their salesforces, and what households gain from adopting. The first contains details on the installing firm, price, and characteristics of every residential solar installation in the territories of California's three largest investor-owned utilities, which is publicly available from the California Public Utilities Commission. The second, which I obtain through a public records request to the Contractors State License Board, is a novel dataset on the complete history of firm–agent relationships.[4] The third is administrative electricity billing and usage data from the three utilities for a large sample of households with and without solar, with the adopters matched to their installations in the first dataset.

I begin by documenting three features of the California market. First, during my main analysis period from 2017 to 2022, the number of door-to-door sales agents grew much faster than the number of installations: installations more than doubled while registered sales agents more than tripled. Second, a small set of large firms, defined as the largest 20 percent of firms by average annual installations, make 96 percent of installations and employ 97 percent of sales agents. These large firms compete with one another alongside a long tail of small, local installers, so markets—the median of which has around a hundred

[3]In the United States, some localities have banned unsolicited door-to-door sales since the 1930s under so-called "Green River" ordinances, which the Supreme Court upheld in *Breard v. Alexandria* (1951). Subsequent courts have struck down bans deemed more restrictive on commercial speech than necessary to protect residents, as in *Project 80's, Inc. v. City of Pocatello* (1988) (Vile, 2024). In solar, recent regulation has primarily taken the form of disclosure and licensing requirements for installing firms and their sales agents (Nevada Legislature, 2023; Washington State Legislature, 2024; Texas Department of Licensing and Regulation, 2025).

[4]California law requires door-to-door salespeople to register with the state and firms to report the start and end dates of any selling relationship. I describe the data in Section 2.

installing firms—are generally unconcentrated by antitrust standards. Even among large firms, agent use is selective: half of large firms do not use agents, and firms that use agents do so in only about a third of the markets that they operate in statewide. Third, firms that use agents charge higher prices than their rivals, though this could reflect higher installation costs, higher markups, or both.

Guided by these patterns, I develop a model of demand and supply that accounts for sales agents. On the demand side, households choose whether to adopt solar, from which firm, and what size of system, following the random coefficients discrete choice structure of Berry, Levinsohn and Pakes (1995). Households differ observably in income and in how much they save on their electricity bills from adoption. An unobserved taste for solar flexibly captures how many households are on the margin of adoption and therefore how much market expansion versus business stealing agents cause. Consumer demand for a firm's systems depends on the net cost of the system (the present value of electricity bill savings less the upfront installation prices net of the tax credit), on the firm's local salesforce, and on other characteristics. As in Hastings, Hortaçsu and Syverson (2017), agents shift demand directly and have diminishing returns. I build on their paper with a richer microfoundation of the demand specification, which also aids in identification. I also allow the effectiveness of agents in shifting demand to vary with households' income and bill savings as a reduced-form way to capture heterogeneity in information, search frictions, and targeting.

I estimate the demand model by GMM, using a combination of excluded instruments to address the endogeneity of prices and salesforces, and micromoments that identify heterogeneous preferences for solar (Conlon and Gortmaker, 2026). The administrative billing data contribute to estimation in two ways: I use double/debiased machine learning to directly estimate how much households would save on their electricity bills with solar (Semenova and Chernozhukov, 2021; Chernozhukov et al., 2025), and I construct micromoments from the relationships between these estimated bill savings and the likelihood of adoption, the prices households pay, and the salesforces of the firms they choose. These billing moments join micromoments from several other sources to identify a rich set of parameters in the demand specification that capture important heterogeneity in the market.

On the supply side, large firms choose prices and salesforces in each market in Nash–Bertrand competition alongside a monopolistically competitive fringe of small, local installers that sell without agents. Firms differ in their marginal *installation* costs and in their marginal *selling* costs—the cost of hiring agents. These selling costs capture agent

salary and commissions, as well as the overhead of recruiting, training, and managing a salesforce. Employing salespeople incurs a marginal cost per agent, rather than per installation, which separately identifies installation costs from selling costs. The firm's first-order condition delivers the classic Dorfman and Steiner (1954) condition from the advertising literature: a firm hires agents until the additional profit earned equals their marginal cost. The model also rationalizes the extensive margin of salesforce use, since a firm will not hire agents if the marginal cost of the first agent exceeds the marginal gain in profit. This has the structure of a censored Tobit model, which I estimate via maximum likelihood (Gouriéroux et al., 1987).

The model estimates show that agents are effective at shifting demand. The median elasticity of a firm's demand with respect to its salesforce is 0.14, about an order of magnitude larger than the median estimate for television advertising in Shapiro, Hitsch and Tuchman (2021). Agents are somewhat more effective with lower-income households and households with higher bill savings, consistent with agents reaching households that face greater search frictions but still targeting those most likely to adopt (Allcott and Sweeney, 2017; Kahn-Lang, 2024). However, these heterogeneous effects are small relative to the overall mean effect of agents. On the supply side, I find that firms with agents have higher installation costs than other firms, so their higher markups reflect households' stronger preferences for their products rather than lower costs. I also estimate that the cost of using agents sharply declined in 2020, during the COVID-19 pandemic, and remained lower through the end of the sample in 2022.

I use these estimates to compare the observed equilibrium with two simulated counterfactuals: an equilibrium that bans agents and a semi-collusive equilibrium where firms choose salesforces to maximize joint industry profits, thus internalizing the business stealing externality. I find that compared to the ban, agents raise installations by 17 percent, so the market expansion effect is large. Yet much of the salesforce exists because of business stealing: salesforces that maximized joint industry profits would be 55 percent smaller and install only 4 percent fewer systems.

Industry reports often suggest that one cause of high installation prices is the use of salespeople (Connelly, 2021; Ramasamy et al., 2025), but my estimates do not support this claim. Relative to equilibrium under the ban, agents raise prices by only 1.1 percent. Most of this increase comes from steering households toward higher-cost firms, which lowers productive efficiency, rather than from softer price competition. A ban on door-to-door sales would therefore reduce adoption substantially without lowering prices by much.

To speak to environmental and subsidy policy, I measure the *additionality* of agents as

a technology for raising adoption: what firms spend on agents per watt of solar that agents add (Aspelund and Russo, 2026). On average, agents cost $0.75 per added watt, less than the $1.01 in environmental benefits of each watt. But the agents that firms hire because of business stealing cost $1.40 per added watt, more than the benefits they bring. I also find that agents and the ITC reinforce each other. Without the ITC, firms would hire 78 percent fewer agents, which speaks to the role of subsidies in shaping the widespread use of agents. In turn, agents extend the reach of the subsidy, so that the ITC adds 22 percent more installations with agents than without.

To speak to regulator concerns around consumer protection and financing, I measure each household's financial benefits from electricity bill savings by the internal rate of return (IRR) on its system. Unlike consumer surplus, this return does not depend on assumptions about how agents enter utility (Dixit and Norman, 1978; Grossman and Shapiro, 1984). I evaluate these returns against the range of solar loan rates reported by Consumer Financial Protection Bureau (2024). Including dealer fees and other terms, a typical loan costs about 5.5 percent a year in real terms, and about 12 percent at the high end. I find that the households brought into the market by agents earn similar returns to those of inframarginal households, but whether they gain depends crucially on financing: at 5.5 percent, 83 percent would make money over 25 years, while at 12 percent, 86 percent would lose money.

Finally, I measure the effect of agents on social welfare, summing changes in consumer surplus, producer surplus, government spending on the ITC, environmental benefits, and the cost that net metering shifts onto other ratepayers. Because net metering credited exported generation at retail rates well above marginal cost, each added installation lowers utility revenues, which utilities recover by raising rates on other ratepayers (Borenstein, 2024). Agents raise consumer surplus, producer surplus, and environmental benefits, but the ITC and the cost shift outweigh these gains: agents lower social welfare by $1.7 billion, mainly because the installations they add shift $2.5 billion over their lifetimes onto other ratepayers.

Related literature and contributions. This paper contributes to three main strands of the literature. First is the study of competition through selling. A large marketing literature studies how firms manage their salesforces (see Misra, 2019).[5] Empirical studies have measured how much selling expands demand and how much it takes from rivals (Shapiro, 2018; Sinkinson and Starc, 2019; Grennan et al., 2025), how it changes price

[5]This includes studies on the design of incentive-compatible compensation contracts (Larkin, 2014; Allcott and Sweeney, 2017; Kim, Sudhir and Uetake, 2022).

competition (Hastings, Hortaçsu and Syverson, 2017; Dubois, Griffith and O'Connell, 2018; Kahn-Lang, 2024), and how far it departs from what would maximize industry profits or social surplus (Hsieh and Moretti, 2003; Barwick and Pathak, 2015; Shapiro, 2018). I model prices and salesforces as endogenous firm choices in equilibrium, which lets me measure all these effects in a single market and show that selling can substantially expand the market while business stealing still drives much of the salesforce.

Second, I contribute to the literature on residential solar and clean-technology adoption, which has examined subsidies, learning-by-doing, rate design, and search (Hughes and Podolefsky, 2015; De Groote and Verboven, 2019; Feger, Pavanini and Radulescu, 2022; Langer and Lemoine, 2022; Gerarden, 2023; Dorsey, 2024; Bradt, 2026), as well as how sales agents affect take-up of energy-efficient durables (Allcott and Sweeney, 2017). This paper is the first to account for salespeople in an equilibrium model of the solar market, which allows me to inform several policy debates that interact with solar sales: restrictions on door-to-door sales, consumer protection in solar financing, and green energy subsidies.[6]

Third, I contribute to the empirical industrial organization literature on demand estimation with micromoments, which uses additional information about consumers to identify heterogeneous preferences (Petrin, 2002; Berry, Levinsohn and Pakes, 2004; Grieco, Murry and Yurukoglu, 2024; Conlon and Gortmaker, 2026). With administrative billing data similar to that previously used to study electricity rate design (Borenstein, 2017; Gillingham, Kirkpatrick and Bollinger, 2025; Snashall-Woodhams, 2025), I estimate how much households save from adopting solar using double/debiased machine learning (Semenova and Chernozhukov, 2021; Chernozhukov et al., 2025), and bring these estimates into demand both as observable heterogeneity and as micromoments. This approach illustrates how machine-learning summaries of heterogeneity can enrich structural models in industrial organization and marketing.

[6]Selling and its regulation has been studied in other markets, such as fiduciary standards for financial advisors (Bhattacharya, Illanes and Padi, 2025), auto dealer loan markups, which the U.K. Financial Conduct Authority banned in 2021 (Grunewald et al., 2023), pharmaceutical payments to physicians (Grennan et al., 2025), and commissions and steering in residential brokerage (Barwick, Pathak and Wong, 2017).

2 The California Residential Solar Market

2.1 Setting

Policy context. I study California's residential solar market during the second net energy metering tariff (NEM 2.0) policy period from 2017 to 2022.[7] Under NEM 2.0, households were placed on time-of-use rates and exported generation was credited at the retail rate. During NEM 2.0, with the exception of a few small, low-income solar programs, there were no state or local installation subsidies.[8] The California Solar Initiative program, which paid upfront rebates, ended in 2016. Households could also claim the federal Investment Tax Credit (ITC) against the upfront system price, which was 26 percent in 2020 and 2021 and 30 percent in other years.[9]

On December 15, 2022, the California Public Utilities Commission adopted a successor tariff, commonly called NEM 3.0, for interconnection applications submitted on or after April 15, 2023 (California Public Utilities Commission, 2022). The new tariff credits exports at the utility's avoided cost rather than the retail rate. The average export credit fell by approximately 75 percent, substantially lowering the value of exported electricity, which is typically around or under half of a system's generation. In anticipation of this change, the announcement produced a rush of applications in early 2023 to be grandfathered into the old NEM 2.0 tariff. To avoid confounding the effect of salespeople with this anticipatory adoption, I end my analysis sample in 2022. The number of registered salespeople also rose by 38 percent in 2023, consistent with salespeople marketing the deadline to households.

The period that I study also differs from the earlier era of solar adoption in the evolution of prices. Figure 1 shows that the average installed price of a host-owned system fell by 41 percent from \$8.95 per watt in 2010 to \$5.31 in 2016, largely tracking declines in the cost of hardware and installation labor.[10,11] By contrast, from 2017 to 2022, prices fell only 13 percent, from \$4.98 to \$4.34 per watt. Hardware and labor costs were flat over this period. The remainder, termed "soft costs" in the industry, cover design, permitting,

[7] NEM 1.0 began in 1996. San Diego Gas & Electric moved new applications to NEM 2.0 in June 2016, Pacific Gas & Electric in December 2016, and Southern California Edison in July 2017.

[8] The largest, the Single-Family Affordable Solar Homes program, covered at most about a thousand systems per year (GRID Alternatives, 2022).

[9] The ITC is a non-refundable tax credit, so households could only claim the credit if they had sufficient tax liability. This means the ITC likely was a greater incentive for higher income households (Borenstein and Davis, 2025). However, households could carry forward unused credits into future years. In my analysis, I assume that households claim the full ITC.

[10] Unless otherwise noted, all dollar figures are deflated to 2023 dollars using the CPI-U.

[11] Gerarden (2023) and Bradt (2026) study the role of production innovation and learning-by-doing in contributing to cost declines during this earlier period.

business overhead, sales and marketing, and direct firm profit. While soft costs accounted for 40 percent of the upfront price paid by households in 2010, by 2022 they were over 60 percent. The rise in soft costs has coincided with the growth of sales agents, and the industry has cited persistently high customer acquisition costs as a potential barrier to lowering prices and raising adoption (Connelly, 2021; Ramasamy et al., 2025).

Installing firms and salespeople. Installing firms are licensed contractors that design and install rooftop systems. They are responsible for managing interconnection applications with the utility and obtaining work permits from local authorities. Contractors buy equipment from manufacturers and distributors.[12] Firms acquire potential customers, or "leads," through several channels. In industry terms, these channels are divided into *inside sales*—leads generated through referrals, websites, telemarketing, and advertising that are handled by representatives in a firm's own office—and *outside sales*—leads generated by door-to-door salespeople employed by the firm to visit customers at their homes. Third-party organizations also generate leads and sell them to installing firms. From the firm's perspective these are inside sales, and my analysis captures them insofar as they ultimately involve the firm's own agents following up on them.

California requires anyone who sells home improvement contracts for a licensed contractor away from the contractor's place of business to register with the Contractors State License Board (CSLB) as a Home Improvement Salesperson. Rooftop solar and door-to-door sales agents fit into this category. Each contractor must notify the CSLB when it begins and stops working with a registered salesperson. Appendix Figure A3 shows the registration and notification forms. Selling without an active registration is illegal; individuals can be fined or worse, contracts can be voided, and firms knowingly violating these requirements can be sanctioned by the CSLB. A sales visit may include sharing information about the installation process, walking through a household's electricity bills and potential bill savings, providing a quote for the system, and discussing financing options.

Regulatory concern has focused on high-pressure tactics, misleading claims about bill savings and subsidies, and contracts with opaque financing terms signed with little time to compare alternatives (Khan, 2024). Adopter surveys suggest that households reached through door-to-door sales search less: among California adopters with a door-to-door contact, 43 percent talked to only one installer, compared with 4 percent of those who

[12]Even the select few vertically integrated installing firms like SunPower or Tesla Energy used outside equipment. This occurred both directly, under partnerships with manufacturers, and indirectly, by white-labeling third-party equipment.

found their installer through their own research. Appendix A.4 describes the survey, and Appendix Figure A4, panel (b), shows these patterns. Panel (c) of the same figure shows that door-to-door sales is also the most common way adopters met their chosen installer in every income group, but its share falls from 62–64 percent below $100,000 to 44 percent at $150,000 or more.

Job postings describe door-to-door sales as a full-time position, paid through base pay, commissions, or both, with advertised annual earnings from tens of thousands to hundreds of thousands of dollars (Revelio Labs, 2026).[13] I abstract from the design of compensation contracts in my analysis.

2.2 Data

I draw on three main datasets to study salespeople in the California solar market. Appendix A provides additional details on the data and sample construction.

Interconnected applications. The first dataset is the Interconnected Applications dataset from the California Public Utilities Commission (CPUC). When the three major investor-owned utilities—Pacific Gas & Electric (PG&E), Southern California Edison (SCE), and San Diego Gas & Electric (SDG&E)—interconnect a solar system to the grid, they report the application to the CPUC, which maintains a public database of these applications. Together, these three utilities serve approximately three-quarters of residential electricity demand in California.

Each record is an interconnection application submitted by the installing firm. It contains the date of application, date of permission-to-operate, the household's utility and zip code, the name and CSLB-issued license number of the installing firm, and characteristics of the system such as upfront price, installed capacity, and equipment used.[14] I date each installation by its application date rather than permission-to-operate date. The application date is the earliest recorded date and likely the closest to when a household entered into a contract with the installing firm, while the permission-to-operate date is a function of construction and administrative processing time. In practice, the median time between the two dates is 7 days, though the 90th percentile is two months.

I assign each installation to a market using its zip code and application date. A market is a region in a year, where regions are groups of counties following Census core-based

[13] Appendix A.5 describes common features of job postings using data from Revelio Labs.

[14] I merge this dataset with Lawrence Berkeley National Laboratory's Tracking the Sun data, which cleans and standardizes the CPUC data. The raw CPUC data contains additional variables that are not included in the Tracking the Sun dataset, the most important of which is the firm's CSLB license number.

statistical areas (CBSAs).[15] Appendix Figure A1 shows a map of the 30 regions; there are 180 markets (30 regions × 6 years) in total.

When I turn to the demand model, I will estimate local area fixed effects within each market. I define these local areas using the Census public use microdata areas (PUMAs), which are geographic areas of at least 100,000 residents, mapping installations to PUMAs using their zip codes. There are 215 of these local areas in the sample. For example, the San Francisco–Oakland–Berkeley CBSA contains 34 PUMAs. I exclude PUMAs that fall outside the IOU service areas. Where present, I assume that firms and sales agents operate in an entire market. Since there are many firms and PUMAs are granular areas, a firm need not install in every local area within the market that it operates in.

I restrict the data to residential, single-family, rooftop solar systems between 1 and 25 kW. I exclude systems that were self-installed, part of new construction, or co-installed with a battery. I use the CPI-U to deflate prices to 2023 dollars. I treat real prices per watt above the 99.5th and below the 0.5th percentile, weighted by capacity, as missing. Prior work has documented that prices for third-party owned systems (leases and power purchase agreements) generally reflect the taxable basis of the system rather than the total cost for households (Barbose and Darghouth, 2019). I retain these third-party owned systems—approximately 30 percent of the sample—when computing market shares but treat their price as missing.

Salesperson registrations. The second dataset is the CSLB registry of Home Improvement Salespersons, which I obtain through a public records request. Each record is a relationship spell between a registered salesperson and a licensed firm, which reports the salesperson's registration number and address on file, the firm's CSLB license number, and the start and end dates of the relationship.[16] I geocode the addresses and identify door-to-door sales agents as those with residential addresses. Agents with commercial addresses are more likely to be office-based sales representatives and managers. Linking the two datasets on the firm's CSLB license number allows me to observe, for every firm, the number of local door-to-door sales agents that they employ in each market.[17]

Electricity billing and usage records. The third dataset is administrative billing and usage data from PG&E, SCE, and SDG&E. Through each utility's Energy Data

[15] I combine adjacent rural counties that do not fall within a CBSA into their own regions. This applies to Glenn and Colusa counties and Amador and Calaveras counties.

[16] The recorded address is the one on file with the CSLB when the database was pulled in May 2025.

[17] CPUC Decision 21-06-026 required installers to report the registration number of any sales agent involved in a contract to the utilities as part of the interconnection application, but this occurred late in the sample period and a data request for these records was unsuccessful.

Request Program, which grants researchers access to utility data for approved projects, I obtain billing and usage records from 2020 to 2024 for a stratified random sample of solar adopting and non-adopting households. Adopters are households that adopted solar between 2021 and 2023; non-adopters had no solar as of 2024. Each record is a household bill, which reports the billing period, electricity usage in kilowatt-hours, the total bill amount, the rate schedule, the household's zip code, the climate region, and enrollment in utility programs such as low-income rate assistance.

As part of the data request, the utilities matched adopters to their application IDs in the Interconnected Applications, which allows me to observe characteristics of their installed system and the installing firm. I use this information to estimate the annual electricity bill savings per kilowatt of installed solar capacity for each household, which I use as an input into the demand model and as a measure of consumer benefits in the counterfactual analysis. The relationships between bill savings and adoption rates, upfront prices, and the number of sales agents at the chosen firm also serve as micromoments in demand estimation. After cleaning, I am left with 115,564 solar households and 73,409 non-solar households. Appendix A describes the sample construction.

2.3 Market Descriptives and Characterizing Agent-Using Firms

Figure 2 shows the annual number of installations and registered salespeople in the full data from 2010 to 2023. Over the analysis period from 2017 to 2022, installations more than doubled from 92,000 to 198,000, while the number of registered salespeople more than tripled from 2,000 to 7,000. This growth did not come from net firm entry. Although firms enter and exit, the number of active firms is roughly constant at approximately 1,800 each year, in contrast to the earlier period of firm entry studied by Bradt (2026).

Installations are concentrated among a minority of firms. Table 1 reports descriptive statistics about firms and markets in the sample. Sorted by average number of installations per active year, the largest 20 percent of firms account for 96 percent of installations. Accordingly, I classify these firms as *large firms* and the remaining 80 percent as the *fringe*.[18] The median large firm installs 38 systems per year across 3 markets. By contrast, the median fringe firm installs 2 systems per year and operates in a single market. Even collectively, fringe firms together account for a median of 5 percent of installations in a market and never more than 38 percent. Judging by their names, many of these firms appear to be small, local general contractors or sole proprietorships by master electricians.

[18]This split falls at 15 statewide installations per active year.

Markets are nonetheless unconcentrated by usual antitrust standards: the median market has 104 firms, and Appendix Figure A2 shows that the median Herfindahl–Hirschman Index of installation shares across markets is 665.

Salespeople are even more concentrated: large firms employ 97 percent of all registered salespeople. Nor did their growth come from more firms using them. As Figure 2, panel (d), shows, the share of firms each year with a registered salesperson only increased from 15 to 19 percent. Instead, the growth in salespeople was concentrated among firms that built very large salesforces: the average statewide salesforce among firms with salespeople grew from 8 to 31 agents, while the median only increased from 2 to 3.

The large firms that use salespeople are often the largest firms in their markets: large firms with salespeople make up only 23 percent of large firm–market observations but account for a median of 55 percent of installations in those markets. The decision to use agents nonetheless varies across firms and markets. Only 53 percent of large firms register a salesperson anywhere, and those that do use salespeople in only a third of their markets on average.

Finally, firms that use salespeople also charge more per watt. In the median market, large firms with salespeople charge \$4.86 per watt on average, compared with \$4.61 for large firms without salespeople and \$4.15 for the fringe. The lower fringe price is consistent with the idea that these firms act as a competitive, local alternative. Higher prices at agent-using firms could reflect higher costs, higher markups, or both. The structural model in Section 5 will allow me to decompose the price difference into these components.

These patterns guide several assumptions in the structural model. Because the number of firms is stable across years, I treat firm presence as exogenous and do not model firm entry and exit. Because fringe firms are numerous, local, and individually negligible within a market, I treat them as monopolistically competitive and aggregate them into a single composite firm in each market. Because few fringe firms use salespeople, I simplify the fringe to having zero salesforce, so that only large firms face a strategic choice over hiring sales agents. By contrast, I model large firms as oligopolists that internalize their effect on prices and rivals, simultaneously choosing their salesforce and prices in Nash–Bertrand competition.

3 Structural Model of Demand

This section covers the demand model. Section 3.1 microfounds how agents enter demand, Section 3.2 presents the full demand specification, and Section 3.3 describes estimation.

3.1 A Sales Offer Microfoundation

Setup and notation. I microfound a discrete choice demand model of solar adoption where firms sell solar systems through agent and non-agent channels, and households encounter offers through these channels that are heterogeneous in "quality"—how they shift their demand for solar. It differs from the microfoundation in Hastings, Hortaçsu and Syverson (2017) by incorporating non-agent channels, which is important in the solar setting given that many firms do not employ agents and rely on traditional marketing channels. It also relaxes their assumption that households are aware of all agent offers in the market.

Firms f have products $\mathcal{J}_f$, indexed by j, and A_f sales agents. For each product j, a household has indirect utility

$$u_{ij} = \bar{u}_{ij} + \zeta_{ij} \tag{1}$$

where $\bar{u}_{ij}$ is the systematic component of indirect utility and ζ_{ij} is an i.i.d. preference shock.

Receiving offers. A household hears about a product j through offers o. These come independently from the firm's non-agent channels—referrals, websites, and advertising—or directly from the firm's sales agents. They follow a Poisson arrival process with relative arrival intensities $\lambda_{\text{non-agent}}$ and λ_{agent}. Not all pitches are equal, so offers vary in quality and are right-tailed. I specify a particular exponential distribution of offer quality such that, given their preference u_{ij}, the expected number of offers above a threshold v ("acceptable offers") is given by (Resnick, 1987, Section 4.3):[19]

$$\begin{aligned} \mathbb{E}\left[\, N_{ij}(v) \,\right] &= \mathbb{E}\left[\, N_{ij,\text{non-agent}}(v) + N_{ij,\text{agent}}(v) \,\right] \\ &= (\lambda_{\text{non-agent}} + \lambda_{\text{agent}}) \exp\left(-\frac{v - u_{ij}}{\gamma_c}\right). \end{aligned} \tag{2}$$

The partitioning of total offers into non-agent and agent sources follows from the additive property of independent Poisson processes. The perceived dispersion in quality is captured by γ_c, which varies with the household's demographic cell c, and in particular the household's income and electricity bill savings from solar. Equation (2) shows that a household receives more acceptable offers when u_{ij} is higher or, for thresholds above u_{ij}, when γ_c is higher, both of which increase the likelihood of receiving a high draw.

[19] Appendix B.1 gives the exponential form of offer quality.

Without information on the total amount of non-agent marketing that firms do, I cannot separately identify $\lambda_{\text{non-agent}}$ and λ_{agent}. Instead, I will identify the relative arrival intensity of agent offers, $\lambda_{\text{agent}}/\lambda_{\text{non-agent}}$. Without loss of generality, normalize $\lambda_{\text{non-agent}} = 1$ (the level is absorbed into $\bar{u}_{ij}$). With agent density $a_f = A_f/N$, where A_f is the firm's number of agents and N is the number of potential customers, we can define the relative arrival intensity of agent offers as $\lambda_{\text{agent}} = \phi a_f$, where ϕ is the reach of agents relative to non-agent channels. The total offer intensity is then $\lambda_{\text{non-agent}} + \lambda_{\text{agent}} = 1 + \phi a_f$.

Proposition 1. *Let V_{ij} be the indirect utility of the best offer for product j. Under the exponential distribution of offer quality in equation (2), V_{ij} is type-1 extreme value with location $\bar{u}_{ij} + \gamma_c \log(1 + \phi a_f) + \zeta_{ij}$ and scale γ_c.*

Proof. The best offer is worse than v exactly when no offer is acceptable at v. For a Poisson process, $\mathbb{P}\left(N_{ij}(v) = 0\right) = \exp\left(-\mathbb{E}\left[N_{ij}(v)\right]\right)$, so the CDF of V_{ij} is

$$\begin{aligned}
\mathbb{P}\left(V_{ij} \leq v\right) &= \mathbb{P}\left(N_{ij}(v) = 0\right) = \exp\left(-\mathbb{E}\left[N_{ij}(v)\right]\right) \\
&= \exp\left(-(1+\phi a_f)\exp\left(-\frac{v - (\bar{u}_{ij} + \zeta_{ij})}{\gamma_c}\right)\right) \\
&= \exp\left(-\exp\left(-\frac{v - (\bar{u}_{ij} + \gamma_c \log(1+\phi a_f) + \zeta_{ij})}{\gamma_c}\right)\right),
\end{aligned}$$

which is the type-1 extreme value distribution with this location and scale. □

Further, if ζ_{ij} follows the particular Cardell (1997) distribution with parameter $\gamma_c \in (0, 1]$, then the indirect utility of the best offer for product j yields a mixed logit with systematic component $\bar{u}_{ij} + \gamma_c \log(1 + \phi a_f)$ and i.i.d. type-1 extreme value error ε_{ij}.[20]

$$V_{ij} = \bar{u}_{ij} + \gamma_c \log(1 + \phi a_f) + \underbrace{\zeta_{ij} + \gamma_c \epsilon_{ij}}_{\varepsilon_{ij}}, \qquad \varepsilon_{ij} \sim \text{T1EV}(0, 1). \tag{3}$$

An alternative interpretation of this requirement that $\gamma_c \in (0, 1]$ is that households choose the best offer consistent with the assumption of random utility maximization (McFadden, 1978).

The two agent parameters have distinct roles. The *reach* of agents, ϕ, governs how many offers a firm's salesforce generates relative to its other channels. The *effectiveness* of agents, γ_c, governs how much those offers raise utility, and varies with the household's demographics. The specification has diminishing returns to agents, and a firm without

[20] Galichon (2022) provides the form of this distribution as $\zeta_{ij} = \gamma_c \log Z_{ij}$ where Z_{ij} is Positive Stable with parameter γ_c. This result is more commonly used to derive the nested logit representation.

agents still sells through its other channels. A firm's agents raise demand only for its own products. This rules out the positive spillovers that Shapiro (2018) finds for drug advertising, where an ad can send a patient to a physician who prescribes a rival's drug. It fits door-to-door sales, where an agent's contact rarely leads a household to seek other quotes: as Section 2.1 shows, 43 percent of adopters contacted by an agent talk to only one installer, against 4 percent of those who find their installer through their own research.

In the next section, I specify the systematic component of indirect utility $\bar{u}_{ij}$ as a function of price and other product characteristics.

3.2 Demand for Solar

Markets, products, and households. A market is a region m in year t: 30 regions over 2017–2022, or 180 markets. Within each market, there are N_{mt} potential adopting households. Each household belongs to a demographic cell $c = (\ell, y, r)$, where ℓ is one of 215 local areas, y is one of 12 income bins, and r is one of 10 deciles of expected bill savings, to be estimated in Section 4. Each local area has its own distribution of income and expected bill savings. The share of a market's potential adopters in each cell is ω_{cmt}.

A product is a firm-size combination $j = (f, q)$, where q indexes the capacity of the system. I discretize capacity into five bins with representative size G_q, $q = 1, ..., 5$. There are two types of firms: large installers $f = 1, ..., F$ and a fringe composite $f = F + 1$ of all other local installers.[21] Table 2 summarizes the demand data, counting the fringe composite as one firm in each market. The outside option of no installation, $j = 0$, has utility $V_{i0mt} = \varepsilon_{i0mt}$, which is i.i.d. standard type-1 extreme value.

Agents and effective price. A firm's salesforce enters through its agent density in the market, $a_{f(j)mt} = A_{f(j)mt}/N_{mt}$, as in Section 3.1.[22] The fringe has no agents.

The total upfront price of each installation option is P_{jmt}. While households pay this price at the time of installation, two implicit financial benefits are available: the federal Investment Tax Credit (ITC), τ_t, and net energy metering, which provides annual electricity bill savings over the life of the system, R_{cjt}. Thus, the effective price of each

[21] Appendix B.2 derives a Melitz (2003)-style result that differentiated fringe firms in monopolistic competition with common installation costs can be equivalently modeled as a representative firm in demand.

[22] In estimation, I measure density in agents per 100,000 potential adopting households.

option is minus the net present value of the investment

$$\begin{aligned}\tilde{P}_{cjmt} &= (1-\tau_t)P_{jmt} - R_{cjt}, \\ R_{cjt} &= \sum_{h=1}^{H} \frac{[(1+g)(1-d)]^{h-1}}{(1+\delta_y)^h}(1+g)^{-(2024-t)}\rho_r G_q.\end{aligned} \tag{4}$$

The present value of savings varies across cells with expected annual per-kW bill savings ρ_r, which I estimate from the billing data in Section 4. I also allow for heterogeneity across income bins in the implicit discount rate δ_y. Gillingham, Kirkpatrick and Bollinger (2025) estimate heterogeneous discount rates by tercile of wealth: 17.2%, 15.6%, and 10.9% for low, middle, and high wealth households, respectively, which I apply to the approximate terciles of income in my data.[23] These implicit discount rates reflect, for example, borrowing costs, credit constraints, behavioral frictions, information, and the resale value of the home, all of which lower the value that households place on future bill savings. Properly accounting for heterogeneity in the future value of solar is important for recovering credible estimates of price sensitivity in my static model (Gowrisankaran and Rysman, 2012). Typical engineering estimates place the life of a solar system at $H = 25$ years with $d = 0.75\%$ annual degradation (Jordan et al., 2022). Finally, I take the average real increase in retail electricity prices across all three IOUs from 2016 to 2022, which comes out to $g = 4.03\%$ annualized, assuming households expect this growth to continue over the life of the system. The factor $(1+g)^{-(2024-t)}$ converts savings at the 2024 retail electricity rates observed in the billing data to the lower retail rates in the year of adoption.

Indirect utility. The indirect utility of household i's best offer for product j is

$$V_{ijmt} = -\alpha_{c(i)}\tilde{P}_{c(i)jmt} + \gamma_{c(i)}\log(1+\phi a_{f(j)mt}) + \boldsymbol{\beta}_i'\boldsymbol{X}_{jmt} + \xi_{jmt} + \varepsilon_{ijmt}. \tag{5}$$

Unobserved preferences for products are represented by ξ_{jmt}, and ε_{ijmt} is an i.i.d. standard type-1 extreme value error. Recall from the offer model that ϕ is the reach of agents per unit of density, while γ_c is their effectiveness, which varies across demographic cells. For price sensitivity, I follow Miravete, Seim and Thurk (2023) to allow for flexible income effects:

$$\alpha_c = \alpha_0 \left(\frac{y_c}{\bar{y}}\right)^{-\alpha_y}, \tag{6}$$

[23] De Groote and Verboven (2019) estimate an average rate of 15.0% across households in Flanders over an earlier time period.

where y_c is income and $\bar{y}$ is a reference income, the mean income of potential adopters, so that α_0 is the price sensitivity of a household with mean income. The specification of price sensitivity nests both quasi-linear (Nevo, 2000) when $\alpha_y = 0$ and Cobb-Douglas (Berry, Levinsohn and Pakes, 2004) preferences when $\alpha_y = 1$, without imposing either. This generates flexible demand curvature, which governs how equilibrium prices and markups respond when agents shift demand in counterfactuals.[24]

For the effectiveness of agents, I allow heterogeneity on the basis of income and potential savings:

$$\gamma_c = \gamma_0 + \gamma_y z_y + \gamma_r z_r, \tag{7}$$

where z_y and z_r are standardized z-scores of log income and expected bill savings, respectively, so that γ_0 is the effectiveness of agents for a household with mean log income and savings.

The term $\boldsymbol{\beta}_i' \boldsymbol{X}_{jmt}$ represents preferences for observable characteristics:

$$\begin{aligned}\boldsymbol{\beta}_i' \boldsymbol{X}_{jmt} = \sigma_\nu \nu_i + \beta_q G_q + \beta_E \text{Eff}_{jmt} + \beta_{\text{ILR}} \log \text{ILR}_{jmt} + \beta_M \text{Microinv}_{jmt} \\ + \kappa_\ell + \kappa_t + \kappa_f + \eta_y z_y + \eta_q z_y G_q.\end{aligned} \tag{8}$$

The exogenous characteristics are system size, panel efficiency, log inverter loading ratio, and the share of systems of that size with microinverters. I include interactions of income with a constant and system size. I also include local area, year, and firm fixed effects to capture unobserved preferences for solar that vary across space, time, and firm. Intuitively, κ_ℓ captures a local area's average unobserved preference for solar relative to its region, which may reflect, for example, political leanings or environmental attitudes. κ_t and κ_f capture common time-varying preferences for solar and market-wide preferences for specific firms, respectively.

Finally, the random coefficient $\nu_i \sim N(0, 1)$ on all solar products captures unobserved heterogeneity in whether households adopt at all. The potential market includes all single-family, owner-occupied households, but only a subset of them is marginal to adoption, so

[24] One could further allow α_c to vary with the number of agents in the market, as in Hastings, Hortaçsu and Syverson (2017), to capture the idea that agents influence price sensitivity directly through salience and obfuscation. When I include this term in demand, I estimate it to be small and not statistically significant. In their setting, identification is aided by their data on individual variation in costs at the same banks, which allows them to separate how bank agents shift demand from how agents rotate demand via the interaction of agents and price sensitivity. Since I define average prices at the firm–market level, I lack this variation in the model. Further, this term complicates the supply side equilibrium because agents generate spillover effects on other firms. Therefore, I omit it from the baseline specification.

this term is important for separating the market expansion effects of agents from business stealing.

Market shares. Given its taste ν_i, household i in cell c chooses product j with the logit probability

$$s_{ijmt} = \frac{\exp\left(\bar{u}_{cjmt} + \gamma_c \log(1 + \phi a_{f(j)mt}) + \sigma_\nu \nu_i\right)}{1 + \sum_{k \in \mathcal{J}_{mt}} \exp\left(\bar{u}_{ckmt} + \gamma_c \log(1 + \phi a_{f(k)mt}) + \sigma_\nu \nu_i\right)}, \tag{9}$$

where $\bar{u}_{cjmt}$ collects the terms of equation (5) other than the agent term, $\sigma_\nu \nu_i$, and ε_{ijmt}. Integrating over the taste distribution gives the share of cell c's households that choose product j:

$$s_{cjmt} = \int s_{ijmt} \, d\Phi(\nu_i). \tag{10}$$

Aggregating over cells gives the market share:

$$s_{jmt} = \sum_c \omega_{cmt} \, s_{cjmt}. \tag{11}$$

3.3 Estimation

I estimate demand using the GMM estimator following Berry, Levinsohn and Pakes (1995), which consists of an outer search over the nonlinear parameters $\Theta = (\alpha_0, \alpha_y, \gamma_y, \gamma_r, \phi, \eta_y, \eta_q, \sigma_\nu, \kappa_\ell)$ and a Berry inversion step that solves for the product mean utilities that exactly match the observed market shares s_{jmt} in equation (11) in each market. The remaining linear parameters are estimated by IV-GMM. I use four sets of moments to identify the parameters.

Excluded instruments. The first set of moments addresses the endogeneity of price and salesforce to unobserved preferences, ξ_{jmt}. Demand shocks raise a firm's markups, and with them both its price and, as Section 5 derives, its incentive to employ agents, which biases α_0 toward zero and γ_0 upward. I use 17 excluded instruments Z_D satisfying $\mathbb{E}\left[\, Z_D' \xi(\Theta) \,\right] = 0$, where $\xi(\Theta)$ is the residual implied by the Berry inversion at a given Θ: five cost shifters, including input prices and an indicator for markets other than the firm's headquarters; six Berry, Levinsohn and Pakes (1995) sums of rival characteristics; and six Gandhi and Houde (2019) differentiation instruments. Three of these measure differentiation in predicted prices and are interacted with the income composition of the market, which identifies how price sensitivity varies with income (Miravete, Seim and

Thurk, 2023). Appendix D.1 defines each instrument.

Local adoption counts. The second set of moments identifies the local area fixed effects κ_ℓ. Local area firm shares are sparse and overwhelmingly zero, so a Berry inversion at the local level is impossible. Following De Groote and Verboven (2019), I instead match predicted to observed counts of adopters in each local area over the sample period, normalizing the local effects to average zero within each region, which leaves 185 independent count moments for the 215 local areas.

Micromoments. The third set of moments are micromoments, which discipline the household heterogeneity that determines whom agents bring into the market (Conlon and Gortmaker, 2026). The first source is external: from the LBNL's report on the income of solar adopters across Census tracts (Forrester et al., 2024), I calculate the share of adopters in each income bin within the three IOU service territories, pooled across 2017–2022. The second source is the billing data, which link each household's estimated bill savings to whether it adopts, the price it pays, and the agent density of the firm it chooses; I describe these moments in detail in the next section. These moments describe what aggregate market shares reveal only indirectly: how adoption varies with savings, and which households buy from firms with agents.

Identification is joint, but the billing micromoments are especially informative about two features of demand. The first is the dispersion of the unobserved taste for solar, σ_ν. Bill savings lower the effective price of every solar product, so at a given price sensitivity, households with higher savings should adopt more often. How much more often depends on how many households are near the margin of adoption, which σ_ν determines. When σ_ν is small, many households are close to indifferent, and higher savings move many of them into adoption; when it is large, most households are dispersed far from the margin, and the same bill savings moves fewer households. Because ν_i shifts the utility of every solar product equally, it governs substitution between solar and not adopting rather than among installers, and so separates market expansion from business stealing in counterfactuals: holding the effect of agents fixed, a larger σ_ν leaves fewer households near the margin of adoption, so more of the households that agents bring to a firm would otherwise have bought from another installer.[25] The second is how the effectiveness of agents varies with income and savings, γ_y and γ_r: the agent density of the chosen firm by income bin and savings decile are the only moments that relate salesforces to household characteristics.

[25] Berry and Waldfogel (1999) and Sinkinson and Starc (2019) use nested logit models for the same reason, in their studies of entry in radio broadcasting and of pharmaceutical advertising.

Agent reach. To identify the agent reach parameter ϕ I use the Department of Energy's SEEDS survey, which asked a sample of California adopters whether they first met their installer through a sales agent or through other channels, such as referrals, advertising, or their own research. In this survey, the share of adopters who met their installer through an agent is 0.432. In the offer model in Section 3.1, the probability that an adopter with firm f met the firm through an agent is $\phi a_f/(1+\phi a_f)$. The aggregate survey share of 0.432 should then equal the average of this probability over all firms, weighted by their installation shares, given a candidate value of ϕ. I compute this moment using firms' observed agent densities and market shares in 2013–2015 within my data. This share separates the reach of agents from their effectiveness, much as Allen, Clark and Houde (2019) use survey data on how many lenders borrowers consult to separate search costs from preferences. Appendix D.2 gives details.

Weighting and inference. I use a two-step efficient GMM estimator, weighting each set of moments by the inverse of its uncertainty in the first step so that no one block dominates, and report sandwich standard errors, clustered by market, that account for the uncertainty in each set of moments. Appendix D.1 describes how I estimate the covariance of each set of moments.

4 Consumer Heterogeneity in Bill Savings

The savings term in the demand model depends on expected annual per-kilowatt bill savings, ρ_r, in each savings decile. In this section, I use the billing data to estimate heterogeneity in the potential bill savings from solar adoption. Because I estimate savings from realized bills, the estimates reflect the tariffs households actually pay and any rebound in their consumption after adoption, which approaches based on engineering models and counterfactual bill calculators must instead assume or estimate (Feger, Pavanini and Radulescu, 2022; Gillingham, Kirkpatrick and Bollinger, 2025; Bradt, 2026).

Many factors contribute to heterogeneity in savings, including patterns of electricity usage, rate schedules, low-income assistance, and location. These can enter into predicted savings in a complex, non-linear way. To account for this high-dimensional set of observables and their interactions, I employ recent advances in machine learning methods to estimate how per-kilowatt savings varies with these observed characteristics (Semenova and Chernozhukov, 2021; Chernozhukov et al., 2025).

4.1 Setup and Identification

As a reminder, the data contain a random sample of households in each of the three IOUs that adopted between 2021 and 2023, plus a random sample of non-adopters. For all households, I observe baseline billing and usage covariates X_i in 2020 and billing outcomes in 2024. Appendix A.2 describes the sample and variable construction in more detail.

A household's solar adoption status is $D_i \in \{0, 1\}$. Adoption is a bundle of two choices: whether to install, and what size to install. Let $G_i(1) > 0$ denote the size the household installs if it adopts. In potential outcomes notation, $B_i(0)$ is the 2024 bill if the household does not adopt and $B_i(1, G)$ is the bill if it adopts size G, with $B_i(1) = B_i(1, G_i(1))$ the bill at its chosen size. Observed size and bills are $G_i = D_i G_i(1)$ and $B_i = (1 - D_i)B_i(0) + D_i B_i(1)$. The estimand is written in terms of the population conditional expectations

$$\begin{aligned} \mu_0(x) &= \mathbb{E}\left[\, B_i(0) \mid X_i = x \,\right], \qquad \mu_1(x) = \mathbb{E}\left[\, B_i(1) \mid X_i = x \,\right], \\ p(x) &= \mathbb{P}\left(\, D_i = 1 \mid X_i = x \,\right), \quad \mu_G(x) = \mathbb{E}\left[\, G_i(1) \mid X_i = x \,\right], \end{aligned} \tag{12}$$

where $\mu_0(x)$ and $\mu_1(x)$ are the expected bills without and with adoption, $p(x)$ is the propensity score, and $\mu_G(x)$ is the expected adoption size, all conditional on covariates $X_i = x$. Collectively, denote these functions as $\Psi = (\mu_0, \mu_1, p, \mu_G)$.

Denoting total savings by $\Delta B(x) = \mu_0(x) - \mu_1(x)$, the estimand is total savings divided by expected system size,

$$\rho(x) = \frac{\Delta B(x)}{\mu_G(x)}, \tag{13}$$

in dollars per kW per year.

To identify $\rho(x)$ in the data, I make use of two assumptions:

Assumption 1 (Selection on observables)**.** $(B_i(0), B_i(1), G_i(1)) \perp\!\!\!\perp D_i \mid X_i$.

This assumption says that conditional on observables, adoption is as good as random with respect to the potential outcomes. Adoption can still depend on unobservables, such as financing, sales effort, preferences for installers, environmental attitudes, or costs of installation, as they do in the demand model. However, these factors must be independent of the potential outcomes after conditioning on X_i. The principal threat to this assumption is that households have private information about their rooftop's solar potential or anticipated changes in their electricity usage that is not captured by current and past usage patterns. Because most solar return on investment calculators are

themselves based on past electricity bills, the richness of the billing data and the ability of machine learning estimators to recover information beyond simple linear relationships make this a plausible assumption; Appendix C.1 discusses this in more detail.

Assumption 2 (Overlap)**.** Propensity scores are bounded away from zero and one: $0 < \underline{p} \leq p(x) \leq \bar{p} < 1$ for all x.

This yields the following identification result for the per-kilowatt savings function:

Proposition 2. *Under Assumptions 1 and 2, the data identify the per-kilowatt savings function:*

$$\rho(x) = \frac{\mathbb{E}\left[\, B_i \mid X_i = x, D_i = 0 \,\right] - \mathbb{E}\left[\, B_i \mid X_i = x, D_i = 1 \,\right]}{\mathbb{E}\left[\, G_i \mid X_i = x, D_i = 1 \,\right]}. \tag{14}$$

Proof. Provided in Appendix C.1. □

Proposition 2 identifies average savings at the sizes households would choose, but the demand model requires the savings that each household would earn at every system size it could choose. Two additional assumptions extend this to average savings at any possible size.

Assumption 3 (No sorting on per-kW gains)**.** $\mathbb{E}\left[\, \frac{B_i(0) - B_i(1,G)}{G} \;\middle|\; X_i = x,\, G_i(1) = G \,\right] = \mathbb{E}\left[\, \frac{B_i(0) - B_i(1,G)}{G} \;\middle|\; X_i = x \,\right]$ for every supported $G > 0$.

Assumption 4 (Proportional savings)**.** $\mathbb{E}\left[\, B_i(0) - B_i(1,G) \mid X_i = x \,\right] / G$ is the same for every supported $G > 0$.

Assumption 3 says that among households with the same billing and usage patterns, those who choose larger or smaller systems do so on the basis of factors like installation costs or preferences that are unrelated to their per-kW savings. Assumption 4 says that savings are proportional to system size. Under NEM 2.0, each kilowatt-hour a system generates offsets usage or is credited at close to the retail rate, so savings are close to proportional unless a system is substantially oversized. Because the demand model offers every household every size, this assumption overstates the savings that a household with low usage would earn from a large system. Under the two assumptions, average savings per kW is constant across sizes, equal to $\rho(x)$.

4.2 Estimating Savings by Decile

To summarize how per-kilowatt savings varies across households, I follow the generic machine learning inference approach of Chernozhukov et al. (2025). With high-dimensional covariates, inference on the full function $\rho(x)$ is generally not possible. Machine learning predictions of $\rho(x)$ are also possibly biased and noisy, so their distribution across households misstates the distribution of savings. However, features of $\rho(x)$, such as group averages, lend themselves to estimation and inference. In what follows, I take deciles of predicted savings as the groups.

Within each decile I estimate the sorted group average treatment effect (GATES),

$$\rho_r = \mathbb{E}\left[\, \rho(X_i) \mid r_i = r \,\right], \tag{15}$$

where r_i is household i's decile of predicted savings. Substituting machine learning estimates of the functions in Ψ into $\rho(x)$ as a plug-in estimate would inherit two biases: regularization bias, because flexible ML estimators trade bias for lower variance, and overfitting bias, because the estimates would be evaluated on the same households as were used to fit them (Ahrens et al., forthcoming). Forming deciles from predictions fit to the same households would also select households on noise, ranking high those whose outcomes happened to be high and exaggerating the differences between deciles even if savings did not vary.

I follow the standard approach in the literature and address these issues with sample splitting and orthogonalization:

Estimation procedure.

1. *Split.* I randomly split the sample into an auxiliary half and a main half.
2. *Fit.* On the auxiliary half, I estimate the functions in Ψ using machine learning.
3. *Evaluate.* On the main half, I evaluate these estimates at each household's covariates X_i. This addresses the overfitting bias because the functions are never evaluated on the households used to fit them.
4. *Sort.* I sort the main-half households into deciles of predicted savings—their plug-in estimate of $\rho(X_i)$ from the functions in Ψ. This addresses the selection-on-noise problem because the deciles are formed from predictions that depend only on X_i and not on the household's own outcomes (Chernozhukov et al., 2025).
5. *Average.* Within each decile, I estimate the GATES as the average of a *Neyman orthogonal signal* (shown below) for $\rho(X_i)$, computed from the estimated functions and the household's outcomes, with the usual confidence interval for a mean (Semenova

and Chernozhukov, 2021). Using the orthogonal signal addresses the regularization bias because it is locally robust to small errors in the estimated functions (Ahrens et al., forthcoming).

6. *Repeat.* I repeat this procedure over 150 random splits and report the medians of the estimates and the medians of the lower and upper confidence bounds. This addresses the dependence of the results on a single, arbitrary split (Chernozhukov et al., 2025).

Each signal is constructed with the general recipe in Ahrens et al. (forthcoming, Appendix B). For total savings and expected system size, this gives signals

$$\begin{aligned}\psi_i^{\Delta} &= \Delta B(X_i) + \frac{1 - D_i}{1 - p(X_i)}\{B_i - \mu_0(X_i)\} - \frac{D_i}{p(X_i)}\{B_i - \mu_1(X_i)\},\\ \psi_i^{G} &= \mu_G(X_i) + \frac{D_i}{p(X_i)}\{G_i - \mu_G(X_i)\},\end{aligned} \tag{16}$$

and for per-kilowatt savings it gives

$$\psi_i^{\rho} = \rho(X_i) + \frac{\psi_i^{\Delta} - \Delta B(X_i)}{\mu_G(X_i)} - \rho(X_i)\,\frac{\psi_i^{G} - \mu_G(X_i)}{\mu_G(X_i)}. \tag{17}$$

Under Assumptions 1 and 2, each signal is unbiased for its target, so that $\mathbb{E}\,[\,\psi_i^{\rho} \mid X_i\,] = \rho(X_i)$ (Semenova and Chernozhukov, 2021).

Because r_i depends on X_i alone, by the law of iterated expectations the GATES is the average of the signal within deciles, $\rho_r = \mathbb{E}\,[\,\psi_i^{\rho} \mid r_i = r\,]$. Predicted savings can therefore form the deciles even though it is a biased estimate of $\rho(X_i)$, because its errors only move households between deciles. That said, worse predictions mean deciles will only weakly separate high- and low-savings households, so the GATES will understate true heterogeneity in savings. Therefore, to make the deciles as informative as possible, I choose the machine learning method for predicted savings based on how much of the variation in savings its groups explain, following Chernozhukov et al. (2025); Appendix C.1 gives details.

The covariates in X_i, which Appendix Table C1 lists, are measures of a household's 2020 billing, load profile, program enrollment, and local climate, as well as indicators for the household's rate schedule, baseline territory, and local area. The split is stratified by utility and adoption status, and the functions in Ψ are estimated separately by utility, so that each can differ with the utility's tariffs. For each function in Ψ, I choose the machine learning method by cross-validated prediction error, and Appendix Table C2 reports the

methods chosen.

4.3 Estimates and Their Use in Demand

Appendix Table C3 shows that the random forest and XGBoost perform about equally well, so I use the average of their predictions, which outperforms both. Appendix Table C5 reports the results by decile. Savings per kilowatt rise steadily from $57 per year in the lowest decile to $376 in the highest: the same kilowatt of solar is worth 6.6 times as much to a household in the top decile as to one in the bottom. High-savings households also install larger systems, so total annual savings differ even more, from $314 to $3,375. Households with the largest predicted savings are those with large bills and summer-heavy usage: the average 2020 bill in the top decile is $3,995, against $621 in the bottom decile, and the ratio of summer to winter bills is 1.65, against 1.13. Appendix Table C6 reports the same results by quintile.

The shares of households in each savings decile vary widely across local areas and, as Appendix Figure C4 shows, line up with where solar has been adopted: across local areas, each additional percentage point of households in the top savings quintile is associated with 0.24 percentage points more cumulative adoption over 2017–2022, and each additional point in the bottom quintile with 0.34 points less.

The billing estimates enter the demand model in three ways. First, under Assumptions 3 and 4, households in savings decile r save $\rho_r G_q$ a year on average at 2024 rates from a system of size G_q, which equation (4) adjusts to the year of adoption at the real growth rate of retail rates, g. Second, the local shares of households in each savings decile, averaged across splits and assumed constant over time, are the observed savings demographics in the demand model. Income and savings are positively correlated, so the demand cells must account for their joint distribution in each local area. The billing data do not record income, so I predict each household's income from its billing and location characteristics with a model estimated on the American Community Survey. The cell weights ω_{cmt} match each local area's income distribution and savings shares while staying as close as possible to the joint distribution these predictions imply. Appendix C.2 describes the income model and how the weights are formed. In these weights, the average correlation between log income and per-kW savings within local areas is about 0.25.

Finally, on the 2021–2022 billing sample that overlaps the demand data, I compute 38 micromoments, each a difference relative to the lowest savings decile or income bin: in the annual adoption rate and the post-ITC price of the chosen product by savings decile, and in the agent density of the chosen installer by savings decile and by income bin.

Appendix C.3 describes how they are computed, and Appendix Table C7 reports them. Households in the top savings decile adopt at a rate 7.0 percentage points a year higher than those in the bottom decile and pay $7,700 more for their systems after the ITC, and lower-income adopters buy from firms with more agents. As in Grieco, Murry and Yurukoglu (2024), matching differences rather than levels requires the model to reproduce only how these outcomes vary with savings and income, while the market shares determine their levels.

5 Structural Model of Supply

I model the supply side as a simultaneous Nash–Bertrand game in which large firms choose prices and salesforces to maximize profits subject to separate installation costs and salesforce costs. The fringe competes in monopolistic competition over price with no salesforce. The model captures both the extensive and intensive margins of salesforce choice; heterogeneous salesforce costs create selection into using agents on the basis of unobserved (to the econometrician) cost draws.

5.1 Firm Objective

Large firms. Define the gross profit per potential household of large firm $f = 1, ..., F$, before accounting for salesforce costs, as a function of the vector of market prices $\boldsymbol{P}_{mt}$ and salesforces $\boldsymbol{a}_{mt}$:

$$\pi_{fmt}(\boldsymbol{P}_{mt}, \boldsymbol{a}_{mt}) = \sum_{j \in \mathcal{J}_{fmt}} (P_{jmt} - C^I_{jmt}) s_{jmt}(\boldsymbol{P}_{mt}, \boldsymbol{a}_{mt}) \tag{18}$$

where P_{jmt} is the price of the firm's product, $\mathcal{J}_{fmt}$ is the set of firm options, C^I_{jmt} is the marginal *installation* cost, and $s_{jmt}(\boldsymbol{P}_{mt}, \boldsymbol{a}_{mt})$ is the market share in equation (11), which depends on the prices and salesforces of all firms in the market.

Sales agents do not change the installation cost. Rather, the firm incurs a constant *selling* cost C^A_{fmt} per agent.[26] This captures the direct salary and commissions cost of agents, as well as the overhead of recruiting, training, and managing the salesforce even if

[26] One might worry that a component of marginal installation cost is actually commissions paid to agents. This does not change the first-order condition for the firm's optimal salesforce choice that recovers C^A_{fmt}. However, it does change the interpretation of this cost. To test for this, I regress the recovered installation costs on the density of agents in the market. If these costs are positively correlated with the use of agents, that would provide evidence for this concern. Appendix D.4 presents this test, which fails to reject the null of no relationship.

they do not generate any sales. It also maps onto a standard accounting treatment of the cost of goods sold (COGS) versus selling, general, and administrative (SG&A) expenses.

Since the total number of agents is $A_{fmt} = a_{fmt}N_{mt}$, the firm chooses price and salesforce to maximize total profits net of salesforce costs:

$$\max_{P_{jmt},\, a_{fmt} \geq 0} \pi_{fmt}(\boldsymbol{P}_{mt}, \boldsymbol{a}_{mt})N_{mt} - C^A_{fmt} a_{fmt} N_{mt}. \tag{19}$$

As in Goeree (2008)'s study of advertising, I model salesforce costs as log-normal, with a cost draw that firms observe but the econometrician does not:

$$\log C^A_{fmt} = \boldsymbol{\beta}'_A \boldsymbol{W}_{fmt} + \sigma_{fmt}\iota_{fmt}, \qquad \iota_{fmt} \sim N(0,1). \tag{20}$$

I specify $\boldsymbol{W}_{fmt}$ to contain region and year fixed effects, the log wage of sales-related occupations in the market, the firm's years in the market, and an indicator for whether the firm's headquarters is outside the market. The scale of the cost shock is given by σ_{fmt}, the standard deviation of log agent cost.

Firms with agents reveal their marginal agent cost exactly through the salesforce first-order condition, so the costs I recover for them do not depend on the distribution of C^A_{fmt}. Rather, specifying the distribution of agent costs is only necessary to describe the costs of firms that do not hire agents, which allows me to characterize selection into using agents and simulate counterfactuals that change the number of firms with agents.

Fringe firms. I assume the fringe is composed of small local installers that do not employ sales agents and engage in monopolistic competition. I derive their problem and solution in Appendix B.2. It is the same solution one would get from a single-product firm in Bertrand competition as it becomes infinitesimally small relative to the market. This is the substance of the monopolistic competition assumption, whereby fringe firms act as price takers that do not internalize their effect on the market. Large firms take this fringe behavior as given when choosing their own price and salesforce.

5.2 Profit-Maximizing Price and Salesforce

While firms choose prices and salesforces simultaneously, the optimal choice of price and salesforce can be solved separately.

Prices. The first order condition for price yields the standard multiproduct optimal markup equation for each firm-size option j evaluated at the optimal price and salesforce:

$$s_{jmt} + \sum_{k \in \mathcal{J}_{fmt}} (P^*_{kmt} - C^I_{kmt}) \frac{\partial s_{kmt}}{\partial P_{jmt}} = 0 \tag{21}$$

Stacking these conditions for all large firm-size options gives the following matrix form:

$$\boldsymbol{P}^*_{mt} - \boldsymbol{C}^I_{mt} = -\boldsymbol{\Delta}^{-1}_{mt} \boldsymbol{s}_{mt}, \tag{22}$$

where $\boldsymbol{\Delta}_{mt}$ is the matrix of $\partial s_{kmt}/\partial P_{jmt}$ derivatives of the market shares with respect to prices, interacted with product ownership.[27] Inverting this condition allows installation costs to be recovered from observed prices, salesforces, and demand elasticities.

Agents. Given a candidate choice of agents, firms set prices to maximize gross profit before salesforce costs, $\pi^*_{fmt}(a_{fmt}) = \pi_{fmt}(P^*(a_{fmt}), a_{fmt})$. The remaining choice for the firm is then to pick a salesforce to maximize profits net of salesforce costs. This is represented by the following KKT conditions, omitting subscripts for readability and with λ the Lagrange multiplier on the non-negative agent constraint:

$$\begin{aligned} a^* \geq 0, \quad \lambda \geq 0, \quad & \text{(primal and dual feasibility)} \\ a^* \lambda = 0, \quad & \text{(complementary slackness)} \\ \underbrace{\pi^{*'}(a^*)}_{\substack{\text{marginal} \\ \text{gross profit}}} - \underbrace{C^A}_{\substack{\text{marginal} \\ \text{agent cost}}} + \lambda = 0. \quad & \text{(stationarity)} \end{aligned} \tag{23}$$

Stationarity corresponds to the Dorfman and Steiner (1954) condition in the advertising literature. It equates the marginal benefit of salesforce to its marginal cost at the interior, when $\lambda = 0$. Firms compare their best interior point to the alternative of zero salesforce. If marginal cost exceeds the marginal benefit, then $\lambda > 0$ represents this wedge and the firm hires no agents.

By the envelope theorem, the effect of agents on the firm's optimal prices drops out, so the marginal benefit of salesforce is the markup on the additional demand that agents create:

$$\pi^{*'}(a) = \sum_{j \in \mathcal{J}_f} (P^*_j(a) - C^I_j) \frac{\partial s_j(P^*(a), a)}{\partial a}, \tag{24}$$

[27] Any element j, k of $\boldsymbol{\Delta}_{mt}$ is zero if j, k are not options at the same firm, and is $\partial s_{kmt}/\partial P_{jmt}$ otherwise.

Agents are most valuable to firms with high markups in markets where households have high demand for solar.

5.3 Estimation

Given demand, I estimate supply in two steps. First, as is standard, I recover marginal installation costs by inverting equation (22) at observed prices, demands, and salesforces. This yields gross profits for each firm and their derivatives with respect to firm salesforce. In the second step, I use these objects to estimate agent costs. The intuition is that a firm hires no agents whenever its first agent would cost more than it earns, so observed salesforces censor the distribution of marginal agent costs, as in a Tobit model. I estimate these agent costs by maximum likelihood.

The marginal profit of an additional unit of a_{fmt} is $m_{fmt}(a) = \pi^{*'}_{fmt}(a)$. Define the standardized cost draw $\iota_{fmt} = (\log C^A_{fmt} - \boldsymbol{\beta}'_A \boldsymbol{W}_{fmt})/\sigma_{fmt}$, which given the distribution of C^A_{fmt} is i.i.d. standard normal. When gross profit is concave in salesforce, m_{fmt} is decreasing in a_{fmt}, and the KKT conditions imply a threshold $\bar{\iota}_{fmt}$ for the cost draw

$$\bar{\iota}_{fmt} = \frac{\log m_{fmt}(0) - \boldsymbol{\beta}'_A \boldsymbol{W}_{fmt}}{\sigma_{fmt}},$$

such that the firm chooses a positive salesforce if and only if $\iota_{fmt} \leq \bar{\iota}_{fmt}$. The threshold depends on the marginal profit of the first agent, $m_{fmt}(0)$, which I compute by re-solving the firm's profit-maximizing price at zero agents, holding rivals' prices and agents at their observed values. It is defined whenever $m_{fmt}(0) > 0$, which empirically holds for all firm–markets in the sample. Agent-using firms therefore reveal exactly their cost draw, $\iota_{fmt} = (\log m_{fmt}(a_{fmt}) - \boldsymbol{\beta}'_A \boldsymbol{W}_{fmt})/\sigma_{fmt}$, while non-agent firms only reveal that their cost draw exceeds the threshold $\bar{\iota}_{fmt}$.

Intuitively, the scale is identified by how agent use varies with estimated returns to the first agent: the more dispersed agent costs are, the less agent use varies with $m_{fmt}(0)$.[28] In the main specification, I allow the scale of the cost shock to differ in markets other than the firm's headquarters, $\sigma_{fmt} = \sigma_{\text{HQ}}\, 1\{\text{HQ}\}_{fmt} + \sigma_{\text{non-HQ}}\, 1\{\text{non-HQ}\}_{fmt}$, which captures differences in familiarity with local markets and ability to manage a remote salesforce.

[28] While Goeree (2008) normalizes this scale to one, my setting has sufficient variation to identify it directly.

Likelihood function. The log-likelihood of the recovered log marginal agent costs sums over the respective contributions of agent and non-agent firms:

$$\log \mathcal{L} = \sum_{fmt} \underbrace{1\{a_{fmt} > 0\} \left(\log \varphi(\iota_{fmt}) - \log \sigma_{fmt}\right)}_{\text{Contribution of agent-using firms}} + \underbrace{1\{a_{fmt} = 0\} \log\left(1 - \Phi(\bar{\iota}_{fmt})\right)}_{\text{Contribution of non-agent firms}}. \quad (25)$$

The first term corresponds to the density of the cost draw for agent-using firms, while the second term corresponds to the probability that non-agent firms' cost draws exceed the threshold. The resulting first-order conditions set the average standardized cost residual to zero and the average of its square to one, matching the mean and variance of the standard normal cost draw. In these conditions, agent-using firms contribute their revealed residuals, and non-agent firms contribute the expected residual given that their cost draw exceeds the threshold, which Gouriéroux et al. (1987) term generalized residuals. Appendix D.3 derives these conditions.

I compute standard errors using two-way clustering by firm and by market to account for correlation in the residuals across markets for the same firm and across firms in the same market.

5.4 Demand and Supply Equilibrium

Equilibrium. An equilibrium in market mt is a vector of prices $\boldsymbol{P}^*_{mt}$ and salesforces $\boldsymbol{a}^*_{mt}$ such that, given the prices and salesforces of its rivals, each firm's prices satisfy its pricing first-order conditions and each firm's salesforce satisfies the KKT conditions (the fringe faces no choice of salesforce). In counterfactuals, I hold fixed the product menus, product qualities ξ_{jmt}, local area effects, installation costs, and agent costs, set salesforces at their counterfactual values, and solve for the new price equilibrium, as described in Appendix D.5.

I verify that every equilibrium I compute satisfies each firm's second-order conditions. With multiproduct firms and heterogeneous consumers, there is no guarantee that the equilibrium is unique (Morrow and Skerlos, 2011). In practice, I do not encounter multiple pricing equilibria from different starting values.

6 Model Estimates

6.1 Demand

Table 3 reports the demand estimates. Households are price sensitive, and less so at higher incomes. Appendix Figure D1, panel (a), shows the distribution of own-price elasticities across products, with a median of −4.85. This is in line with prior estimates from California and other markets.[29]

Agents are effective at shifting demand. Appendix Figure D1, panel (b), shows that among products of firms with agents, the elasticity of demand with respect to the firm's agent density has a median of 0.14, about an order of magnitude larger than the median of recent estimates for television advertising (Shapiro, Hitsch and Tuchman, 2021). Given the high-touch nature of in-person sales, this is not surprising. Firms choose their marketing mix as a tradeoff between more expensive but more effective agents against cheaper but less influential traditional marketing channels. To get a sense of how this compares to the effect of price, a 10 percent larger salesforce raises a firm's demand as much as a 0.3 percent price cut.

Agents are more effective with households that save more from solar ($\gamma_r > 0$) and with lower-income households ($\gamma_y < 0$), but these differences are small: γ_c ranges from 0.13 to 0.20 across demographic cells.[30] Prior literature has reported similar findings. Kahn-Lang (2024) finds that lower-income households are more receptive to in-person marketing from retail electricity providers, while Allcott and Sweeney (2017) find evidence that retail sellers of appliances selectively target information about energy efficiency to potential customers who seem receptive while not discussing it with those who are not, which could waste the seller's time and lower the likelihood of a sale.

I estimate the agent reach parameter to be $\phi = 2.50$. This can be interpreted as follows: in the offer model, households receive a firm's offers at a rate proportional to $1 + \phi a_{fmt}$, so $\phi = 2.50$ implies that each agent per 100,000 potential adopters adds 2.50 times as many offers as all of the firm's other channels combined. The median firm with agents, at 1.6 agents per 100,000 potential adopters, therefore makes about 5 times as many offers as it would without them ($1 + 2.50 \times 1.6 \approx 5$).

[29] Alongside his primary estimate of −3.46, Bradt (2026) reports a summary of existing estimates that range between −0.65 (Gillingham and Tsvetanov, 2019) to −6.6 (De Groote and Verboven, 2019). Estimating a more similar demand model to the one in this paper, Wang and Houde (2026) estimate mean own-price elasticities of −3.93 to −4.15 across specifications.

[30] Recall that for the microfoundation in Section 3.1 to be consistent with the assumption of random utility maximization, it must be that $\gamma_c \in (0, 1]$. Note that the estimates satisfy this condition, even though I do not impose it in estimation.

The unobserved taste for solar has a standard deviation of $\sigma_\nu = 3.25$, so households differ widely in how close they are to the margin of adopting. Appendix Figure C8 compares the model with the billing data. The model reproduces how adoption rises with bill savings, from 0.7 percent a year in the lowest decile to 7.7 percent in the highest in the data, against 0.7 to 8.3 percent in the model. It also reproduces how prices and system size, which estimation does not use, rise with savings, although it understates both in levels.

Table 3, panel (b), reports tests for weak instruments following Lewis and Mertens (2026), who extend first-stage F statistics and the critical values of Stock and Yogo (2005) to several endogenous regressors with clustered standard errors. Because the full model is nonlinear, I apply the tests to an auxiliary linear first stage of price and $\log(1 + \hat{\phi}a)$ on the excluded instruments. At the 5 percent level, the tests reject weak instruments for price and for agents separately at a 10 percent bias tolerance. A joint test fails to reject at 10 percent bias tolerance but does reject at 15 percent. Weak instruments are therefore unlikely to bias the estimates substantially.

6.2 Supply

The distribution of marginal installation cost estimates is shown in Figure 3, panel (a). The average marginal installation cost is \$3.62 per watt, weighted by watts of installed capacity. This implies average markups of \$0.98 per watt, or 21 percent of price. Firms with agents have higher installation costs and markups than the fringe and, by a smaller margin, large firms without agents. Firms with agents have an average marginal installation cost of \$3.68 against \$3.15 for the fringe and \$3.55 for large firms without agents. In markups, the comparison is \$1.00 against \$0.88 and \$0.94 per watt.

This ordering of costs and markups is not guaranteed. Since an agent earns the markup on each sale it brings in, agents pay off for firms with high markups, which could come from low costs or from households' stronger preferences for their products. Had they come from low costs, agents would raise productive efficiency. Instead, firms with agents have higher costs, so their higher markups reflect those preferences, which implies that agents lower productive efficiency.

Table 4 reports the agent cost estimates, with the primary specification in column (2). The cost shifters have the expected signs, though they are imprecisely estimated: agent costs rise with the local sales wage and fall with the firm's years in the market. Agent costs also fell over the sample period, most sharply during and after the COVID-19 pandemic: relative to 2017, the year effects imply that the marginal cost of an agent was

57 percent lower (0.84 log points) in 2020 and 47 percent lower (0.64 log points) in 2022. This suggests door-to-door sales was less costly while more households were at home.

Away from headquarters, agent costs differ mainly in their dispersion. With a common scale, in column (1), agents appear cheaper away from headquarters. Once the scale may differ, in column (2), this shift disappears: the median agent cost is about the same away from headquarters, but costs are more dispersed, with a scale of 2.36 compared with 1.76 in the headquarters market. This is consistent with firms facing more varied costs of recruiting and managing a salesforce outside their home market. These estimates are robust to market fixed effects, shown in column (3). Finally, the fit of the model on the extensive margin is good, predicting that 22.5 percent of firm–markets hire agents, compared with 23.0 percent in the data.

Among firms that hire agents, the marginal agent cost in Figure 3, panel (b), has a median of \$9,700 per year and a mean of \$21,400, and 10 percent of marginal agent costs exceed \$50,000, reflecting a long right tail of firms for which agents are costly yet still profitable. This dispersion mirrors the wide range of earnings that firms advertise in the job postings for door-to-door sales described in Appendix A.5. Of course, it is important to recognize that these estimated economic costs may differ from accounting costs for several reasons, in particular the marginal sales agent may bring in less revenue than the average agent, which under commission pay would cost the firm little, and sales agents may work less than full-time, so their annual cost may be less than the compensation posted for a productive, full-time salesperson.

7 Sales Agents and Competition

This section measures the two ways that agents raise demand, market expansion and business stealing, what competition between salesforces does to the number of agents that firms hire and to prices, and how agents interact with environmental policy. Section 7.1 asks how much agents expand the market, how much of what they sell is taken from rivals, and how far business stealing leads firms to hire past the salesforces that maximize joint industry profits. Section 7.2 asks whether firms that could not compete on their salesforces would compete harder on price, and how agents change which firms sell. Section 7.3 asks what the adoption that agents add costs relative to its environmental benefits, and how agents and the ITC interact.

7.1 Market Expansion and Business Stealing

Table 5, columns (1) and (2), compare the observed equilibrium with a counterfactual equilibrium in which every salesforce is zero. Agents raise installations by 17 percent, or 103,000 installations, as firms with agents gain 129,000 installations while other installers lose 26,000. Agents add 626 MW (DC) of capacity over the six years of the sample, which produces around 1.0 TWh of electricity per year. For comparison, this is more total generation than what the largest utility-scale solar farms in California provide. Thus, the presence of sales agents in the residential solar market has had a large effect on the amount of solar capacity installed in California.

Despite all this additional capacity, however, the increase in industry profits is small. Firms with agents earn $0.86 billion more in gross profit, but this comes at a high direct sales cost, as they spend $0.47 billion on agents, for a net gain of only $0.39 billion. Meanwhile, firms without agents lose $0.17 billion in gross profit, 43 percent of the net gain of firms with agents. Industry profit therefore rises by only $0.22 billion.

Rivals' losses do not enter a firm's decision to hire agents: a firm keeps hiring until its last agent pays for itself, no matter if those returns come at the expense of other firms. This is a classic prisoner's dilemma, as firms hire more agents than would maximize joint industry profits (Sinkinson and Starc, 2019). How far firms over-hire, and what the extra agents buy, matters for policy toward door-to-door sales. If the agents that firms hire past the joint-profit level mainly bring in households that would not otherwise adopt, restricting salesforces would mainly reduce adoption. If they mainly move households between installers, salesforces could shrink substantially with little loss of adoption.

To measure how much of the last agent's return comes at rivals' expense, I decompose the installations that one more agent adds to its firm into those new to solar and those taken from rivals. Let q_f be firm f's installations and $Q = \sum_{f'} q_{f'}$ total installations. One more agent at firm f brings dq_f/dA_f installations:

$$\frac{dq_f}{dA_f} = \underbrace{\left(\frac{dQ}{dA_f}\right)}_{\text{market expansion}} + \underbrace{\left(-\sum_{f' \neq f} \frac{dq_{f'}}{dA_f}\right)}_{\text{business stealing}}. \tag{26}$$

The total derivatives account for the equilibrium price response.

At the margin, much of what an agent earns for its firm is taken from rivals. Table 6 shows that the last agent brings 1.4 households a year into the market, less than half the 3.5 that the average agent brings. A firm hires until the last agent's gross profit covers its cost, $16,000 a year, but industry gross profit rises by only $9,200, so rivals lose 43

cents of each dollar that the last agent earns for its firm. This business-stealing externality is similar in size to the one that Sinkinson and Starc (2019) find for statin advertising, where one more ad for one of the two advertised drugs takes 27 to 43 percent as many new patients from the other as it brings to its own.

To measure how much this externality inflates salesforces, I compute the salesforces that maximize joint industry profits. This is a semi-collusive equilibrium, in which firms choose their salesforces jointly but still compete on price, similar to the exercise in Shapiro (2018) for drug advertising. Suppressing market subscripts, each firm's salesforce satisfies the modified first-order condition from equation (23):

$$\underbrace{\pi_f^{*\prime}(a^*)}_{\substack{\text{marginal}\\ \text{gross profit}}} + \underbrace{\sum_{f' \neq f} \sum_{k \in \mathcal{J}_{f'}} \left(P_k^* - C_k^I\right) \frac{\partial s_k}{\partial a_f}}_{\substack{\text{effect on rivals'}\\ \text{gross profit}}} - \underbrace{C_f^A}_{\substack{\text{marginal}\\ \text{agent cost}}} + \lambda_f = 0, \tag{27}$$

where the second term internalizes the effect of firm f's salesforce on rivals' gross profits.

Maximizing joint industry profits would cut salesforces by more than half but installations by little. Table 5, column (3), shows that firms would hire 55 percent fewer agents, yet installations would fall by only 4 percent, from 715,000 to 686,000. This point bears emphasis: one might think that cutting the number of agents in the market in half would dramatically reduce the number of new adopters brought into the market, but this is not the case, as 72 percent of the increase in adoptions relative to the world without agents would remain. The reason is that the agents that firms would drop bring few households in from non-adoption: only 1.8 each, against 3.5 for the average agent. Much of the salesforce therefore exists because of business stealing rather than market expansion. This contrasts with the positive advertising spillovers in Shapiro (2018), where antidepressant ads raise rivals' sales, so that an industry cooperative would advertise four times as much as competing firms. Coordination would also raise the industry's profit gain from agents by more than 30 percent, from 5.7 to 7.5 percent of industry profit without agents, and firms with agents would earn more, because they would save more on agents than they would lose in gross profit.

7.2 Effects on Prices

I find that compared to the equilibrium without agents, prices are only slightly higher with agents: an increase of 5.1 cents per watt, or 1.1 percent. Figure 4 decomposes this increase. Most of it, 4.5 cents, comes from shifting sales toward installers with higher

installation costs and higher markups, and only 0.6 cent comes from firms raising the prices of the same products.

Agents raise prices mainly by shifting sales toward installers with higher installation costs. Installation costs account for 3.2 of the 4.5 cents, and the higher markups of these installers for the rest. Part of their higher markups reflects the demand that agents create, but even without agents these firms' products have higher perceived quality. Because firms with agents have higher installation costs, as Section 6.2 shows, this reallocation also lowers productive efficiency: households who would adopt anyway switch to costlier installers, which costs $55 million in real resources.[31]

Selling costs barely reach prices, and price competition changes little. Firms spend $0.10 per installed watt on agents across the market, yet holding the product fixed, agents raise prices by only 0.6 cent per watt. Appendix D.4 shows that a firm's recovered installation costs are no higher in the markets where it uses agents, so what firms pay their agents does not enter the cost that drives their prices. Agents raise concentration, but Appendix Figure A2, panel (b), shows that the median market's Herfindahl–Hirschman Index is still only about 730 with agents, against 560 without, and the households that agents bring in are more price-sensitive, which offsets part of the softer competition.

These results cut against implicit arguments that high customer acquisition costs are a direct cause of high installation prices, and that prices would be substantially lower without agents (Connelly, 2021; Ramasamy et al., 2025). Like Dubois, Griffith and O'Connell (2018), who analyze a counterfactual ban on advertising for potato chips, I find that a ban lowers prices modestly, but for a different reason. In their setting, prices fall by 4 percent because demand becomes more sensitive to price and price competition intensifies. Here prices fall without agents mainly because sales reallocate toward lower-cost installers, with little change in price competition. These differences demonstrate the importance of modeling price competition and market heterogeneity in evaluating the effects of sales.

7.3 Environmental Benefits and Subsidies

Rooftop solar is often promoted on environmental grounds. Agents can be viewed as a technology for increasing solar adoption that firms voluntarily invest in. To evaluate how business-stealing incentives affect the environmental returns to agents, I follow Sexton et al. (2021) and value each watt by the damages from carbon dioxide and local air

[31] To separate households who would adopt anyway from new adopters, I assume that each household's idiosyncratic preference shocks are the same in both equilibria. The probability that a household moves from one choice to another then has a closed form (de Palma and Kilani, 2011).

pollution that its generation avoids, which vary across locations.[32] Each watt of solar has an environmental benefit of $1.01 on average.

Spread over all installations, agent costs appear small at only $0.10 per installed watt (Appendix Figure D2, panel (a)), but this measure is misleading because most of these installations are inframarginal and would have happened without agents. The relevant measure is *additionality*: what agents cost per watt that they add (Aspelund and Russo, 2026). Relative to a ban, firms spend $0.75 on agents per added watt (panel (b)), still less than the $1.01 in benefits.

This average cost obscures the fact that business stealing pushes firms to hire agents well past the point where they cost less per added watt than the environmental benefits they bring. Under salesforces that maximize joint industry profits, firms would spend only $0.47 on agents per added watt, while the 16,000 agents that firms hire beyond those salesforces cost $1.40 per added watt. As Section 7.1 shows, later agents bring few new households into the market, but firms hire them anyway because much of what they earn is taken from rivals. Thus, the last agent costs $1.88 per added watt, nearly twice its environmental benefit.

I also consider the interaction between agents and subsidy policy via the ITC, and find that agents and subsidy policy reinforce each other. By reducing upfront costs, the ITC raises demand for solar and the return to agents. Table 5, column (4), shows that without the ITC, firms would hire 6,500 agents instead of 29,300, 78 percent fewer. Agents in turn extend the reach of the subsidy. Comparing column (1) with column (4)—observed equilibrium versus no ITC—shows that the ITC adds 413,000 installations; comparing column (2) with column (5)—no agents versus no ITC and no agents—shows that the ITC adds only 338,000 installations when firms cannot use agents. With agents, the subsidy is able to add 22 percent more installations. The ITC costs taxpayers about the same per added installation in both cases, $15,100 with agents against $15,800 without, so agents amplify the reach of the subsidy without raising its cost per added installation.

8 Households and Welfare

This section asks whether the households that agents bring into the market gain financially, and combines the effects of agents into social welfare.

[32] I use the California Public Utilities Commission's social cost of carbon, $62.40 per metric ton in 2023 dollars, and its 3 percent real discount rate (California Public Utilities Commission, 2024) to value future emissions reductions. Appendix E.2 further describes the calculation.

8.1 Financial Returns to Households

I measure households' financial returns by the internal rate of return on each system: the real discount rate at which the present value of the household's bill savings over the 25-year life of the system equals the price it pays after the ITC. I focus on these returns for two reasons. First, they directly measure the financial harm that primarily concerns regulators, that is, whether households lose money at typically observed lending rates. A regulator may have a mandate to protect households from financial harm even if adoption also gives them some value beyond its financial return, such as a warm glow or a sense of energy independence. Second, financial returns do not depend on how agents enter utility or on households' discount rates (Dixit and Norman, 1978; Grossman and Shapiro, 1984), so they can be compared directly to the interest rates at which households borrow, much as abatement costs are compared to a social cost of carbon (Gillingham and Stock, 2018). Consumer surplus, by contrast, rests on the demand model's implicit discount rates, which rationalize households' adoption decisions and are well above market interest rates because they absorb behavioral frictions, beliefs, and credit constraints. Under revealed preference, a household that adopts at a financial loss must have preferences for solar that outweigh this loss, so consumer surplus cannot treat its adoption as a mistake. I return to consumer surplus in Section 8.2.

The households that agents bring into the market earn returns similar to those of households who would adopt anyway. Figure 5, panel (a), shows that the average agent-induced adopter earns 8.4 percent, against 8.9 percent for inframarginal adopters. Returns rise with income, and panel (b) shows that agents sell slightly worse deals at every income, about 0.35 percentage points less than inframarginal adopters at the same income. The gap mostly reflects where agent-induced adopters buy: nearly all of them buy from firms with agents, which sell lower-return systems to all of their customers.

Whether these returns make households better off depends on the rate at which they borrow. Over this period, only about a fifth of U.S. residential solar systems were bought with cash; the rest were financed with loans, leases, or power purchase agreements (O'Shaughnessy and Barbose, 2026). Figure 6 plots the share of adopters whose return exceeds each real borrowing rate. Solar loans carry dealer fees that make them cost more than their stated rates (Consumer Financial Protection Bureau, 2024), and a typical solar loan costs about 5.5 percent in real terms.[33] At this rate, 83 percent of agent-induced

[33]Lenders add a dealer fee to the loan principal, so borrowers repay more than the cash price they receive. I compute the effective rate at which the cash price equals the present value of the borrower's payments, assuming a 20 percent fee, a 25-year term, prepayment of the ITC's value by month 18, and repayment after 8 years, which is typical. A loan at a stated 3.5 percent then costs 5.5 percent a year in

adopters gain, on average $8,800 over the life of the system.

Many agent-induced adopters lose money at higher rates, especially those with lower incomes. Agents bring disproportionately many lower-income households into the market, as Appendix Figure E1 shows, and at 5.5 percent, 25 percent of agent-induced adopters in the bottom quartile of income lose money, against 17 percent overall. This is a concern if lower-income households also face worse financing terms. At the high end of stated rates and fees, a solar loan costs about 12 percent in real terms, and at this rate 86 percent of agent-induced adopters lose money. Whether agents leave the households they bring in better off therefore depends on how those households finance their systems.

Business stealing also affects households who would adopt anyway. Households that switch to a firm with agents earn 1.0 percentage points less than they would at the firm they would otherwise choose, a loss of $0.11 billion at 5.5 percent. Prices barely change for households who buy the same product with and without agents. Appendix Figure E2 shows that, netting these losses against the $0.90 billion that agent-induced adopters gain, agents raise households' financial returns by $0.79 billion at this rate.

8.2 Social Welfare

I measure the welfare effect of agents as the sum of changes in consumer surplus, producer surplus, government spending, external benefits, and the utility account (Appendix E). Consumer surplus is the expected maximum utility of Small and Rosen (1981), which counts the agent term in utility as value to households, as when agents provide information or lower the hassle of adoption (Butters, 1977; Grossman and Shapiro, 1984).[34] Producer surplus is gross profit net of agent expense. Government spending is the ITC outlay, valued dollar for dollar. External benefits are the avoided environmental damages of Section 7.3. Finally, net metering credits exports at the retail rate, so each installation shifts part of the utility's fixed costs onto other ratepayers (Borenstein, 2024). I count this cost shift as the bill revenue the utility loses net of the cost of the electricity it no longer supplies, discounted at a 5 percent utility cost of capital.

Most of the private gains from agents go to households. Table 5, panel (b), column (2), shows that banning agents would lower consumer surplus by $0.86 billion, close to the $0.79 billion by which agents raise households' financial returns at the 5.5 percent rate of

real terms, at 2 percent expected inflation. At 7 percent with a 30 percent fee, the top of the reported ranges, it costs 11.7 percent.

[34] Under the alternative extreme view that agents only distort households' choices, the agent term is not value to households, and agents instead lower consumer surplus by $0.22 billion.

a typical solar loan, though these measures are not necessarily comparable, as Section 8.1 discusses. Consumer surplus measures the implied preference value that explains purchase decisions, while financial returns are an objective value that is not necessarily decision relevant. By either measure, households gain far more than firms, which gain only $0.22 billion.

Agents nonetheless lower social welfare, because much of what households gain is paid for by others. The ITC on the added installations is a transfer of $0.89 billion from taxpayers. Net metering is a transfer from the utility to adopters: the utility loses $3.16 billion in bill revenue, of which only $0.65 billion is offset by the electricity it no longer supplies, and it must eventually raise rates to cover the rest, the cost shift at the center of the debate over net metering (Borenstein, 2024). Even counting the $0.63 billion in environmental damages that the added installations avoid, agents lower social welfare by $1.68 billion. Without the cost shift, the other components sum to a gain of $0.83 billion.

9 Conclusion

This paper asks whether selling mainly brings new consumers into a market or moves them between firms. In residential solar, the answer is both: the first salespeople a firm hires bring many new households into the market, while the final salespeople bring few, and much of what they earn is taken from rivals. Whether competition then leads firms to sell more or less than they would jointly depends on who captures the households that selling brings in. Unlike advertising, which can raise demand for a whole category, and where not internalizing spillovers leads firms to advertise less than they would jointly (Shapiro, 2018), door-to-door salespeople are tied to a particular firm and work hard to bring households to that firm. Firms therefore hire more salespeople than they would jointly, and competition spends much of the gain from market expansion on selling itself.

These results bear on current debates over policy and consumer protection in solar, as well as tensions therein. The average salesperson costs less than the environmental benefit it adds, but the marginal salespeople cost far more. The ITC sustains much of the salesforce, and salespeople in turn extend its reach. Because net metering shifts costs onto other ratepayers, the adoption that salespeople add lowers social welfare. Reforms that reduce this cost shift, such as California's NEM 3.0 in 2023, do so by lowering households' returns, which raises the risk that the households salespeople bring in lose money under typical financing terms.

More broadly, this paper speaks to the role of sales effort in markets for new, complex

technologies, where many potential customers lack information or face high search costs. The forces underlying this analysis are also present in the markets for heat pumps, electric vehicles, and energy-efficient appliances, where subsidies raise not only what households will pay but also what firms will spend to sell to them. New developments that change how firms reach customers have the potential to transform these markets, such as online platforms that let households compare quotes (Dorsey, 2024) and personalized digital ads that lower the cost of targeting and contacting households. Whether such developments expand markets or mainly reallocate customers across firms, and whether they create more surplus than they cost, are central questions for the design and regulation of these markets.

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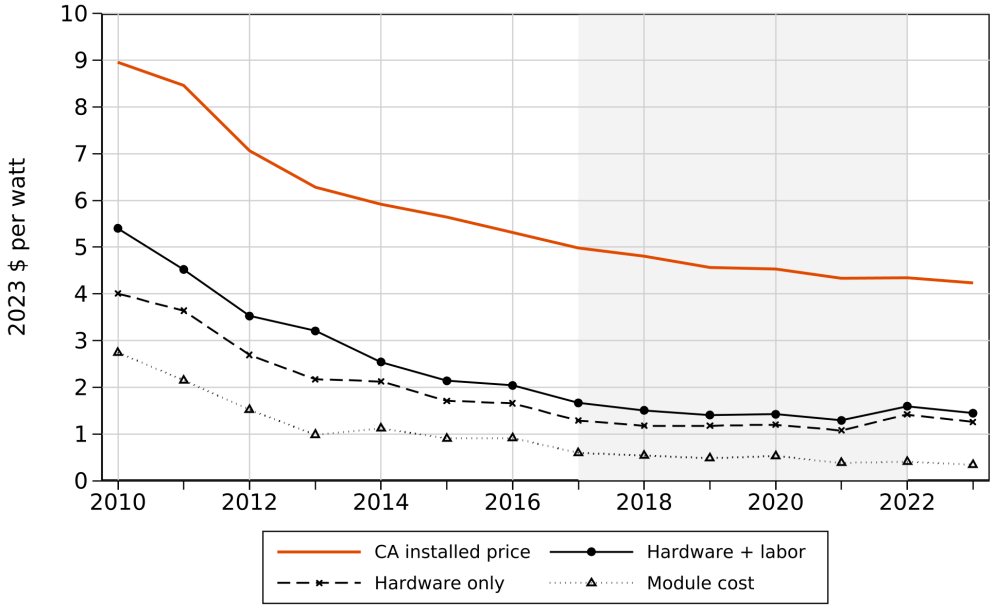
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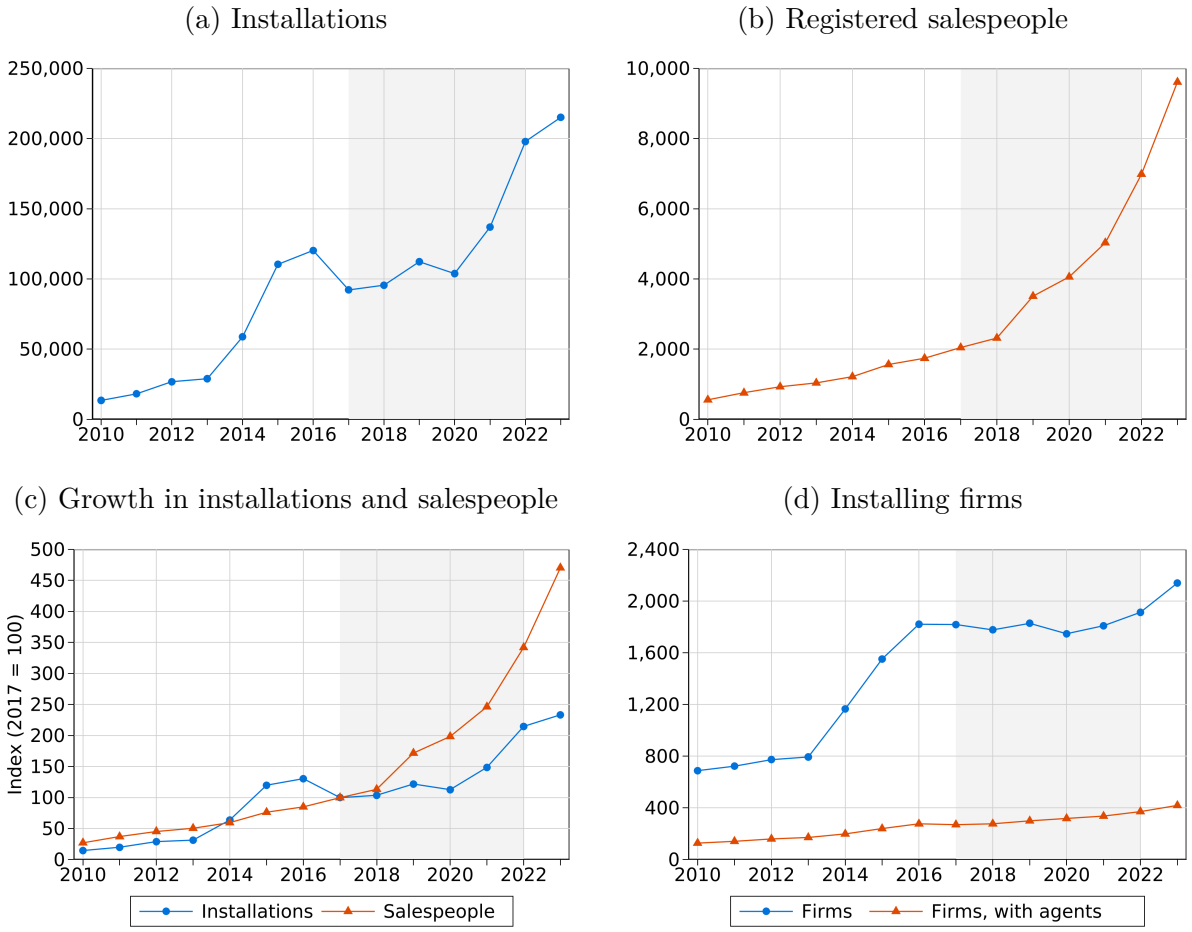
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Figure 1: Installed price and hardware and labor costs of residential solar, 2010–2023



Notes: All series are in real 2023 dollars per watt, deflated with the CPI-U. The California installed price is the size-weighted mean price per watt of host-owned residential systems in the combined Tracking the Sun and California interconnection data, trimmed at the size-weighted 0.5th and 99.5th percentiles; third-party-owned systems and systems installed with batteries are excluded. Hardware plus labor and hardware only are NREL's national residential cost benchmarks (Ramasamy et al., 2025), with the benchmark's module cost replaced by the module cost series. The module cost is the average value of photovoltaic module shipments from EIA Form EIA-63B. The shaded region marks the 2017–2022 sample period.

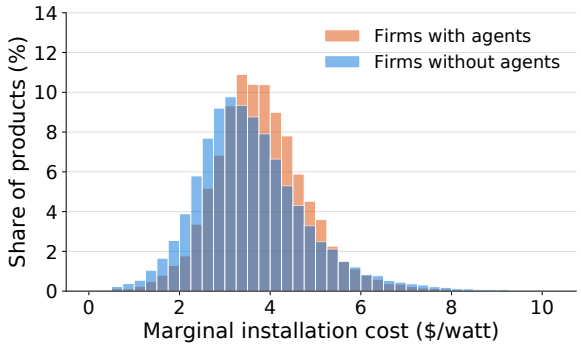
Figure 2: Installations, salespeople, and installing firms in California, 2010–2023



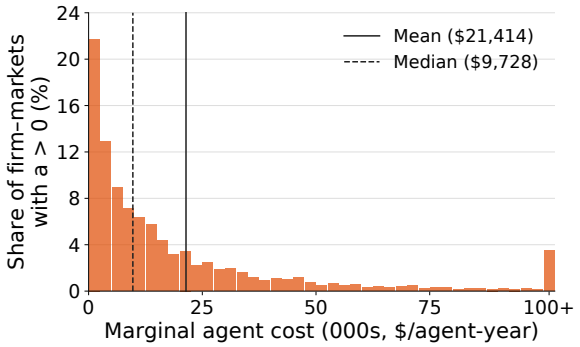
Notes: Residential rooftop installations by identified installers in the 30 regions, dated by application. Salespeople are distinct registered home improvement salespersons, counted once per year, whose home is in one of the regions and who are attached to a firm that has made its first installation. Firms are those with at least one installation in the year; firms with agents are those with at least one registered salesperson in the same year. The shaded region marks the 2017–2022 sample period.

Figure 3: Marginal installation and agent costs

(a) Marginal installation cost

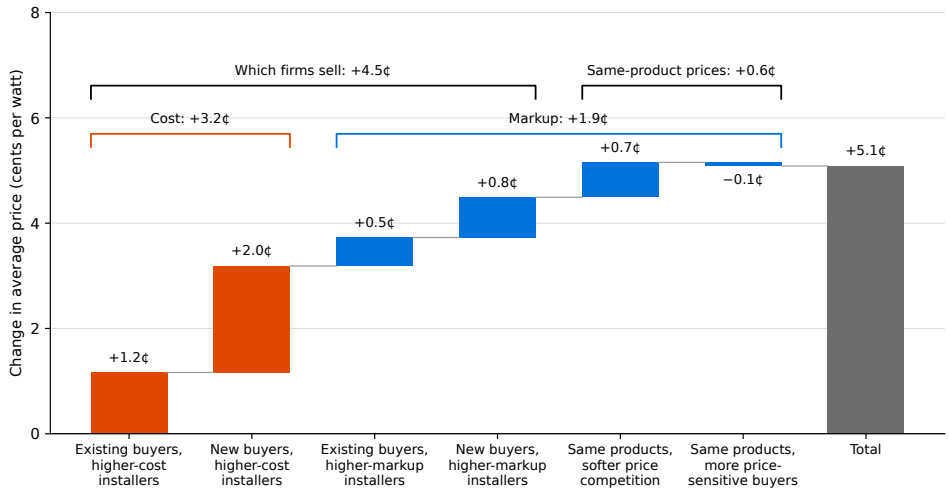


(b) Marginal agent cost



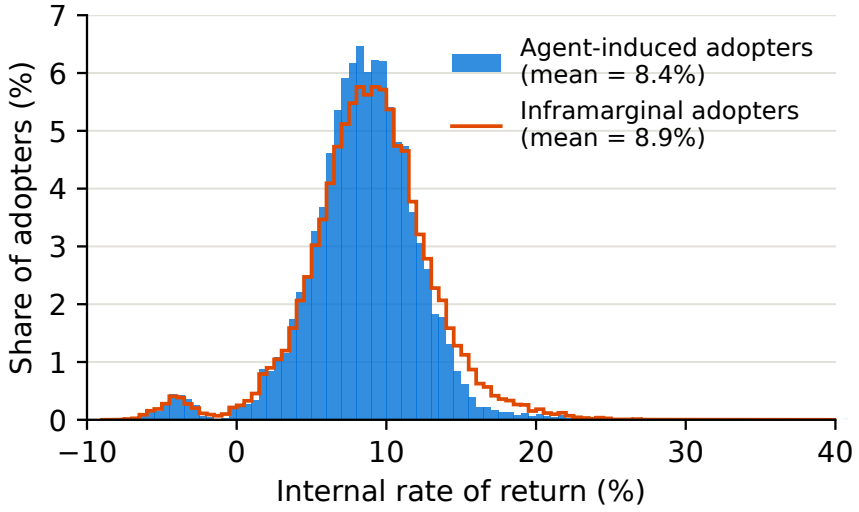
Notes: Real 2023 dollars. Panel (a) is the marginal installation cost recovered from the pricing first-order conditions, per watt of the size bin's representative capacity, in $0.25 bins, separately for the 15,538 products of firms with agents in the market and the 34,494 products of other firms, each weighted equally within its group. Panel (b) is the marginal agent cost revealed by the salesforce first-order condition for the 3,923 firm–markets with agents, each weighted equally, in $2,500 bins; costs of $100,000 or more are pooled in the last bar.

Figure 4: Decomposition of the effect of agents on the average price

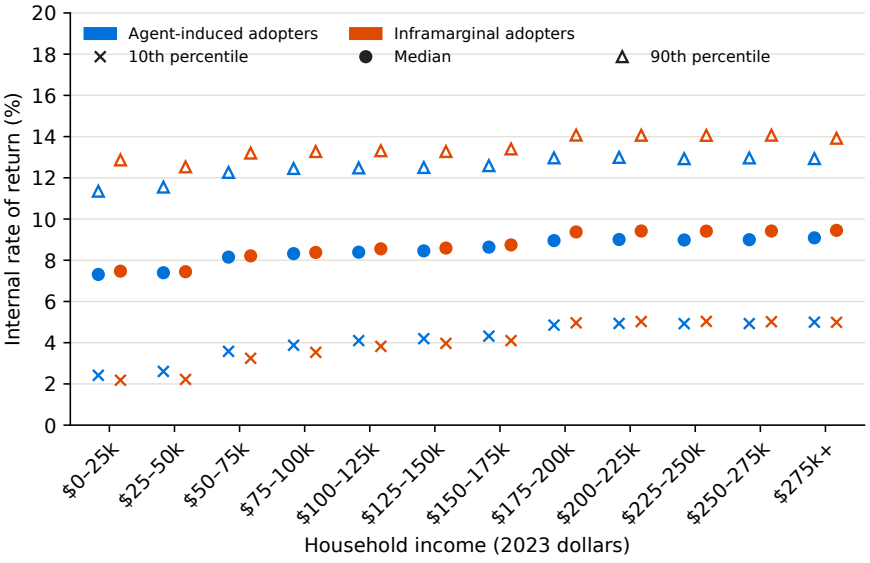


Notes: Change in the capacity-weighted average price per watt before the ITC, the observed equilibrium relative to the counterfactual equilibrium without agents. The cost bars are the change in the average marginal installation cost, and the first two markup bars the change in the average markup, each from households who adopt in both equilibria buying from different installers and from the installers that new adopters buy from. The last two bars are the change in the markups of the same products, from softer price competition and from more price-sensitive buyers. The upper brackets group the bars into those that come from which firms sell and those that come from the prices of the same products.

Figure 5: Internal rates of return of adopters



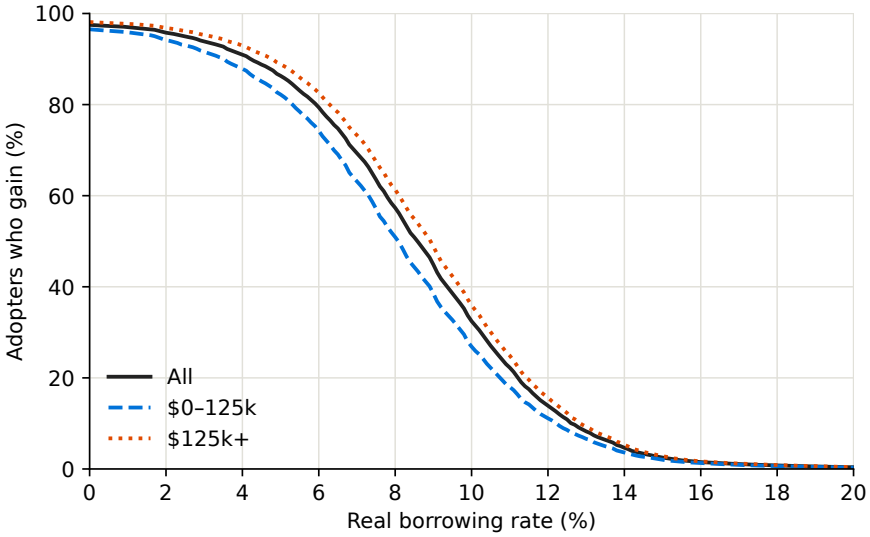
(a) Distribution



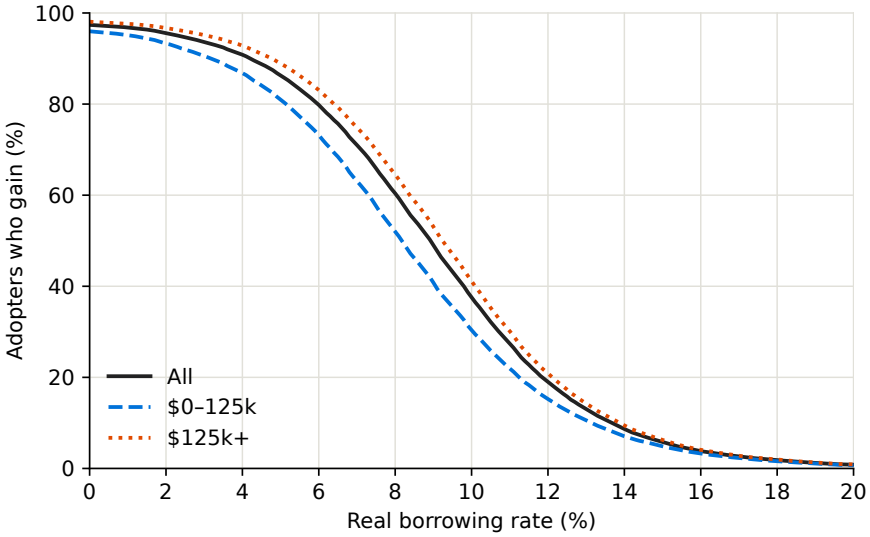
(b) By income

Notes: The internal rate of return is the real discount rate at which the present value of a household's bill savings over the 25-year life of its system equals its price after the ITC. Agent-induced adopters adopt in the observed equilibrium but not in the counterfactual equilibrium without agents; inframarginal adopters adopt in both, and their returns are evaluated at the systems they buy in the observed equilibrium.

Figure 6: Share of adopters who gain at each borrowing rate



(a) Agent-induced adopters



(b) Inframarginal adopters

Notes: Share of adopters whose internal rate of return is at least each real borrowing rate, so that their bill savings repay the system's price after the ITC when they borrow at that rate. Lines split adopters by household income in 2023 dollars. Agent-induced and inframarginal adopters are defined as in Figure 5.

Table 1: Installers in California, 2017–2022

Panel A. Firms *(N = 3,900)*	Fringe	Large	All
Firms	3,118	782	3,900
Share of firms (%)	79.9	20.1	100.0
Share of installations (%)	4.3	95.7	100.0
Share of agents (%)	2.7	97.3	100.0
Median installations per year	2.0	38.1	2.5
Median markets per year	1.0	3.0	1.0
Ever registered an agent (%)	5.6	53.1	15.1
Panel B. Large firm–markets *(N = 17,997)*	Without agents	With agents	All large
Firm–markets	13,943	4,054	17,997
Installations per firm–market	13.4	128.3	39.3
Agents	0.00	7.59	1.71
Agents per 100,000 households	0.00	5.33	1.20
Panel C. Markets *(N = 180)*	Min	Median	Max
Firms	33	104	664
Fringe	4	28	358
Large firms without agents	12	66	211
Large firms with agents	0	15	104
Share of installations (%)			
Fringe	0.5	4.8	37.5
Large firms without agents	11.5	37.0	89.5
Large firms with agents	0.0	54.9	85.9
Price per watt ($)			
Fringe	3.09	4.15	5.44
Large firms without agents	3.31	4.61	5.82
Large firms with agents	4.13	4.86	6.43
HHI of installations	306	665	2,281

Notes: Residential rooftop installations, 2017–2022. A market is a region in a year (30 regions, 180 markets). The fringe is firms averaging 15 or fewer installations per active year. A large firm is with agents in a market if a registered salesperson attached to it lives there, so a firm can fall in different columns of Panel B in different markets; Panel B reports means. Shares of agents count a firm's salespeople in each market where it installs. Prices are real 2023 dollars per watt for host-owned systems. Panel C summarizes each market-level statistic across markets: installation shares are group totals and sum to 100 within a market, and prices are installation-weighted group means, defined for large firms with agents in the 172 markets where they sell a host-owned system.

Table 2: Summary statistics of demand data

	N	Mean	SD	Min	Max
Panel A. Firm–markets					
Installations	17,210	41.6	163.7	1	5,459
Installed capacity (kW)	17,210	277.9	1,043.6	1.2	40,393.8
Market share (% of potential households)	17,210	0.03	0.10	0.00	2.59
Inside share (% of installations)	17,210	1.05	2.93	0.00	45.66
Price per watt before ITC ($/W)	17,210	4.69	1.21	1.20	11.17
Price per watt after ITC ($/W)	17,210	3.35	0.86	0.84	8.06
Panel B. Markets					
Potential households (thousands)	180	146.1	244.1	6.7	1,121.7
Installations	180	3,973	6,017	80	35,177
Adoption rate (% of potential households)	180	3.14	2.13	0.48	11.40
Firms	180	95.6	65.3	12	304

Notes: Residential rooftop installations, 2017–2022, in 180 markets (30 regions by year). A firm–market is a large installer, or the fringe composite of all other installers, in a market. Price per watt is the mean over a firm–market's host-owned installations with a price, in real 2023 dollars. Prices after the ITC apply the federal rate of 30 percent in 2017–2019, 26 percent in 2020–2021, and 30 percent in 2022. The adoption rate is the market's installations as a percent of its potential households, and firms count the fringe composite as one firm. Market share is installations as a percent of the market's potential households, and inside share is installations as a percent of the market's installations. I drop firm–size products with no observed price for any installation, 23,362 installations (3.2 percent), and count them as part of the outside option.

Table 3: Demand estimates

Panel A. Parameter estimates

Parameter	Description	Estimate	Std. Err.
Price			
α_0	Price sensitivity at mean income	2.152	(0.089)
α_y	Box–Cox income parameter	0.230	(0.013)
Agents			
γ_0	Common effect	0.171	(0.110)
γ_y	Income interaction	−0.002	(0.001)
γ_r	Savings interaction	0.018	(0.005)
ϕ	Agent reach relative to non-agent channels	2.497	(0.438)
Heterogeneous preferences			
σ_ν	Random intercept std. dev.	3.252	(0.205)
η_y	Income × constant	−0.759	(0.116)
η_q	Income × size	0.058	(0.018)
Mean preferences			
β_q	Size	0.045	(0.015)
β_E	Panel efficiency	0.039	(0.012)
β_{ILR}	log(Inverter loading ratio)	−0.201	(0.095)
β_M	Microinverter share	0.170	(0.037)
Mean price elasticity		−5.060	
Mean agent elasticity ($a > 0$)		0.129	
Range of γ_c over support		[0.131, 0.202]	

Panel B. Lewis–Mertens robust weak IV tests

		5% critical value	
Test	Statistic	10% bias	15% bias
$g_{\min}$ (joint)	23.8	29.3	21.5
LRR1 (price)	25.2	16.8	12.6
LRR1 (agents)	39.2	16.8	12.5

Notes: Standard errors are clustered by market (180 markets). Prices are in units of \$10,000; β_q and η_q are per kW, and β_E is per percentage point of panel efficiency. Agent density is measured in agents per 100,000 potential adopting households. Firm and year fixed effects are included. Local area fixed effects are normalized to sum to zero, weighted by potential local adopters, in each region. The mean own-price and agent elasticity is the unweighted mean across products. The agent elasticity is computed over products of firms with agents. The range of γ_c is taken over the discrete support of income and savings. Weak instrument tests are from an auxiliary first stage of price and $\log(1 + \hat{\phi}a)$; statistics and critical values follow Lewis and Mertens (2026).

Table 4: Marginal agent cost estimates

	Common σ	Interacted σ	
	(1)	(2)	(3)
Non-HQ market	−0.405	0.025	0.024
	(0.132)	(0.184)	(0.183)
Years in market	−0.016	−0.015	−0.015
	(0.018)	(0.017)	(0.017)
log(Wage, related sales occupations)	0.140	0.138	–
	(0.470)	(0.449)	
σ	2.198	–	–
	(0.083)		
$\sigma \times 1\{\text{HQ}\}$	–	1.756	1.733
		(0.076)	(0.076)
$\sigma \times 1\{\text{non-HQ}\}$	–	2.360	2.340
		(0.103)	(0.103)
$t = 2017$	–	–	–
$t = 2018$	−0.272	−0.257	–
	(0.100)	(0.092)	
$t = 2019$	−0.492	−0.449	–
	(0.125)	(0.116)	
$t = 2020$	−0.884	−0.837	–
	(0.162)	(0.149)	
$t = 2021$	−0.761	−0.708	–
	(0.184)	(0.167)	
$t = 2022$	−0.713	−0.643	–
	(0.193)	(0.180)	
Region FE	Yes	Yes	No
Year FE	Yes	Yes	No
Market FE	No	No	Yes
Firm–markets	17,030	17,030	16,893
Observed share with agents	0.230	0.230	0.230
Predicted share with agents	0.225	0.225	0.226
Log likelihood	−12,737	−12,676	−12,592

Notes: Censored log-normal likelihood estimates of the marginal agent cost, $\log C^A_{fmt} = \boldsymbol{\beta}'_A \boldsymbol{W}_{fmt} + \sigma_{fmt}\iota_{fmt}$. Column (1) restricts the scale of the cost shock to be the same in all markets; column (2), the main specification, allows separate scales in the firm's headquarters market and in other markets. Column (3) replaces the region and year effects of column (2) with market fixed effects, which absorb the sales wage and the year effects, and omits the 8 markets (137 firm–markets) in which no firm hires agents, where market fixed effects are not defined. In columns (1) and (2), $t = 2017$ is the reference year. Standard errors in parentheses are two-way clustered by firm and by market. The observed and predicted shares with agents are over all 17,030 firm–markets; the predicted share is the mean fitted probability of hiring any agents.

Table 5: Market outcomes and welfare in counterfactual equilibria

	(1) Observed	(2) No agents	(3) Joint-profit salesforces	(4) No ITC	(5) No ITC, no agents
Panel A. Market outcomes					
Agents (000s)	29.3	0.0	13.3	6.5	0.0
Installations (000s)	715.2	612.2	686.1	302.7	274.3
Firms with agents	507.2	378.5	471.6	202.8	168.7
Other installers	208.0	233.7	214.5	99.9	105.6
Capacity (MW)	4,714	4,088	4,528	1,694	1,557
Price before the ITC ($/W)	4.60	4.55	4.59	4.10	4.08
Installation cost ($/W)	3.62	3.59	3.61	3.27	3.26
Agent expense ($/W)	0.10	0.00	0.05	0.05	0.00
Gross profit ($ bn)	4.62	3.93	4.43	1.40	1.27
Firms with agents	3.30	2.44	3.07	0.95	0.78
Other installers	1.32	1.49	1.36	0.46	0.48
Agent expense ($ bn)	0.47	0.00	0.21	0.09	0.00
Industry profit ($ bn)	4.15	3.93	4.22	1.31	1.27
Firms with agents	2.83	2.44	2.86	0.85	0.78
Other installers	1.32	1.49	1.36	0.46	0.48
Panel B. Welfare, change from column (1) ($ billion)					
ΔCS	–	−0.86	−0.25	−3.29	−3.50
ΔPS	–	−0.22	0.07	−2.84	−2.89
Gross profit	–	−0.69	−0.19	−3.22	−3.35
Agent expense	–	0.47	0.26	0.38	0.47
ΔG (ITC outlay)	–	0.89	0.25	6.23	6.23
ΔE	–	−0.63	−0.19	−3.04	−3.18
Carbon dioxide	–	−0.53	−0.16	−2.54	−2.65
Local air pollution	–	−0.11	−0.03	−0.51	−0.53
ΔF (net-metering cost shift)	–	2.51	0.75	12.07	12.63
Bill revenue (transfer)	–	3.16	0.94	15.22	15.93
Avoided electricity costs	–	−0.65	−0.19	−3.16	−3.30
ΔW	–	1.68	0.63	9.12	9.28

Notes: Totals over all markets, 2017–2022, in real 2023 dollars. Column (1) is the observed equilibrium. Column (2) sets all salesforces to zero. In column (3), firms choose their salesforces to maximize joint industry profits. Column (4) removes the ITC, and column (5) removes both the ITC and agents. In every counterfactual, prices are re-solved and salesforces, where present, respond. Per-watt values are capacity-weighted averages over all installed watts. Firms with agents are those with agents in the observed equilibrium. Gross profit is revenue net of installation costs, and industry profit is gross profit net of agent expense, all of which firms with agents pay. ΔCS is the change in consumer surplus. ΔPS is the change in industry profit, and ΔG is minus the change in the ITC outlay. ΔE is the change in avoided damages over each system's 25-year life from carbon dioxide, at $62.40 per metric ton, and from local air pollution, at the marginal damages of Sexton et al. (2021), discounted at 3 percent to each installation year. ΔF is minus the change in the utility's lifetime shortfall, discounted at 5 percent to each installation year: the bill revenue the utility loses, a transfer to adopters, net of the cost of the electricity it no longer supplies.

Table 6: Effects per agent

	Average agent	Last agent
Installations per agent-year		
Firms with agents	4.40	1.67
Firms without agents	−0.88	−0.30
All firms (new to solar)	3.52	1.37
Industry profit per agent ($ thousand)		
Gross profit	23.6	9.2
Agent expense	16.0	16.0
Industry profit	7.6	−6.8
Agent cost per added watt ($)	0.75	1.88

Notes: The average agent is the observed equilibrium minus the counterfactual equilibrium without agents, divided by the observed agent-years. The last agent is the effect of one more agent at the observed equilibrium, from equation (26) with prices re-equilibrating, averaged across firms weighted by their agents. Firms with agents are those with agents in the observed equilibrium; for the last agent, their row is the agent's own firm's gain net of what other firms with agents lose. Gross profit is the change in all firms' revenue net of installation costs, agent expense is the cost of the agent, and industry profit is gross profit net of agent expense. Agent cost per added watt divides agent expense by the change in installed DC watts. In both columns, each added watt avoids $1.01 in environmental damages, valuing carbon dioxide at $62.40 per metric ton and local air pollution at the marginal damages of Sexton et al. (2021), discounted at 3 percent.

A Data Appendix

A.1 Interconnected Applications

I link the Interconnected Applications to Tracking the Sun on the utility's application ID. The applications provide the application and permission-to-operate dates, the installer's name and CSLB license number, the household's ZIP code, the ownership type, and the listed equipment; Tracking the Sun provides standardized capacity, prices, customer segments, and equipment characteristics. I define a firm as a CSLB license number. Where the license number is missing or invalid, I recover it from other applications that report the same installer name, and by hand for a small number of installers. I locate each firm's headquarters at the ZIP code it most often reports.

I keep residential, single-family rooftop systems between 1 and 25 kW, and I drop systems that were self-installed, part of new construction, co-installed with a battery, ground-mounted, on multi-phase service, or interconnected under the Multifamily Affordable Solar Housing program. I assign each installation to a county and a local area through its ZIP code, and I drop the counties and local areas outside the three IOUs' service territories, together with their households. The resulting sample has 738,555 installations from 2017 to 2022.

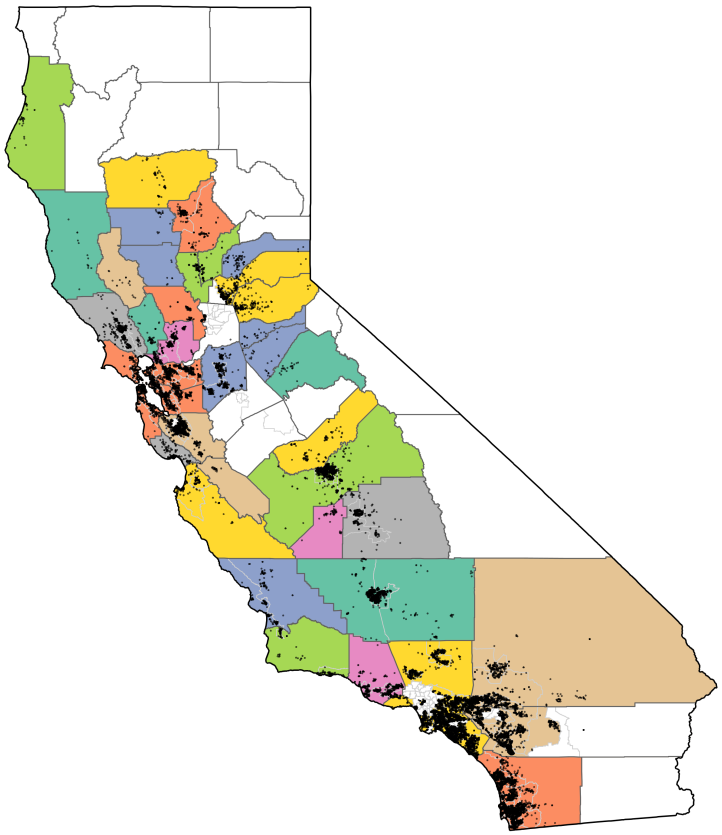
Prices are total installed prices before incentives, divided by DC capacity and deflated to 2023 dollars with the CPI-U. I treat the prices of third-party owned systems as missing, as well as prices outside the 0.5th and 99.5th percentiles of the capacity-weighted distribution. I fill in missing panel efficiencies and inverter capacities from manufacturers' specifications and the equipment listed in the applications.

A.2 Electricity Billing Data

PGE, SCE, and SDGE provided the billing data through their Energy Data Request Programs. Each utility drew samples of residential solar adopters and non-adopters and matched each adopter to its application ID in the Interconnected Applications. A record is a bill for a premise, with the start and end dates of the bill interval, electricity usage, the bill amount, the rate schedule, the baseline territory, the ZIP code, and indicators for CARE, FERA, all-electric service, and, for some utilities, demand response and community choice aggregation. I use the bills that cover 2020 to 2024.

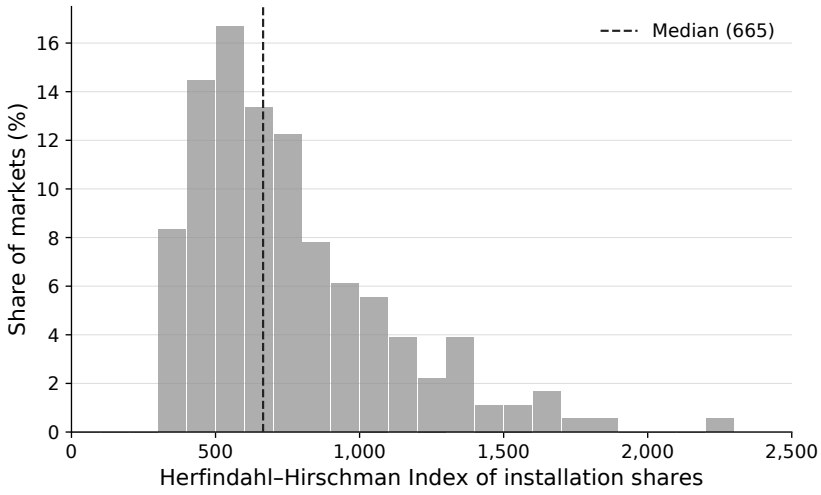
I keep premises that are always billed on a residential rate, and I drop premises that the utility identifies as multifamily or that have no usage. I date adoption at the household's

Figure A1: Solar markets and registered salespeople

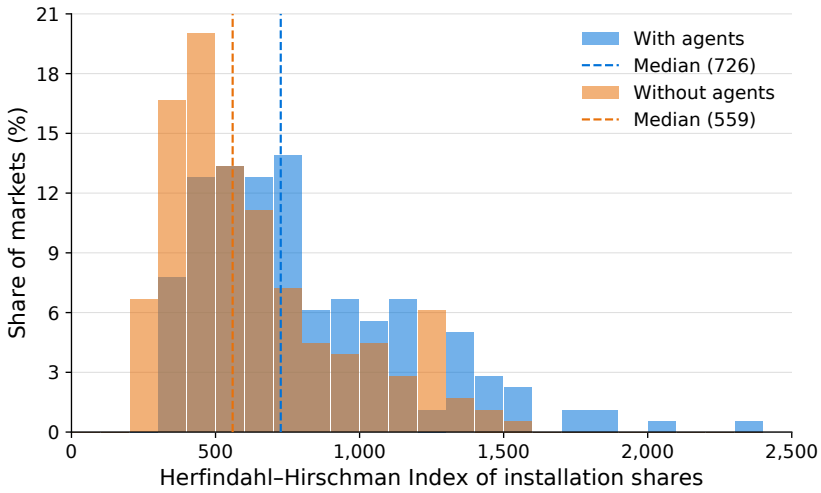


Notes: Each shaded area is one of the 30 regions that, in each year, define a market; adjacent regions have different colors. Gray lines are the boundaries of local areas (public use microdata areas) within regions. White areas are outside the sample: counties served by municipal utilities and areas outside the three investor-owned utilities' service territories. Each dot is the home location of a registered salesperson active between 2017 and 2022.

Figure A2: Market concentration



(a) Observed



(b) Model, with and without agents

Notes: Distribution across the 180 markets (region by year, 2017–2022) of the Herfindahl–Hirschman Index of firms' percent shares of installations, including fringe firms. Bins are 100 points wide. In panel (a), 20 percent of markets are moderately concentrated (HHI 1,000 to 1,800) and 1 percent are highly concentrated (HHI above 1,800); the remaining 79 percent have an HHI below 1,000 (U.S. Department of Justice and Federal Trade Commission, 2023). Panel (b) shows the HHI in the model's observed equilibrium and in the counterfactual equilibrium without agents; in each, the fringe's installations are split across the market's fringe firms in proportion to their observed installations. The median market's HHI is 726 with agents and 559 without.

first bill with net metering or solar generation, or at its permission-to-operate date if that is earlier. Adopters must adopt between 2021 and 2023, have bills before that date, be on net metering 2.0, and have a system that meets the restrictions of Appendix A.1. Non-adopters never appear on net metering. Every premise must have bills in both 2020 and 2024. The cleaned sample has 115,564 adopters and 73,409 non-adopters.

I combine duplicate records and records that cover the same days, and I spread each bill's usage and amount evenly over the days of its interval. An annual total sums usage or bills over the days of a calendar year that bills cover and scales the sum to 365 days. Bills are measured before the California Climate Credit, a lump-sum credit that does not depend on usage, and deflated to 2023 dollars with the annual average CPI-U. The bill savings estimation sample keeps households whose bills cover at least 300 days of both 2020 and 2024 and that have every 2020 covariate, and adopters with a positive system size. It has 110,433 adopters and 64,173 non-adopters. The billing sampling weights scale, within each utility, adopters to the residential solar systems that the utility interconnected between 2021 and 2023 and non-adopters to its owner-occupied, individually metered residential customers without solar.

A.3 Salesperson Registration

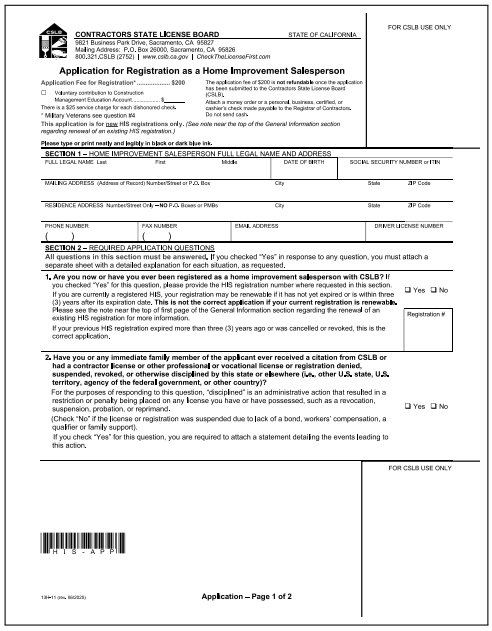
Figure A3 shows the CSLB forms for salesperson registrations. A salesperson registers with the CSLB once, and each contractor files a notification when it begins and when it stops working with a registered salesperson. I geocode each salesperson's address and classify it as residential or commercial with the U.S. Postal Service's Residential Delivery Indicator. I keep salespeople with residential addresses, who are plausibly door-to-door agents, and set aside those with commercial addresses, who are more likely to be office-based sales representatives and managers. I assign each salesperson to the region that contains their address, and count them in a firm's salesforce in every year in which their relationship with that firm is active.

A.4 SEEDS Survey

The Department of Energy's Solar Energy Evolution and Diffusion Studies (SEEDS) fielded household surveys of solar adopters, non-adopters who seriously considered solar, and the general population in California, New Jersey, New York, and Arizona between June 2014 and April 2015 (Sigrin et al., 2017; Moezzi et al., 2017). I use the adopter survey from California, which was fielded between December 2014 and April 2015 on households drawn

Figure A3: Contractors State License Board forms for home improvement salespersons

(a) Registration application, page 1



CONTRACTORS STATE LICENSE BOARD
STATE OF CALIFORNIA
9821 Business Park Drive, Sacramento, CA 95827
Mailing Address: P.O. Box 26000, Sacramento, CA 95826
800.321.CSLB (2752) | www.cslb.ca.gov | CheckTheLicenseFirst.com

FOR CSLB USE ONLY

Application for Registration as a Home Improvement Salesperson

Application Fee for Registration* $200
☐ Voluntary contribution to Construction Management Education Account ..... $ _____
There is a $25 service charge for each dishonored check.
* Military Veterans see question #4

The application fee of $200 is **not refundable** once the application has been submitted to the Contractors State License Board (CSLB).
Attach a money order or a personal, business, certified, or cashier's check made payable to the Registrar of Contractors.
Do not send cash.

This application is for new HIS registrations only. *(See note near the top of the General Information section regarding renewal of an existing HIS registration.)*

Please type or print neatly and legibly in black or dark blue ink.

SECTION 1 – HOME IMPROVEMENT SALESPERSON FULL LEGAL NAME AND ADDRESS

FULL LEGAL NAME: Last | First | Middle | DATE OF BIRTH | SOCIAL SECURITY NUMBER or ITIN

MAILING ADDRESS (Address of Record) Number/Street or P.O. Box | City | State | ZIP Code

RESIDENCE ADDRESS Number/Street Only – NO P.O. Boxes or PMBs | City | State | ZIP Code

PHONE NUMBER ( ) | FAX NUMBER ( ) | EMAIL ADDRESS | DRIVER LICENSE NUMBER

SECTION 2 – REQUIRED APPLICATION QUESTIONS

All questions in this section must be answered. If you checked "Yes" in response to any question, you must attach a separate sheet with a detailed explanation for each situation, as requested.

1. Are you now or have you ever been registered as a home improvement salesperson with CSLB? If you checked "Yes" for this question, please provide the HIS registration number where requested in this section. ☐ Yes ☐ No

If you are currently a registered HIS, your registration may be renewable if it has not yet expired or is within three (3) years after its expiration date. **This is not the correct application if your current registration is renewable.** Please see the note near the top of first page of the General Information section regarding the renewal of an existing HIS registration for more information.

If your previous HIS registration expired more than three (3) years ago or was cancelled or revoked, this is the correct application.

Registration #

2. Have you or any immediate family member of the applicant ever received a citation from CSLB or had a contractor license or other professional or vocational license or registration denied, suspended, revoked, or otherwise disciplined by this state or elsewhere (i.e., other U.S. state, U.S. territory, agency of the federal government, or other country)? ☐ Yes ☐ No

For the purposes of responding to this question, "disciplined" is an administrative action that resulted in a restriction or penalty being placed on any license you have or have possessed, such as a revocation, suspension, probation, or reprimand.

(Check "No" if the license or registration was suspended due to lack of a bond, workers' compensation, a qualifier or family support).

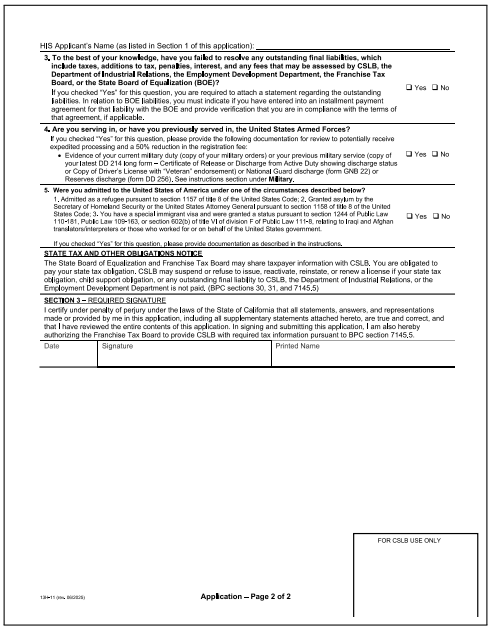
If you check "Yes" for this question, you are required to attach a statement detailing the events leading to this action.

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HIS - APP

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(b) Registration application, page 2



HIS Applicant's Name (as listed in Section 1 of this application): _____

3. To the best of your knowledge, have you failed to resolve any outstanding final liabilities, which include taxes, additions to tax, penalties, interest, and any fees that may be assessed by CSLB, the Department of Industrial Relations, the Employment Development Department, the Franchise Tax Board, or the State Board of Equalization (BOE)? ☐ Yes ☐ No

If you checked "Yes" for this question, you are required to attach a statement regarding the outstanding liabilities. In relation to BOE liabilities, you must indicate if you have entered into an installment payment agreement for that liability with the BOE and provide verification that you are in compliance with the terms of that agreement, if applicable.

4. Are you serving in, or have you previously served in, the United States Armed Forces? ☐ Yes ☐ No

If you checked "Yes" for this question, please provide the following documentation for review to potentially receive expedited processing and a 50% reduction in the registration fee:

- Evidence of your current military duty (copy of your military orders) or your previous military service (copy of your latest DD 214 long form – Certificate of Release or Discharge from Active Duty showing discharge status or Copy of Driver's License with "Veteran" endorsement) or National Guard discharge (form GNB 22) or Reserves discharge (form DD 256). See instructions section under **Military.**

5. Were you admitted to the United States of America under one of the circumstances described below?

1. Admitted as a refugee pursuant to section 1157 of title 8 of the United States Code; 2. Granted asylum by the Secretary of Homeland Security or the United States Attorney General pursuant to section 1158 of title 8 of the United States Code; 3. You have a special immigrant visa and were granted a status pursuant to section 1244 of Public Law 110-181, Public Law 109-163, or section 602(b) of title VI of division F of Public Law 111-8, relating to Iraqi and Afghan translators/interpreters or those who worked for or on behalf of the United States government. ☐ Yes ☐ No

If you checked "Yes" for this question, please provide documentation as described in the instructions.

STATE TAX AND OTHER OBLIGATIONS NOTICE

The State Board of Equalization and Franchise Tax Board may share taxpayer information with CSLB. You are obligated to pay your state tax obligation, CSLB may suspend or refuse to issue, reactivate, reinstate, or renew a license if your state tax obligation, child support obligation, or any outstanding final liability to CSLB, the Department of Industrial Relations, or the Employment Development Department is not paid, (BPC sections 30, 31, and 7145.5)

SECTION 3 – REQUIRED SIGNATURE

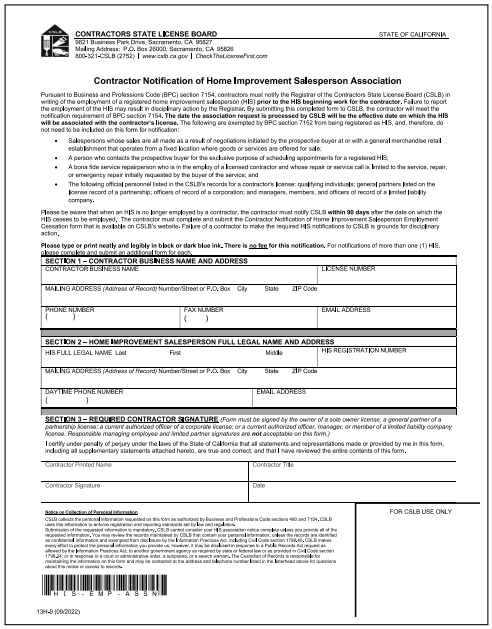
I certify under penalty of perjury under the laws of the State of California that all statements, answers, and representations made or provided by me in this application, including all supplementary statements attached hereto, are true and correct, and that I have reviewed the entire contents of this application. In signing and submitting this application, I am also hereby authorizing the Franchise Tax Board to provide CSLB with required tax information pursuant to BPC section 7145.5.

Date | Signature | Printed Name

FOR CSLB USE ONLY

13H-11 (rev. 06/2025) **Application – Page 2 of 2**

(c) Contractor notification of association



CONTRACTORS STATE LICENSE BOARD
STATE OF CALIFORNIA
9821 Business Park Drive, Sacramento, CA 95827
Mailing Address: P.O. Box 26000, Sacramento, CA 95826
800-321-CSLB (2752) | www.cslb.ca.gov | CheckTheLicenseFirst.com

Contractor Notification of Home Improvement Salesperson Association

Pursuant to Business and Professions Code (BPC) section 7154, contractors must notify the Registrar of the Contractors State License Board (CSLB) in writing of the employment of a registered home improvement salesperson (HIS) **prior to the HIS beginning work for the contractor.** Failure to report the employment of the HIS may result in disciplinary action by the Registrar. By submitting this completed form to CSLB, the contractor will meet the notification requirement of BPC section 7154. **The date the association request is processed by CSLB will be the effective date on which the HIS will be associated with the contractor's license.** The following are exempted by BPC section 7152 from being registered as HIS, and, therefore, do not need to be included on this form for notification:

- Salespersons whose sales are all made as a result of negotiations initiated by the prospective buyer at or with a general merchandise retail establishment that operates from a fixed location where goods or services are offered for sale;
- A person who contacts the prospective buyer for the exclusive purpose of scheduling appointments for a registered HIS;
- A bona fide service repairperson who is in the employ of a licensed contractor and whose repair or service call is limited to the service, repair, or emergency repair initially requested by the buyer of the service; and
- The following official personnel listed in the CSLB's records for a contractor's license: qualifying individuals; general partners listed on the license record of a partnership; officers of record of a corporation; and managers, members, and officers of record of a limited liability company.

Please be aware that when an HIS is no longer employed by a contractor, the contractor must notify CSLB **within 90 days** after the date on which the HIS ceases to be employed. The contractor must complete and submit the Contractor Notification of Home Improvement Salesperson Employment Cessation form that is available on CSLB's website. Failure of a contractor to make the required HIS notifications to CSLB is grounds for disciplinary action.

Please type or print neatly and legibly in black or dark blue ink. There is no fee for this notification. For notifications of more than one (1) HIS, please complete and submit an additional form for each.

SECTION 1 – CONTRACTOR BUSINESS NAME AND ADDRESS

CONTRACTOR BUSINESS NAME | LICENSE NUMBER

MAILING ADDRESS (Address of Record) Number/Street or P.O. Box | City | State | ZIP Code

PHONE NUMBER ( ) | FAX NUMBER ( ) | EMAIL ADDRESS

SECTION 2 – HOME IMPROVEMENT SALESPERSON FULL LEGAL NAME AND ADDRESS

HIS FULL LEGAL NAME: Last | First | Middle | HIS REGISTRATION NUMBER

MAILING ADDRESS (Address of Record) Number/Street or P.O. Box | City | State | ZIP Code

DAYTIME PHONE NUMBER ( ) | EMAIL ADDRESS

SECTION 3 – REQUIRED CONTRACTOR SIGNATURE *(Form must be signed by the owner of a sole owner license; a general partner of a partnership license; a current authorized officer of a corporate license; or a current authorized officer, manager, or member of a limited liability company license. Responsible managing employee and limited partner signatures are not acceptable on this form.)*

I certify under penalty of perjury under the laws of the State of California that all statements and representations made or provided by me in this form, including all supplementary statements attached hereto, are true and correct, and that I have reviewed the entire contents of this form.

Contractor Printed Name | Contractor Title

Contractor Signature | Date

FOR CSLB USE ONLY

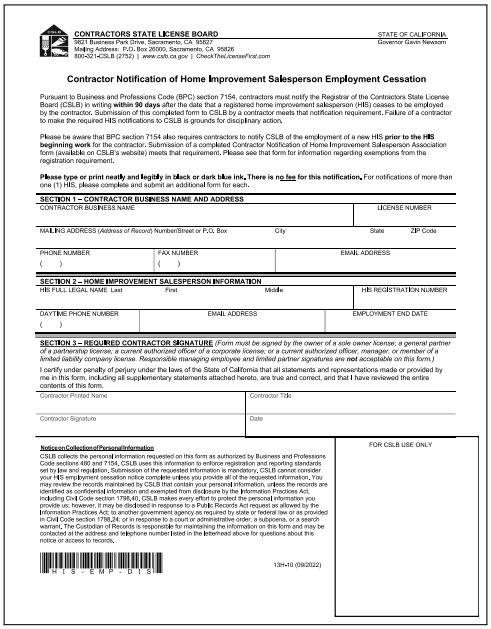
Notice on Collection of Personal Information

CSLB collects the personal information requested on this form as authorized by Business and Professions Code sections 480 and 7154. CSLB uses this information to enforce registration and reporting standards set by law and regulation. Submission of the requested information is mandatory. CSLB cannot consider your HIS employment cessation notice complete unless you provide all of the requested information. You may review the records maintained by CSLB that contain your personal information, unless the records are identified as confidential information and exempted from disclosure by the Information Practices Act, including Civil Code section 1798.40. CSLB makes every effort to protect the personal information you provide us; however, it may be disclosed in response to a Public Records Act request as allowed by the Information Practices Act; to another government agency as required by state or federal law or as provided in Civil Code section 1798.24; or in response to a court or administrative order, a subpoena, or a search warrant. The Custodian of Records is responsible for maintaining the information on this form and may be contacted at the address and telephone number listed in the letterhead above for questions about this notice or access to records.

HIS - EMP - ASSN

13H-9 (09/2022)

(d) Contractor notification of employment cessation



CONTRACTORS STATE LICENSE BOARD
STATE OF CALIFORNIA
Governor Gavin Newsom
9821 Business Park Drive, Sacramento, CA 95827
Mailing Address: P.O. Box 26000, Sacramento, CA 95826
800-321-CSLB (2752) | www.cslb.ca.gov | CheckTheLicenseFirst.com

Contractor Notification of Home Improvement Salesperson Employment Cessation

Pursuant to Business and Professions Code (BPC) section 7154, contractors must notify the Registrar of the Contractors State License Board (CSLB) in writing **within 90 days** after the date that a registered home improvement salesperson (HIS) ceases to be employed by the contractor. Submission of this completed form to CSLB by a contractor meets that notification requirement. Failure of a contractor to make the required HIS notifications to CSLB is grounds for disciplinary action.

Please be aware that BPC section 7154 also requires contractors to notify CSLB of the employment of a new HIS **prior to the HIS beginning work** for the contractor. Submission of a completed Contractor Notification of Home Improvement Salesperson Association form (available on CSLB's website) meets that requirement. Please see that form for information regarding exemptions from the registration requirement.

Please type or print neatly and legibly in black or dark blue ink. There is no fee for this notification. For notifications of more than one (1) HIS, please complete and submit an additional form for each.

SECTION 1 – CONTRACTOR BUSINESS NAME AND ADDRESS

CONTRACTOR BUSINESS NAME | LICENSE NUMBER

MAILING ADDRESS (Address of Record) Number/Street or P.O. Box | City | State | ZIP Code

PHONE NUMBER ( ) | FAX NUMBER ( ) | EMAIL ADDRESS

SECTION 2 – HOME IMPROVEMENT SALESPERSON INFORMATION

HIS FULL LEGAL NAME: Last | First | Middle | HIS REGISTRATION NUMBER

DAYTIME PHONE NUMBER ( ) | EMAIL ADDRESS | EMPLOYMENT END DATE

SECTION 3 – REQUIRED CONTRACTOR SIGNATURE *(Form must be signed by the owner of a sole owner license; a general partner of a partnership license; a current authorized officer of a corporate license; or a current authorized officer, manager, or member of a limited liability company license. Responsible managing employee and limited partner signatures are not acceptable on this form.)*

I certify under penalty of perjury under the laws of the State of California that all statements and representations made or provided by me in this form, including all supplementary statements attached hereto, are true and correct, and that I have reviewed the entire contents of this form.

Contractor Printed Name | Contractor Title

Contractor Signature | Date

FOR CSLB USE ONLY

Notice on Collection of Personal Information

CSLB collects the personal information requested on this form as authorized by Business and Professions Code sections 480 and 7154. CSLB uses this information to enforce registration and reporting standards set by law and regulation. Submission of the requested information is mandatory. CSLB cannot consider your HIS employment cessation notice complete unless you provide all of the requested information. You may review the records maintained by CSLB that contain your personal information, unless the records are identified as confidential information and exempted from disclosure by the Information Practices Act, including Civil Code section 1798.40. CSLB makes every effort to protect the personal information you provide us; however, it may be disclosed in response to a Public Records Act request as allowed by the Information Practices Act; to another government agency as required by state or federal law or as provided in Civil Code section 1798.24; or in response to a court or administrative order, a subpoena, or a search warrant. The Custodian of Records is responsible for maintaining the information on this form and may be contacted at the address and telephone number listed in the letterhead above for questions about this notice or access to records.

HIS - EMP - DISC

13H-10 (09/2022)

Notes: Current blank forms from the Contractors State License Board: Application for Registration as a Home Improvement Salesperson (13H-11, rev. 06/2025), Contractor Notification of Home Improvement Salesperson Association (13H-9, 09/2022), and Contractor Notification of Home Improvement Salesperson Employment Cessation (13H-10, 09/2022).

from installer customer lists. The California adopter sample has 1,176 households.

The survey asks how many installers the household talked to and how it first connected with each. A household that talked to one installer reports how it first connected with that installer; a household that talked to several reports it separately for the installer it chose and for the installers it did not choose. Each question offers ten responses, and respondents may check several. I group the responses into channels:

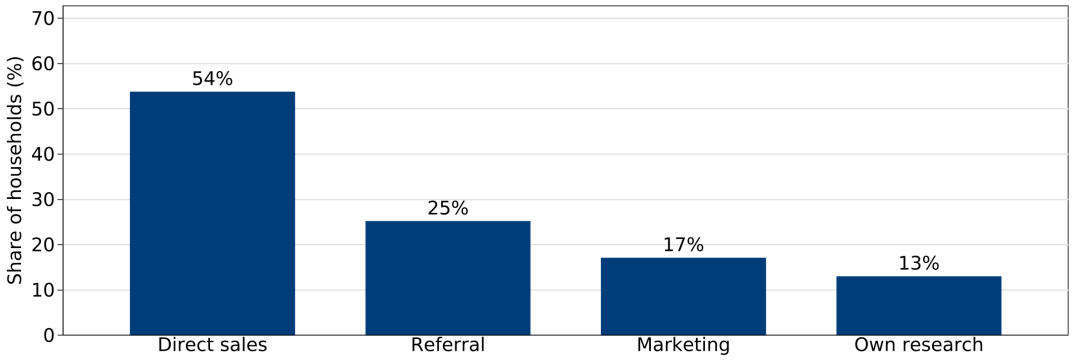
- *Direct sales*: the installer knocked on the door or left a flyer.
- *Referral*: someone recommended the installer, or someone gave the installer the household's name.
- *Marketing*: the household met the installer at a community event or store; found it through an advertisement or the news; was called, emailed, or mailed; or was previously familiar.
- *Own research*: the household filled out a website form, or they found the installer through own research.
- *Other*: any other way.

The direct sales channel is the agent channel of the reach moment in demand estimation.

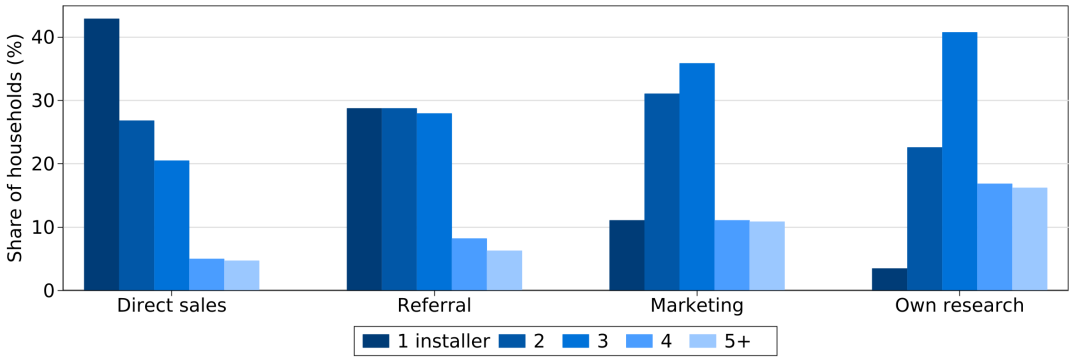
Figure A4 summarizes the responses. Panel (a) shows that, of the 1,175 households that report how they connected with their installers, 54 percent first met the installer they chose through direct sales, 25 percent through a referral, 17 percent through marketing, and 13 percent through their own research. Counting the installers they did not choose, 59 percent have a direct seller in their consideration set, 39 percent an installer they reached through marketing, 31 percent one referred to them, and 27 percent one they found through their own research. Households that considered an installer through direct sales search the least: 43 percent of them talked to only one installer, compared with 29 percent of those with a referred installer, 11 percent of those with an installer reached through marketing, and 4 percent of those with an installer found through their own research. In total, as panel (b) shows, 37 percent of households talked to one installer, 24 percent to two, 24 percent to three, and 14 percent to four or more. Panel (c) shows that direct sales is also the most common way households met the installer they chose in every income group, but its share falls from 62–64 percent below $100,000 to 44 percent at $150,000 or more, while referrals (19 to 29 percent), marketing (15 to 22 percent), and own research (9 to 18 percent) are more common in the top income group than in the bottom one.

Figure A4: How SEEDS adopters reached installers

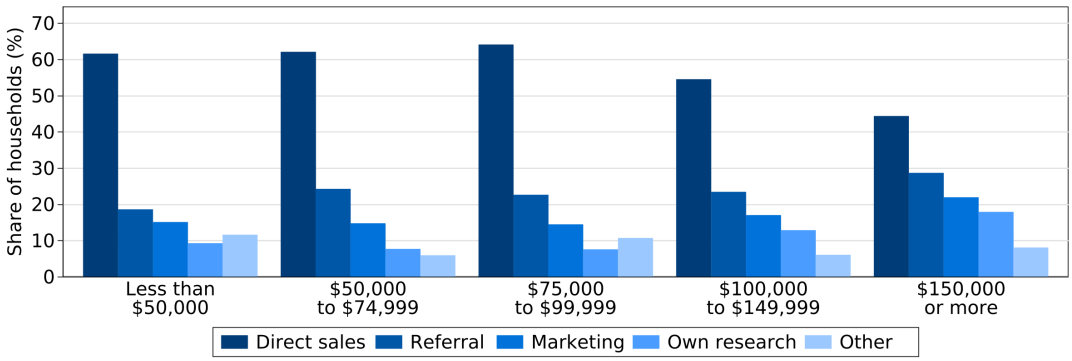
(a) Channel for the chosen installer



(b) Installers considered, by channel in consideration set



(c) Channel for the chosen installer, by household income



Notes: California adopters in the SEEDS adopter survey. A channel is in the consideration set if the household reports first connecting with the chosen installer or any other installer it talked to through that channel. Panel (a) reports the share of the 1,175 households whose chosen installer was first reached through each channel. Panel (b) reports, among households with each channel in the consideration set, the distribution of the number of installers they talked to. Panel (c) reports, for the 901 households that report an income group, the share whose chosen installer was first reached through each channel. Households may check several channels, so shares do not sum to 100 percent. Income is nominal.

A.5 Job Postings for Solar Salespeople

In this section, I briefly characterize solar sales jobs using online job postings from Revelio Labs' COSMOS database, accessed through Wharton Research Data Services (Revelio Labs, 2026). I pull every posting located in California from 2016 to 2024 that Revelio classifies into a solar role and into an O*NET major occupation group for sales (41), construction and extraction (47), or business and financial operations (13), and link each posting to its title and description text. This gives 86,194 postings.

I select outside sales positions from the posting titles. Of the 86,194 postings, 37,866 have a title that names a sales role, such as a sales representative, consultant, advisor, canvasser, setter, or closer. I drop 2,280 whose title names a management, support, or technical role (for example, sales manager, coordinator, inside sales, or sales engineer), 2,104 whose title names a retail or in-store setting, and 563 whose title advertises that the job involves no door knocking. The remaining 32,919 postings form the outside sales sample, of which 26,109 include a description. I report the 2,104 retail postings separately.

Among outside sales postings with a description, 77 percent mention commissions, 24 percent offer commission-only pay, and 21 percent mention a base salary or hourly pay. Retail postings with a description differ: 89 percent mention base or hourly pay and 1 percent offer commission only. Of outside sales postings with a description, 66 percent describe a full-time position, 63 percent mention travel, a vehicle, or a sales territory, 39 percent explicitly mention door-to-door selling or canvassing, and 12 percent mention CSLB registration as a Home Improvement Salesperson. Among retail postings with a description, only 40 percent describe a full-time position. Of the outside sales postings, about 13,000 post a pay range, with median lower and upper ends of $84,000 and $185,000. Across the 18,132 that post any pay, taking the midpoint of each range, the 10th and 90th percentiles of pay are $62,000 and $225,000.

B Model Appendix

B.1 Offer Quality

This appendix gives the distribution of offer quality behind equation (3) and derives the probability that an adopter's best offer came from an agent. Fix a household i and a product j of firm f, with $u_{ij} = \bar{u}_{ij} + \zeta_{ij}$. An offer o for product j gives the household utility $u_{ij} + e_o$, where e_o is the quality of the offer. Offers through each channel $k \in \{\text{non-agent}, \text{agent}\}$

arrive as independent Poisson processes over quality with intensity

$$\frac{\lambda_k}{\gamma_c}\exp\left(-\frac{e}{\gamma_c}\right)de, \qquad e\in\mathbb{R}, \qquad \lambda_{\text{non-agent}}=1, \quad \lambda_{\text{agent}}=\phi a_f.$$

This is the exponential form of offer quality: each channel delivers finitely many offers above any threshold, and the quality of each of these offers in excess of the threshold is exponential with mean γ_c. The two channels differ in how many offers they deliver, not in the quality of those offers.

Offers above a threshold. An offer is acceptable at v if $u_{ij}+e_o>v$. The expected number of acceptable offers from channel k is

$$\mathbb{E}\left[\,N_{ij,k}(v)\,\right]=\int_{v-u_{ij}}^{\infty}\frac{\lambda_k}{\gamma_c}\exp\left(-\frac{e}{\gamma_c}\right)de=\lambda_k\exp\left(-\frac{v-u_{ij}}{\gamma_c}\right),$$

and summing over the two channels gives $\mathbb{E}\left[\,N_{ij}(v)\,\right]=(1+\phi a_f)\exp\left(-(v-u_{ij})/\gamma_c\right)$, the expression in Section 3.1.

Best offer. The number of acceptable offers $N_{ij}(v)$ is Poisson, and the best offer V_{ij} is below v exactly when no offer is acceptable at v, so

$$\mathbb{P}\left(\,V_{ij}\le v\,\right)=\exp\left(-\mathbb{E}\left[\,N_{ij}(v)\,\right]\right)=\exp\left(-\exp\left(-\frac{v-u_{ij}-\gamma_c\log(1+\phi a_f)}{\gamma_c}\right)\right).$$

The best offer is therefore type-1 extreme value with location $u_{ij}+\gamma_c\log(1+\phi a_f)$ and scale γ_c, that is, $V_{ij}=\bar{u}_{ij}+\gamma_c\log(1+\phi a_f)+\zeta_{ij}+\gamma_c\epsilon_{ij}$, where ϵ_{ij} is standard type-1 extreme value and independent across products because offers for different products arrive independently. Multiplying the arrival rate by $1+\phi a_f$ shifts the best offer by $\gamma_c\log(1+\phi a_f)$, so the dispersion of offer quality γ_c also sets how much additional offers raise utility. With ζ_{ij} drawn from the Cardell (1997) distribution, this gives equation (3).

Agent reach. For a firm with agents, the same calculation applied to each channel alone shows that the best agent offer and the best non-agent offer are independent type-1 extreme value with common scale γ_c and locations $u_{ij}+\gamma_c\log(\phi a_f)$ and u_{ij}. The best offer comes from an agent when the best agent offer is the larger of the two, which by the logit formula has probability

$$\frac{\exp\left(\log(\phi a_f)\right)}{\exp\left(\log(\phi a_f)\right)+1}=\frac{\phi a_f}{1+\phi a_f}. \tag{28}$$

As for any maximum of independent type-1 extreme value variables with a common scale, which channel delivers the best offer is independent of the value of that offer. The household's choice depends on its offers for product j only through V_{ij}, so equation (28) is also the probability that an adopter of product j met firm f through an agent. This probability does not depend on γ_c, $\bar{u}_{ij}$, ζ_{ij}, or the household's cell, so the channel through which adopters met their installers identifies agent reach ϕ separately from effectiveness, as Appendix D.2 describes.

B.2 Fringe Firm Composite

I show that fringe firms in monopolistic competition that share an installation cost at each size set a common price for each size, and that they aggregate into one representative fringe firm. The argument parallels Melitz (2003), where heterogeneous firms in monopolistic competition with CES demand, which a logit can represent (Anderson, de Palma and Thisse, 1992), aggregate into a representative firm. Here firms differ in quality rather than cost, and the result holds under the mixed logit because each firm takes the logit denominators as given.

I omit market subscripts. The fringe is a set $\mathcal{F}$ of small local installers n that have no agents. They differ in characteristics and unobserved quality but share the installation cost C_q^I of a size-q system. The systematic utility that a household in cell c gets from size q of fringe firm n is

$$\bar{u}_{cnq} = \delta_{nq} + \mu_{cq} - \alpha_c(1-\tau)P_{nq},$$

where δ_{nq} collects the firm's characteristics and unobserved quality, which enter equation (5) with common coefficients, and μ_{cq} collects the terms that vary across cells: the savings term $\alpha_c R_{cq}$, the income interactions, and the local area effect. These terms depend on the size but not on the fringe firm. Household i in cell c chooses size q of firm n with probability $s_{inq} = \exp(\bar{u}_{cnq} + \sigma_\nu \nu_i)/D_i$, where D_i is the denominator of equation (9), which sums over the products of the large firms and of every fringe firm.

Prices. Under monopolistic competition, each fringe firm takes D_i as given for every household. Firm n's profit then separates by size:

$$\pi_n(\boldsymbol{P}_n) = N \sum_q (P_{nq} - C_q^I) \sum_c \omega_c \int s_{inq}\, d\Phi(\nu_i) = \sum_q \exp(\delta_{nq})\, \pi_q(P_{nq}),$$
$$\pi_q(P) = N(P - C_q^I) \sum_c \omega_c \int \frac{\exp\left(\mu_{cq} - \alpha_c(1-\tau)P + \sigma_\nu \nu_i\right)}{D_i}\, d\Phi(\nu_i).$$

Quality only scales the profit function π_q, which is common to all fringe firms, so every fringe firm sets the same price P_q^* for size q.[35] With the cell share $s_{cnq} = \int s_{inq}\, d\Phi(\nu_i)$ and $s_{nq} = \sum_c \omega_c s_{cnq}$, the first-order condition $\sum_c \omega_c s_{cnq} \left[1 - \alpha_c(1-\tau)(P_q^* - C_q^I)\right] = 0$ gives

$$P_q^* = C_q^I + \left[(1-\tau) \sum_c \frac{\omega_c s_{cnq}}{s_{nq}}\, \alpha_c\right]^{-1}.$$

At a common price, each fringe firm's cell shares are proportional to $\exp(\delta_{nq})$, so the weights $\omega_c s_{cnq}/s_{nq}$ are the same for every fringe firm, and the markup is the inverse of the average price sensitivity of the households that buy from the fringe. The random intercept ν_i does not change a household's price sensitivity, so it enters the markup only through these weights, that is, through which cells buy from the fringe.

This is the limit of Bertrand pricing by a single-product firm. Such a firm sets $s_j + (P_j - C_j^I)\, \partial s_j / \partial P_j = 0$, with

$$\frac{\partial s_j}{\partial P_j} = -(1-\tau) \sum_c \omega_c \alpha_c \int s_{ij}(1 - s_{ij})\, d\Phi(\nu_i).$$

As the firm becomes small relative to the market, $s_{ij} \to 0$ for every household and the condition converges to the fringe condition. The fringe condition omits only the term s_{ij}, through which a firm's price moves the denominator D_i. In the notation of Appendix D.5, the fringe markup is s_j/Λ_{jj}, the price fixed point with Γ_{jk} set to zero for the fringe's products.

[35] This requires π_q to have a unique maximizer. The first-order condition sets $P - C_q^I$ equal to the inverse of $\bar{\alpha}(P)$, the average of $\alpha_c(1-\tau)$ weighted by each cell's demand at P, and $\bar{\alpha}(P)$ falls in P at a rate equal to the weighted variance of $\alpha_c(1-\tau)$. The condition therefore holds at a single price if this variance is below $\bar{\alpha}(P)^2$, which is the case whenever the largest α_c in the market is less than $3 + 2\sqrt{2} \approx 5.8$ times the smallest. At the estimates in Table 3, the largest price sensitivity is 2.2 times the smallest.

Aggregation. At the common prices, the fringe's products of size q enter every household's choice probabilities only through

$$\sum_{n\in\mathcal{F}} \exp\left(\bar{u}_{cnq} + \sigma_\nu \nu_i\right) = \exp\left(\delta_{(F+1)q} + \mu_{cq} - \alpha_c(1-\tau)P^*_q + \sigma_\nu \nu_i\right),$$

$$\delta_{(F+1)q} = \log \sum_{n\in\mathcal{F}} \exp(\delta_{nq}).$$

The fringe is therefore equivalent to a single firm $f = F+1$ with no agents and one product per size, with price P^*_q, mean utility $\delta_{(F+1)q}$, and share $s_{(F+1)q} = \sum_n s_{nq}$. This is the fringe composite of Section 3.2. Its cell shares have the same composition as each member firm's, so with market subscripts restored its price satisfies

$$P_{(F+1)qmt} = C^I_{(F+1)qmt} + \left[(1-\tau_t)\sum_c \frac{\omega_{cmt}\, s_{c(F+1)qmt}}{s_{(F+1)qmt}}\,\alpha_c\right]^{-1}. \tag{29}$$

I use equation (29) to recover the fringe's installation costs from observed prices and to set the fringe's prices in counterfactuals.

C Bill Savings Appendix

C.1 Bill Savings Estimator

Proof of Proposition 2. For $d \in \{0,1\}$, Assumption 2 gives $\mathbb{P}(\, D_i = d \mid X_i = x \,) > 0$, so the conditional expectations are defined. For bills, since $B_i = B_i(d)$ when $D_i = d$,

$$\begin{aligned}
\mathbb{E}[\, B_i \mid X_i = x, D_i = d \,] &= \mathbb{E}[\, B_i(d) \mid X_i = x, D_i = d \,] \\
&= \mathbb{E}[\, B_i(d) \mid X_i = x \,] && \text{(by Assumption 1)} \\
&= \mu_d(x).
\end{aligned}$$

For capacity, since $G_i = G_i(1)$ when $D_i = 1$,

$$\begin{aligned}
\mathbb{E}[\, G_i \mid X_i = x, D_i = 1 \,] &= \mathbb{E}[\, G_i(1) \mid X_i = x, D_i = 1 \,] \\
&= \mathbb{E}[\, G_i(1) \mid X_i = x \,] && \text{(by Assumption 1)} \\
&= \mu_G(x).
\end{aligned}$$

Therefore, the data identifies $\rho(x)$:

$$\frac{\mathbb{E}\left[\, B_i \mid X_i = x, D_i = 0 \,\right] - \mathbb{E}\left[\, B_i \mid X_i = x, D_i = 1 \,\right]}{\mathbb{E}\left[\, G_i \mid X_i = x, D_i = 1 \,\right]} = \frac{\mu_0(x) - \mu_1(x)}{\mu_G(x)} = \rho(x). \quad \square$$

Threats to selection on observables. The principal threat to Assumption 1 is that households have private information about the suitability of their roof for solar, such as shading or orientation, or about anticipated load growth, such as a planned electric vehicle or large appliance that they can offset with solar. The billing data limit this concern because X_i contains the core inputs to the adoption decision. Systems are typically sized to a household's usage, and savings are largely determined by how large the counterfactual bill is: mechanically, a household with a larger bill can offset more of it. Agents may target households with particular usage patterns, which is allowed as long as agents shift the composition of adopters on X_i and do not affect the potential outcomes themselves.

Orthogonal signal. The signals ψ_i^{Δ} and ψ_i^{G} are the examples of Semenova and Chernozhukov (2021) for a conditional average treatment effect and for a regression function with a partially missing outcome. The ratio is not among their examples, so this paragraph shows how the signal ψ_i^{ρ} follows from the recipe of Ahrens et al. (forthcoming, Appendix B). The decile averages of $\rho(X_i)$ in Section 4.2 solve moment conditions of the form

$$\mathbb{E}\left[w(X_i) \left\{ \frac{\Delta B(X_i)}{\mu_G(X_i)} - \theta \right\} \right] = 0$$

for a function w of the covariates, such as a decile indicator. To first order, errors δ_0, δ_1, and δ_G in μ_0, μ_1, and μ_G change this moment by

$$\mathbb{E}\left[w(X_i) \, \frac{\delta_0(X_i) - \delta_1(X_i) - \rho(X_i)\, \delta_G(X_i)}{\mu_G(X_i)} \right].$$

The recipe adds terms that have mean zero at the true functions and offset this effect. Under Assumption 1, the terms $(1-D_i)\{B_i - \mu_0(X_i)\}/\{1-p(X_i)\}$, $D_i\{B_i - \mu_1(X_i)\}/p(X_i)$, and $D_i\{G_i - \mu_G(X_i)\}/p(X_i)$ have conditional mean zero, and an error in μ_0, μ_1, or μ_G lowers the conditional mean of the corresponding term one for one. Adding each term with its weight in the first-order effect above gives the moment $\mathbb{E}\left[\, w(X_i)\{\psi_i^{\rho} - \theta\} \,\right] = 0$. Errors in p have no first-order effect on it, because they multiply terms with conditional mean zero. Since this holds for every w, the conditional mean of ψ_i^{ρ} is $\rho(x)$, and it is insensitive to small errors in the estimated functions. In estimation, I bound the estimated propensity score to $[0.01, 0.99]$.

Sorting. I sort on predicted savings rather than on the signal because the signal contains the noise in B_i, G_i, and D_i, and sorting on it would sort households partly on noise. The decile cutoffs and the decile averages use the billing sampling weights of the main-half households, with robust standard errors.

Covariates. Table C1 lists the covariates. They are 22 to 24 measures, depending on the utility, of a household's 2020 billing, load profile, program enrollment, and local climate, together with indicators for the household's rate schedule, baseline territory, and local area, giving 157, 163, and 59 covariates for PG&E, SCE, and SDG&E, respectively.

Load profiles. The load profile covariates summarize a curve of each household's daily usage over 2020, which I fit to its bill intervals. The curve is a periodic cubic B-spline in the day of the year with 12 evenly spaced knots, dropping February 29, and it measures gross usage, which in 2020 precedes every adopter's system. The bills report only total usage over each interval, so the fit matches the curve's average over the 2020 days of each interval to the interval's average daily usage. I estimate the 12 coefficients by least squares, weighting each interval by its number of days, with a penalty on the squared second differences of neighboring coefficients around the year, and I constrain the coefficients to be nonnegative so that the curve is never negative. The penalty is common to all households, and I choose it by cross-validation. I fit the curve for the 184,543 households, 98 percent of the cleaned sample, whose 2020 intervals with positive usage cover at least 300 days and number at least 11. Figure C1 shows the fitted curve and the bill interval averages for one household in each utility. From the curve, the summer share is the share of annual usage from June through September, the seasonal amplitude is the range of daily usage divided by its mean, and the peak month, the month with the highest daily usage, is represented jointly by its sine and cosine, which accounts for seasonality by, for example, locating December near both January and November. To measure the shape of the profile separately from its level, I average the curve within each calendar month and divide the 12 monthly means by their average. I compute the principal components of these normalized profiles once, pooling the 184,543 households across utilities and weighting by the billing sampling weights, and I keep the first six, which capture 99.6 percent of the weighted variance. The first component, with 60 percent of the variance, contrasts summer with winter usage.

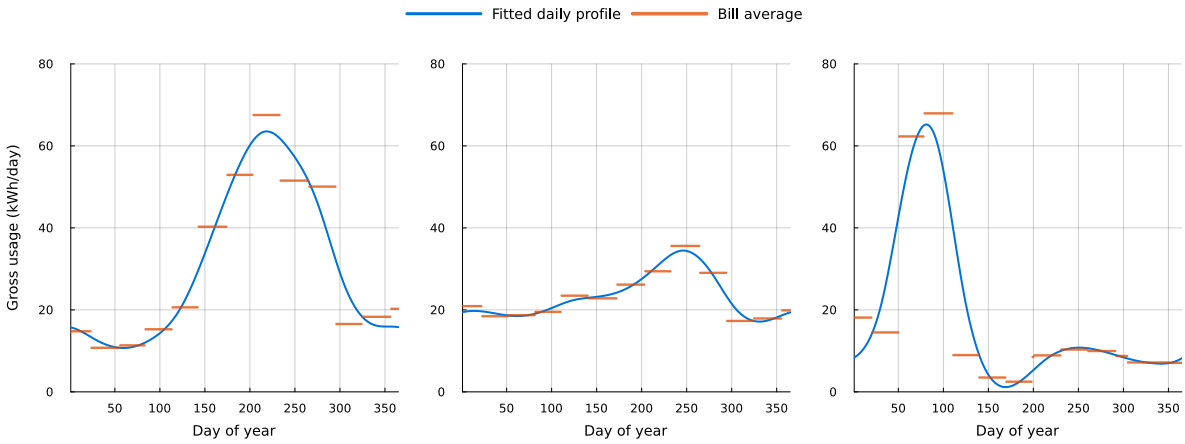
Learners. For the signal, the goal is to predict each function in Ψ as accurately as possible. For the bills and system size, I consider ridge regression, gradient-boosted trees (XGBoost), and a random forest; for the propensity score, ridge logistic regression and

Table C1: Covariates in the savings predictions

Covariate	Type	PGE	SCE	SDGE
Annual bill ($)	Continuous	X	X	X
Annual usage (kWh)	Continuous	X	X	X
Average price ($/kWh)	Continuous	X	X	X
Summer/winter bill ratio	Continuous	X	X	X
Summer share of usage	Continuous	X	X	X
Seasonal amplitude of usage	Continuous	X	X	X
Peak usage month, sine	Continuous	X	X	X
Peak usage month, cosine	Continuous	X	X	X
Load profile principal components	Continuous	(6)	(6)	(6)
Cooling degree days (65°F)	Continuous	X	X	X
Heating degree days (65°F)	Continuous	X	X	X
Irradiance (kWh/kWp)	Continuous	X	X	X
Utility users tax rate	Continuous	X	X	
CARE enrollment	Binary	X	X	X
FERA enrollment	Binary	X	X	X
All-electric service	Binary	X	X	X
Changed tariff schedule in 2020	Binary	X	X	X
Demand response program	Binary		X	X
No utility users tax	Binary	X	X	
Rate schedule	Categorical	(9)	(6)	(6)
Baseline territory	Categorical	(10)	(9)	(4)
Local area	Categorical	(101)	(99)	(27)
Total columns		143	138	59

Notes: All covariates are measured in 2020, before adoption, and bills and prices are in real 2023 dollars. Each prediction is estimated separately by utility; X marks a covariate that varies within the utility's estimation sample and so enters its models. The demand response program is SCE's Summer Discount Plan and SDGE's Plus Service peak-shaving rider. Categorical covariates enter as one indicator per level; parentheses give the number of levels, or the number of columns for the load profile principal components. Twelve local areas straddle two utilities, so the 227 local area columns across utilities correspond to 215 distinct areas.

Figure C1: Fitted usage profiles for three example households



Notes: The fitted daily profile is the household's fitted daily gross usage, and each bill average segment is the household's average daily gross usage over observed bill intervals.

XGBoost. Every adopter's system is between 1 and 25 kW by the sample restrictions in Appendix A.1, so I bound predicted system size below at 1 kW. Within each split and utility, I use whichever method has the lowest cross-validated error in the auxiliary half, weighted by the billing sampling weights: mean squared error for the bills and system size, and log loss for the propensity score. The signal can therefore combine methods across functions, utilities, and splits. Table C2 reports how often each method is chosen. XGBoost is chosen in every split for system size and in at least 89 percent of splits for the propensity score, while the choice for the bills varies by utility.

Fit and overlap. Unlike in the randomized experiments of Chernozhukov et al. (2025), where the propensity score is known, the confidence intervals for the decile averages require the functions in Ψ to be estimated consistently at a sufficiently fast rate (Semenova and Chernozhukov, 2021, Assumption 3.5). Figure C2 plots each household's realized bills and system size against their out-of-sample predictions. The binned means lie close to the 45-degree line for all three predictions. The bill without solar and system size are predicted well, with R^2 between 0.44 and 0.60 across utilities. The bill with solar is harder to predict for individual households, with R^2 of about 0.11, but its predictions remain calibrated. Figure C3 shows the distribution of the estimated propensity score for adopters and non-adopters. The two distributions overlap in every utility, although 11 percent of the weight of SDG&E non-adopters has a propensity score below 0.01. The bound on the propensity score affects 1 percent of the weight overall, and moving it from 0.01 to 0.005 or to 0.05, which affects 23 percent of the weight, changes average savings in every quintile

Table C2: Learners chosen across splits for the signal

Prediction	Utility	XGBoost	Random forest	Ridge
Bill without solar, μ_0	PGE	1.00	0.00	0.00
	SCE	0.84	0.00	0.16
	SDGE	0.52	0.10	0.38
Bill with solar, μ_1	PGE	0.82	0.11	0.07
	SCE	0.34	0.66	0.00
	SDGE	0.39	0.00	0.61
System size, μ_G	All	1.00	0.00	0.00
Adoption probability, p	PGE	0.93	–	0.07
	SCE	0.89	–	0.11
	SDGE	0.97	–	0.03

Notes: Share of the 150 repetitions of the sample split in which each method has the lowest cross-validated error in the auxiliary half, and is therefore used, for the given prediction and utility. Error is weighted mean squared error for the bills and system size and weighted log loss for the adoption probability. For the adoption probability, the ridge column is ridge logistic regression.

by at most \$1.02 per kilowatt.

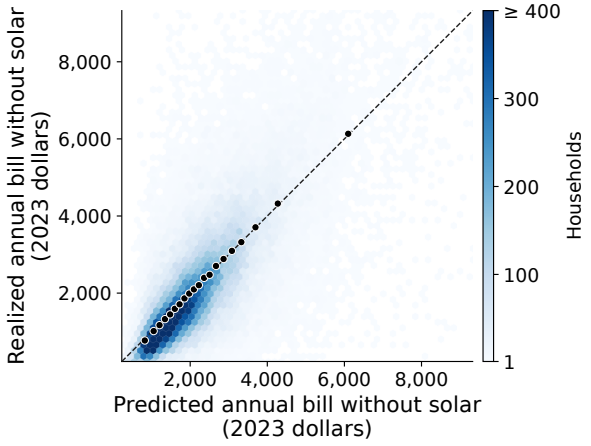
Predicted savings. For predicted savings, the goal is instead to rank households correctly by savings per kilowatt, rather than to predict the levels of the bills. I therefore evaluate candidates directly by how well they capture heterogeneity in the signal on the main half, following Chernozhukov et al. (2025). I consider four candidates: ridge regression, XGBoost, and a random forest, each used for both bills and system size, and a combination of random forest bills and XGBoost system size. For each candidate, I run two regressions on the main half. The first regresses the signal on the centered prediction,

$$\hat{\psi}_i^{\rho} = \beta_1 + \beta_2 \left(\hat{\rho}(X_i) - \overline{\hat{\rho}} \right) + e_i, \qquad i \in \mathcal{M}_s.$$

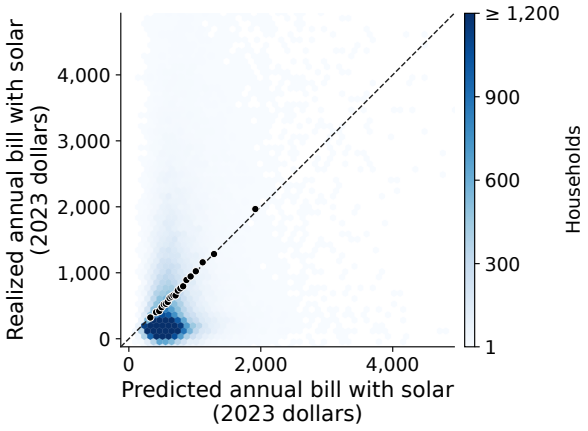
The intercept β_1 is average savings per kilowatt, and the slope β_2 equals one if the prediction is perfectly calibrated and zero if it is unrelated to savings. The second regresses the signal on indicators for $K = 5$ quintile groups of the candidate's prediction, whose coefficients are the group averages of Section 4.2. From these, Chernozhukov et al. (2025) define two measures of how much of the variation in savings a prediction explains,

$$\Lambda^{\mathrm{BLP}} = \beta_2^2 \operatorname{Var}\left(\hat{\rho}(X_i)\right), \qquad \Lambda^{\mathrm{GATES}} = \sum_{r=1}^{K} \mathbb{P}\left(r_{i,s} = r \right) \rho_r^2.$$

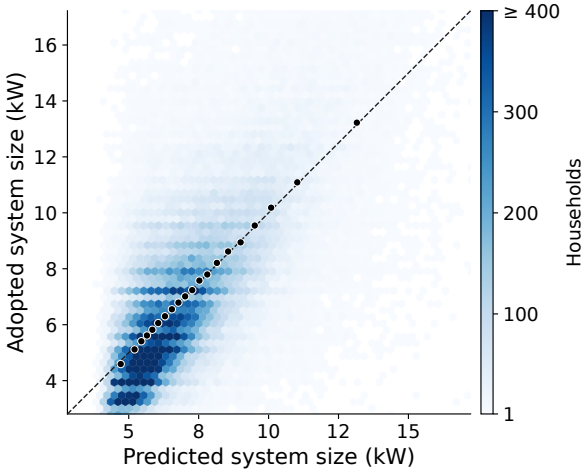
Figure C2: Fit of the out-of-sample predictions



(a) Bill without solar, μ_0



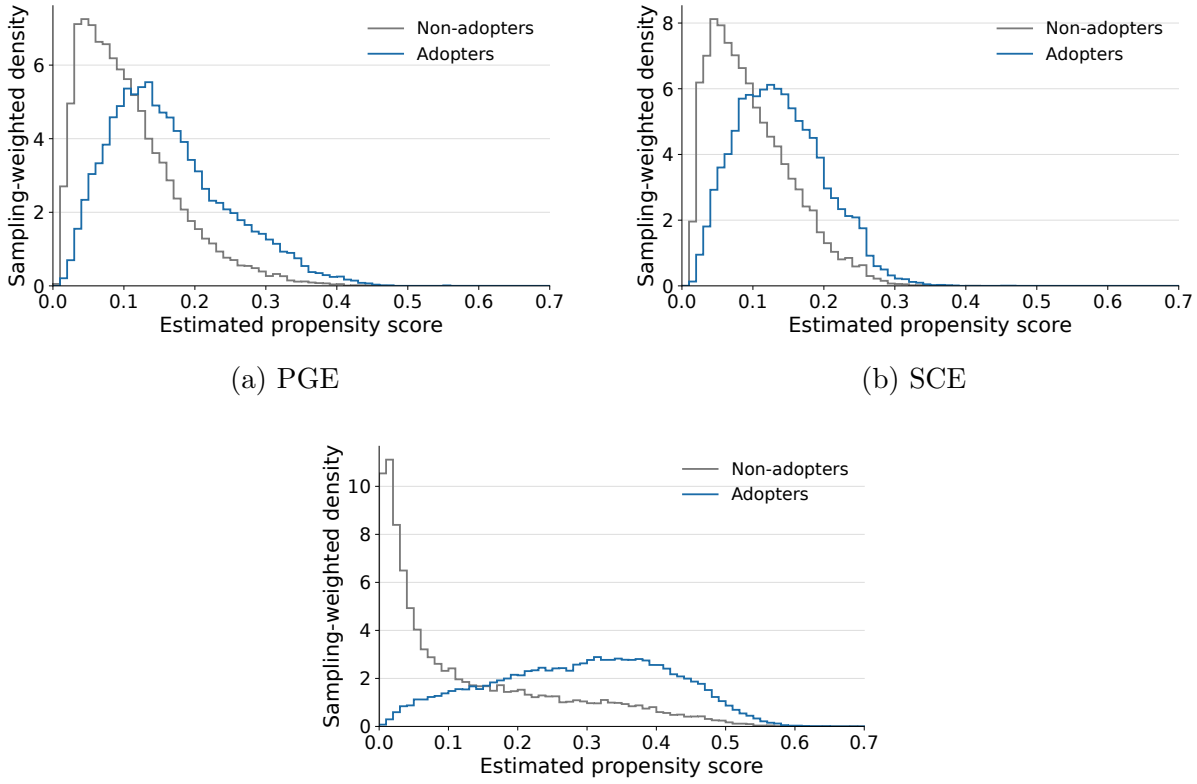
(b) Bill with solar, μ_1



(c) System size, μ_G

Notes: Shading counts households by realized value and out-of-sample prediction, and black markers are weighted means within 20 bins of the prediction, pooled across utilities; the dashed line is the 45-degree line. Predictions are out of sample, averaged over the repetitions of the sample split. Bills are 2024 annual bills in real 2023 dollars. Panel (a) uses the 64,173 non-adopters, and panels (b) and (c) use the 110,433 adopters; about 2 percent of households in each panel lie outside the plotted range. Out-of-sample R^2 for PG&E, SCE, and SDG&E are 0.49, 0.57, and 0.60 for the bill without solar; 0.11, 0.10, and 0.12 for the bill with solar; and 0.44, 0.46, and 0.48 for system size.

Figure C3: Propensity score overlap



(a) PGE

(b) SCE

(c) SDGE

Notes: Weighted densities of the out-of-sample propensity score among adopters and non-adopters in each utility.

Λ^{BLP} is the variation explained by a linear function of the prediction, and Λ^{GATES} is the variation explained by its groups, up to the squared mean savings common to every candidate.

Table C3 reports the results. All candidates capture heterogeneity: every slope is far from zero. The tree-based candidates have slopes close to one, while ridge regression has a slope of 0.67 and explains about half as much variation linearly. The random forest and XGBoost perform about equally well on both measures. Chernozhukov et al. (2025) allow aggregating methods instead of choosing one (Algorithm 6.1), so I use the average of the two predictions within each split, which outperforms both: it has the highest median of both Λ^{BLP} and Λ^{GATES}, and the highest Λ^{GATES} in 97 percent of splits.

Table C3: Fit of candidate savings predictions

Prediction	Heterogeneity slope β_2	Λ^{BLP}	Λ^{GATES}	Ranked first (share of splits)
Random forest	0.931 [0.892, 0.971]	7,697	66,936	0.01
XGBoost	0.929 [0.885, 0.975]	7,580	66,991	0.00
Random forest bills, XGBoost size	0.940 [0.901, 0.981]	7,628	66,877	0.03
Ridge regression	0.666 [0.602, 0.736]	3,756	64,062	0.00
Average of random forest and XGBoost	0.975 [0.934, 1.020]	8,011	67,245	0.97

Notes: Medians across 150 repetitions of the sample split. The heterogeneity slope is from a regression of the signal on the candidate's centered prediction in the main half; brackets contain 95 percent confidence intervals formed from the medians of the split-level bounds. The intercept, average savings per kilowatt, is \$245 for every candidate. Λ^{BLP} and Λ^{GATES} are the fit measures of Chernozhukov et al. (2025), with Λ^{GATES} computed on quintiles. The last column is the share of splits in which the candidate has the highest Λ^{GATES}. The last row is the equal-weight average of the random forest and XGBoost predictions within each split; I use it as the savings prediction, a choice fixed in advance.

Characteristics by savings group. Table C4 compares the 2020 characteristics of households in the highest and lowest savings quintiles and deciles. Households in the top decile had 2020 bills of \$3,995, against \$621 in the bottom decile, and used more than four times as much electricity. They were less often enrolled in CARE (2 against 38 percent) or all-electric (4 against 22 percent), and they used a larger share of their electricity in summer. Irradiance is nearly the same in the two groups.

Table C4: Baseline characteristics in the highest and lowest savings groups

	Savings quintile			Savings decile		
Covariate	5	1	Difference	10	1	Difference
Annual bill ($)	3,529	782	2,746 [2,694, 2,800]	3,995	621	3,366 [3,279, 3,454]
Annual usage (kWh)	11,170	3,188	7,981 [7,816, 8,139]	12,048	2,514	9,517 [9,270, 9,757]
Summer/winter bill ratio	1.644	1.182	0.463 [0.445, 0.481]	1.650	1.128	0.525 [0.501, 0.549]
Average price ($/kWh)	0.315	0.264	0.051 [0.048, 0.055]	0.328	0.275	0.053 [0.046, 0.060]
Changed tariff schedule in 2020	0.073	0.136	−0.063 [−0.072, −0.054]	0.071	0.149	−0.077 [−0.090, −0.064]
CARE enrollment	0.062	0.385	−0.322 [−0.334, −0.309]	0.016	0.382	−0.366 [−0.382, −0.349]
FERA enrollment	0.009	0.004	0.005 [0.002, 0.007]	0.007	0.003	0.005 [0.001, 0.008]
All-electric service	0.056	0.201	−0.146 [−0.156, −0.136]	0.041	0.224	−0.184 [−0.198, −0.170]
Summer Discount Plan	0.014	0.018	−0.004 [−0.008, 0.000]	0.006	0.014	−0.008 [−0.012, −0.003]
Plus Service	0.001	0.001	0.000 [0.000, 0.001]	0.002	0.001	0.000 [0.000, 0.001]
Summer share of usage	0.428	0.356	0.072 [0.069, 0.075]	0.430	0.349	0.082 [0.078, 0.086]
Seasonal amplitude of usage	0.866	0.982	−0.115 [−0.133, −0.097]	0.844	1.003	−0.160 [−0.186, −0.133]
Peak usage month, sine	−0.586	−0.261	−0.324 [−0.343, −0.305]	−0.602	−0.209	−0.395 [−0.422, −0.369]
Peak usage month, cosine	−0.302	0.125	−0.426 [−0.447, −0.405]	−0.344	0.148	−0.490 [−0.520, −0.461]
Load profile PC 1	0.199	−0.375	0.575 [0.551, 0.599]	0.212	−0.435	0.653 [0.620, 0.685]
Load profile PC 2	−0.071	0.033	−0.104 [−0.120, −0.088]	−0.086	0.019	−0.107 [−0.130, −0.083]
Load profile PC 3	−0.054	0.013	−0.067 [−0.077, −0.057]	−0.070	0.011	−0.081 [−0.096, −0.066]
Load profile PC 4	0.027	−0.029	0.057 [0.047, 0.067]	0.033	−0.040	0.074 [0.059, 0.089]
Load profile PC 5	−0.010	−0.014	0.003 [−0.002, 0.008]	−0.017	−0.028	0.011 [0.003, 0.019]
Load profile PC 6	−0.010	−0.002	−0.008 [−0.013, −0.004]	−0.013	−0.010	−0.003 [−0.010, 0.004]
Utility users tax rate	0.016	0.021	−0.005 [−0.006, −0.004]	0.015	0.021	−0.006 [−0.007, −0.004]
No utility users tax	0.702	0.634	0.067 [0.051, 0.083]	0.706	0.637	0.069 [0.047, 0.092]
Cooling degree days	1,048	899	150 [138, 162]	1,050	878	171 [154, 189]
Heating degree days	1,373	1,194	179 [165, 193]	1,400	1,195	203 [182, 223]
Irradiance (kWh/kWp)	1,796	1,791	4 [2, 6]	1,791	1,791	0 [−3, 4]

Notes: Weighted means of 2020 characteristics among main-half households in the highest and lowest quintiles and deciles of predicted savings per kW, the average of the random forest and XGBoost predictions. Estimates are medians across 150 repetitions of the sample split; brackets contain 95 percent confidence intervals for the differences, formed from the medians of the split-level bounds. Bills and prices are in real 2023 dollars. Binary covariates are shares of households.

Table C5: Bill savings, adoption, and system size by savings decile

Savings decile	Savings per kW ($/kW-yr)	Adoption rate (2021–2023)	Adopter system size (kW)	Total savings ($/yr)
1	57	0.024	6.09	314
	[44, 71]	[0.022, 0.026]	[5.89, 6.29]	[234, 399]
2	146	0.040	5.71	814
	[134, 158]	[0.038, 0.043]	[5.57, 5.85]	[751, 880]
3	199	0.060	5.83	1,138
	[190, 210]	[0.056, 0.063]	[5.71, 5.94]	[1,083, 1,195]
4	226	0.081	6.25	1,381
	[216, 235]	[0.077, 0.085]	[6.14, 6.35]	[1,323, 1,442]
5	247	0.100	6.60	1,600
	[238, 256]	[0.095, 0.105]	[6.50, 6.69]	[1,542, 1,659]
6	266	0.117	6.84	1,808
	[257, 275]	[0.112, 0.122]	[6.75, 6.93]	[1,747, 1,868]
7	287	0.134	7.17	2,059
	[278, 296]	[0.129, 0.140]	[7.08, 7.25]	[1,996, 2,125]
8	308	0.154	7.65	2,365
	[300, 317]	[0.148, 0.160]	[7.57, 7.73]	[2,299, 2,431]
9	334	0.176	8.01	2,707
	[326, 342]	[0.169, 0.183]	[7.94, 8.09]	[2,633, 2,782]
10	376	0.212	8.68	3,375
	[367, 385]	[0.204, 0.219]	[8.62, 8.74]	[3,284, 3,473]

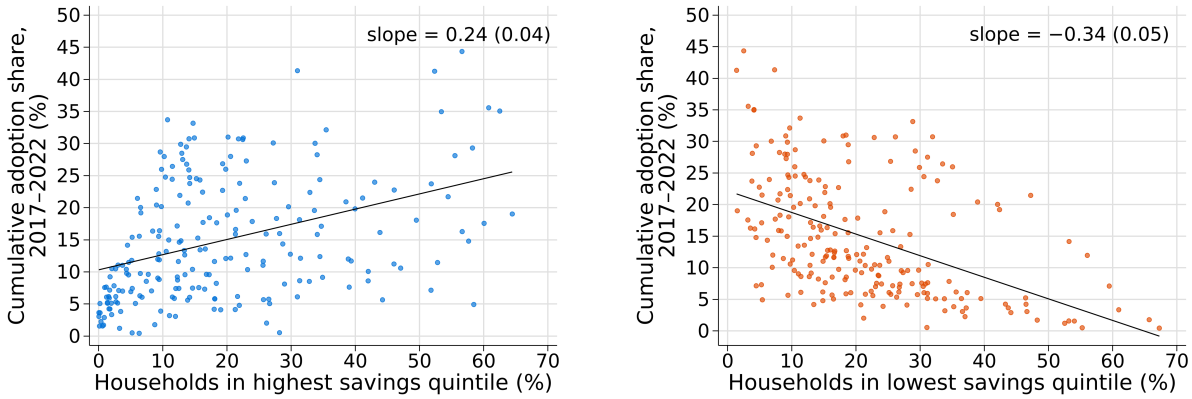
Notes: Deciles of predicted savings per kW, the average of the random forest and XGBoost predictions. Estimates are medians across 150 repetitions of the sample split procedure; brackets contain 95 percent confidence intervals formed from the medians of the split-level bounds. Savings per kW and total savings are group averages of the orthogonal signals measured at each household's own chosen system size, in real 2023 dollars. The adoption rate is the share of households that adopted during 2021–2023. Adopter system size is the mean installed capacity among adopters. Decile 1 has the lowest predicted savings.

Table C6: Savings, adoption, and system size by savings quintile

	Savings quintile				
	1 (lowest)	2	3	4	5 (highest)
Savings per kW ($/kW-yr)	102	213	256	298	355
	[93, 111]	[206, 220]	[250, 263]	[292, 304]	[349, 361]
Adoption rate (2021–2023)	0.032	0.070	0.108	0.144	0.194
	[0.030, 0.034]	[0.068, 0.073]	[0.105, 0.112]	[0.140, 0.148]	[0.189, 0.199]
Adopter system size (kW)	5.85	6.07	6.73	7.42	8.37
	[5.74, 5.97]	[5.99, 6.14]	[6.66, 6.79]	[7.37, 7.48]	[8.33, 8.42]
Total savings ($/yr)	569	1,259	1,704	2,214	3,045
	[517, 622]	[1,218, 1,301]	[1,662, 1,746]	[2,167, 2,261]	[2,987, 3,106]

Notes: Quintiles of predicted savings per kW. Estimates are medians across 150 repetitions of the sample split procedure. Brackets contain 95 percent confidence intervals formed from the medians of the split-level bounds. Savings per kW and total savings are group averages of the orthogonal signals measured at each household's own chosen system size, in real 2023 dollars. The adoption rate is the share of households that adopted during 2021–2023. Adopter system size is the mean installed capacity among adopters.

Figure C4: Local shares of high- and low-savings households and solar adoption



(a) Highest savings quintile

(b) Lowest savings quintile

Notes: Each point is one of the 215 local areas. The horizontal axis is the share of the local area's households in the highest or lowest quintile of predicted savings per kW, averaged across the 150 repetitions of the sample split, and the vertical axis is cumulative adoption over 2017–2022 as a share of the households that had not adopted solar by 2017. Lines are least-squares fits weighted by those households, with heteroskedasticity-robust standard errors in parentheses.

C.2 Income and Bill Savings

The demand cells need the joint distribution of income and savings in each local area and year. The billing data do not record income, so I predict it from covariates that the billing data share with the American Community Survey (ACS), in which I use owner-occupied single-family households throughout. The prediction is a function of covariates already in X_i, so it adds nothing to the savings predictions, but its averages within savings deciles identify the joint distribution under the assumption below.

Joint distribution. Let $Y_i \in \{1, \ldots, 12\}$ be household i's income bin, $r_i \in \{1, \ldots, 10\}$ its savings decile, and X_i^A the covariates common to the two data sets: the annual electricity bill, CARE status, electric heating, and the household's PUMA. In local area ℓ and year t, the joint distribution is $\omega_{\ell t}(y, r) = \mathbb{P}_{\ell t}\left(\, Y_i = y, r_i = r \,\right)$, with income margin $\omega_{\ell t}^Y(y)$ and savings margin $\omega_{\ell t}^R(r)$. Let $h_t(y \mid x) = \mathbb{P}_t\left(\, Y_i = y \,\middle|\, X_i^A = x \, \right)$ be the probability of income bin y given the common covariates.

Assumption 5 (Income prediction within savings deciles)**.** For all y, r, ℓ, and t,

$$\mathbb{E}_{\ell t}\left[\, 1\{Y_i = y\} - h_t(y \mid X_i^A) \,\middle|\, r_i = r \,\right] = 0.$$

The covariates in X_i^A are a subset of those in X_i, so the assumption says that the error in predicting income from the bill, CARE status, electric heating, and location is unrelated to savings. Other features of usage that predict savings, such as the summer/winter bill ratio and the shape of the load profile, must not predict income given these covariates. Under Assumption 5,

$$\omega_{\ell t}(y, r) = \mathbb{E}_{\ell t}\left[\, 1\{Y_i = y\}\, 1\{r_i = r\} \,\right] = \mathbb{E}_{\ell t}\left[\, h_t(y \mid X_i^A)\, 1\{r_i = r\} \,\right],$$

which combines the savings deciles and covariates of billing households with h_t from the ACS.

Income model. I estimate h_t with a multinomial logit on the pooled 2017–2021 ACS. For ACS household i in year t,

$$\begin{aligned}
\mathbb{P}\left(\, Y_i = y \,\middle|\, X_i^A \, \right) &= \frac{1\{y \in \mathcal{Y}(\mathrm{CARE}_i)\} \exp(v_{ity})}{\sum_{k \in \mathcal{Y}(\mathrm{CARE}_i)} \exp(v_{itk})}, \\
v_{ity} &= \boldsymbol{x}_{it}' \boldsymbol{\beta}_{ty} + \vartheta_{\mathrm{puma}(i),y}, \\
\boldsymbol{x}_{it} &= \Big(1,\ s_t(\mathrm{bill}_i)',\ \mathrm{CARE}_i,\ \mathrm{CARE}_i \times s_t(\mathrm{bill}_i)',\ \mathrm{ElecHeat}_i\Big)',
\end{aligned}$$

where bill_i is the annual electricity bill, CARE_i indicates CARE eligibility, and ElecHeat_i indicates an all-electric home. The function s_t is a cubic spline in the bill, in thousands of dollars, with boundary knots at the year-t weighted 5th and 95th percentiles, interior knots at the quartiles, and linear extrapolation beyond the boundary knots. The PUMA effects $\vartheta_{\text{puma},y}$ are common to all years and absorb local differences such as climate, while the coefficients $\boldsymbol{\beta}_{ty}$ are specific to the year. I determine CARE eligibility in the ACS from the program's income guidelines, which depend on household income and size, and from categorical eligibility through Medi-Cal. Because eligibility is defined by income, every ACS household in the lowest income bin is eligible, so $\mathcal{Y}(1) = \{1, \ldots, 12\}$ and $\mathcal{Y}(0) = \{2, \ldots, 12\}$. The CARE terms of bin 1 are not identified and are omitted, and bin 12 is the reference. I estimate the logit by maximum likelihood with the ACS household weights on the pooled complete-case sample and solve the score equations exactly, so the fitted income shares equal the observed shares in every PUMA and income bin, in every year and income bin, and in every year, CARE status, and income bin. Pooling the years leaves every PUMA with households in every income bin. The 2021 terms apply to 2022.

In the billing data, I evaluate the model at each household's 2020 bill, CARE enrollment, all-electric status, and PUMA, so enrollment stands in for eligibility. The two are close: in 2020, 31 percent of ACS households are eligible and 27 percent of billing households are enrolled, and the two shares have a correlation of 0.85 across PUMAs.

Plug-in estimator. In each split s, households in the main half $\mathcal{M}_s$ are assigned to savings deciles $r_{i,s}$. A household's decile can change from one split to the next, but rankings are fairly stable: on average, a household falls in its most frequent decile in 63 percent of splits and in its most frequent quintile in 81 percent. The billing covariates are fixed in 2020, and I assume the population savings distribution is unchanged over time, so the savings distribution is the same in every year. The plug-in estimator averages the split estimates of the joint distribution,

$$\hat{\omega}_{\ell t}(y, r) = \frac{1}{S} \sum_{s=1}^{S} \frac{\sum_{i \in \ell \cap \mathcal{M}_s} w_i \, 1\{r_{i,s} = r\} \, \hat{h}_t(y \mid X_i^A)}{\sum_{i \in \ell \cap \mathcal{M}_s} w_i},$$

where w_i is the billing sampling weight and $S = 150$. Its savings margin, $\hat{\omega}_\ell^R(r)$, is the local share of households in each decile from Section 4.3.

Minimum-KL adjustment. The income margin of the plug-in estimator averages the ACS model over billing households, so it need not equal the income distribution tabulated directly in the ACS. The difference has two sources: the model's fit in the ACS, and

differences between the covariates of ACS and billing households. Statewide, the two are small. Figure C5 shows that the model reproduces the ACS income shares to within 0.3 percentage points in every year and income bin when evaluated at ACS households, and to within 0.6 percentage points when evaluated at billing households. Within local areas, the differences are larger. Figure C6 decomposes them by income bin, local area, and year: weighting by potential adopters, the mean absolute error is 1.4 percentage points from the model's fit to the one-year ACS and 0.7 percentage points from the difference in covariates.

To match both margins, I take the distribution closest to the plug-in estimator in Kullback–Leibler divergence that has the ACS income margin and the estimated savings margin:

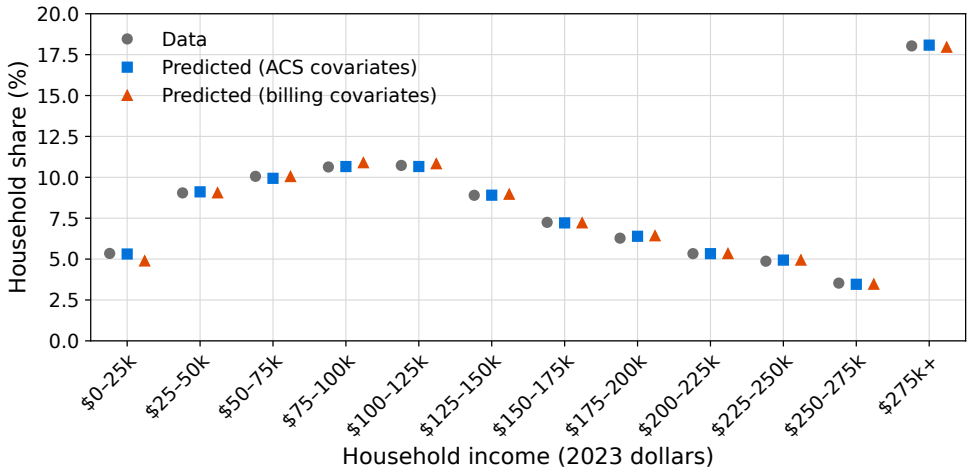
$$\tilde{\omega}_{\ell t} = \arg\min_{\omega \geq 0} \sum_{y,r} \omega(y,r) \log \frac{\omega(y,r)}{\hat{\omega}_{\ell t}(y,r)}$$
$$\text{subject to} \quad \sum_{r} \omega(y,r) = \hat{\omega}^{Y}_{\ell t}(y) \;\; \forall y, \qquad \sum_{y} \omega(y,r) = \hat{\omega}^{R}_{\ell}(r) \;\; \forall r,$$

where $\hat{\omega}^{Y}_{\ell t}(y)$ is the one-year ACS income distribution of owner-occupied single-family households in the local area, using the 2021 survey for 2022. Cells with zero plug-in mass, such as the lowest income bin in a savings decile without CARE households, remain zero. I solve the problem by iterative proportional fitting on the positive cells, and all 1,290 local area and year solutions match both margins to within 10^{-12}. Weighting by potential adopters, the adjustment reassigns 9.8 percent of the joint probability mass in the average local area and year (9.4 to 10.5 percent by year), close to the 9.7 percent by which the plug-in income margin differs from the ACS margin. It raises the mean local correlation between income and savings per kW from 0.240 to 0.253, and Figure C7 shows the distribution of these correlations under both estimators. The demand cell weights are $\omega_{cmt} = (N_{\ell t}/N_{mt})\,\tilde{\omega}_{\ell t}(y,r)$ for $c = (\ell, y, r)$, where $N_{\ell t}$ is the number of potential adopters in local area ℓ.

C.3 Billing Micromoments

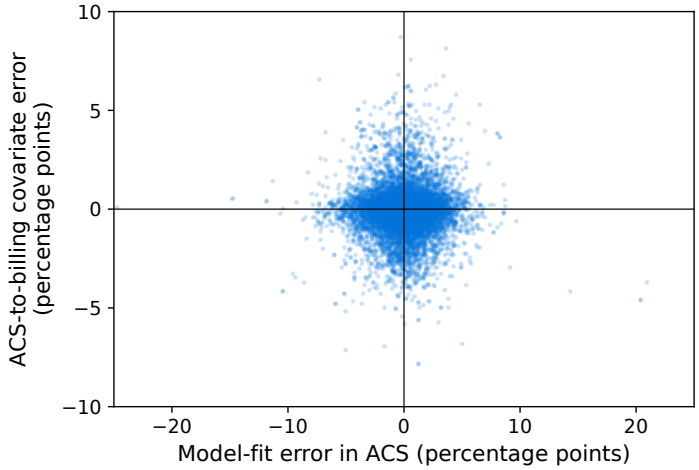
The billing micromoments are 38 differences estimated on billing households in 2021–2022. Table C7 reports them in three panels: the annual adoption rate by savings decile (9 differences), the agent density of the chosen firm by income bin and savings decile (11 and 9 differences), and the post-ITC price of the chosen product by savings decile (9 differences). Each difference is relative to the lowest income bin or savings decile. The income bins are the twelve income bins of the demand cells.

Figure C5: Statewide income shares



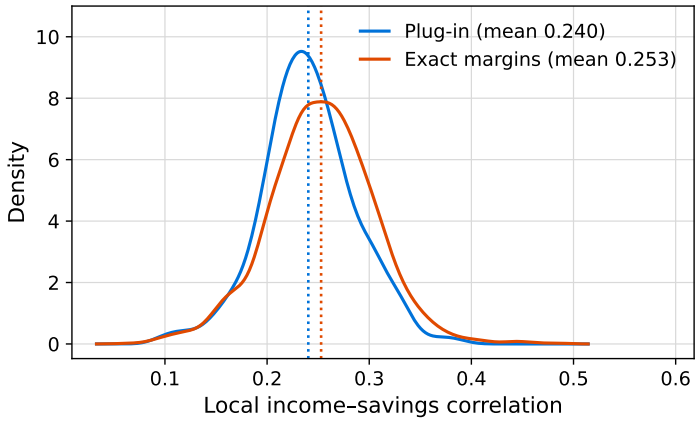
Notes: Income distribution in 2020 over the 215 local areas, weighting local areas by potential adopters. "Data" is the income distribution tabulated directly from the one-year ACS. The two predicted distributions average the fitted income probabilities over ACS and billing households, respectively.

Figure C6: Decomposition of local income share errors



Notes: Each point is one income bin, local area, and year, for the 215 local areas over 2017–2022. The horizontal coordinate is the income share fitted at ACS households minus the observed one-year ACS share. The vertical coordinate is the share fitted at billing households minus the share fitted at ACS households.

Figure C7: Local correlations between income and bill savings



Notes: Weighted kernel densities of the correlation between log income and savings per kW within each local area and year, computed from the representative values of the income bins and savings deciles. The densities pool the 215 local areas over 2017–2022 and weight each local area and year by its potential adopters; dotted lines mark the weighted means. "Plug-in" is the plug-in estimator and "Exact margins" the minimum-KL adjustment.

Households and years. Each billing household contributes one row for each year $t \in \{2021, 2022\}$ in which it has not yet adopted: a household that does not adopt in 2021–2022 contributes both years, and an adopter contributes every year up to and including the year its application is received. Let $D_{it} = 1$ if household i applies in year t for a modeled product, which includes the fringe composite. Row (i, t) carries the weight $\tilde{w}_{it} = k_{g(i,t)}\, w_i$, where w_i is the billing sampling weight and k_g calibrates the rows of pool g to the demand data. The pools are adopters of modeled products, households that adopt in a later year, and households that never adopt, each within a market, and adopters of other installations within a utility and year. The calibrated rows reproduce the potential adopters N_{mt} and the modeled installations of every market. All 60 markets of 2021–2022 have billing households in every pool with a positive target in every split.

Adoption rate. All three panels are computed within each of the 150 sample splits of Section 4.2, on the main half $\mathcal{M}_s$, with calibration factors formed on that half. Households are assigned to savings deciles $r_{i,s}$ at the weighted deciles of predicted savings over these household-years. The annual adoption rate of decile r is the weighted share of these household-years with an application,

$$\bar{D}_{r,s} = \frac{\sum_{i \in \mathcal{M}_s} \sum_t \tilde{w}_{it}\, 1\{r_{i,s} = r\}\, D_{it}}{\sum_{i \in \mathcal{M}_s} \sum_t \tilde{w}_{it}\, 1\{r_{i,s} = r\}},$$

and the moments are the differences $\bar{D}_{r,s} - \bar{D}_{1,s}$, $r = 2, \ldots, 10$, averaged over splits. The model analog of $\bar{D}_{r,s}$ is the annual adoption probability of a cell, the inside share $\sum_{j \neq 0} s_{cjmt}$, averaged over the cells of decile r in the 2021 and 2022 markets with weights equal to their potential adopters, $N_{mt}\,\omega_{cmt}$, the model counterpart of the calibrated row weights.

Chosen price. The price and agent density moments use the adopters of large firms. Let $\mathcal{S}_s$ be these adopters in $\mathcal{M}_s$, with application year $t(i)$, weight $\tilde{w}_i = \tilde{w}_{it(i)}$, and chosen product $j(i) = (f(i), q(i))$, the installing firm and the size bin of the installed system in the household's market $m(i)$. I merge each adopter to the demand data price of its chosen product in that market, $p_i = (1 - \tau_{t(i)}) P_{j(i)m(i)t(i)}$, in \$10,000. The price moment of decile r is the weighted mean $\bar{p}_{r,s} = \sum_{i \in \mathcal{S}_s} \tilde{w}_i 1\{r_{i,s} = r\} p_i \Big/ \sum_{i \in \mathcal{S}_s} \tilde{w}_i 1\{r_{i,s} = r\}$.

Agent density projection. The outcome is the chosen firm's agent density $a_i = a_{f(i)m(i)t(i)}$, in agents per 100,000 potential adopters. Income is not observed in the billing data, so each adopter enters with the predicted probabilities of the income model in Appendix C.2, $\hat{h}_{iy} = \hat{h}_{t(i)}(y \mid X_i^A)$ for the twelve income bins. I expand each adopter into twelve income rows with weights $\tilde{w}_i \hat{h}_{iy}$ and estimate an additive projection on the expanded data by weighted least squares,

$$(\hat{\theta}, \hat{b}^Y, \hat{b}^R)_s = \underset{\theta, b^Y, b^R}{\arg\min} \sum_{i \in \mathcal{S}_s} \sum_{y=1}^{12} \tilde{w}_i \, \hat{h}_{iy} \left(a_i - \theta_{H(i)} - b_y^Y - b_{r_{i,s}}^R \right)^2, \qquad b_1^Y = b_1^R = 0,$$

where $H(i)$ is the adopter's market and size bin. The moments are the income effects $\hat{b}_y^Y$, $y = 2, \ldots, 12$, and the savings effects $\hat{b}_r^R$, $r = 2, \ldots, 10$. These equal the coefficients of the infeasible projection on true income bins when the error in the income prediction is mean independent of the chosen firm's agent density given the covariates, the savings decile, and the fixed effect cell, $\mathbb{E}\left[1\{Y_i = y\} - h_{t(i)}(y \mid X_i^A) \,\middle|\, X_i^A, a_i, r_{i,s}, H(i) \right] = 0$. The model analogs of the price and agent density moments apply the same mean and projection to the model's predicted adoptions of large-firm products, weighting each cell and product by $N_{mt}\,\omega_{cmt}\,s_{cjmt}$ and using the cell's income bin in place of $\hat{h}_{iy}$.

Inference. Each moment is the mean of its split estimates over the 150 splits. Following Fava (2025), the covariance of the 38 moments is the average across splits of the household-level Eicker–Huber–White covariance on the main half, scaled by $\pi(\pi^{-1} + S - 1)/S$ with evaluation fraction $\pi = 1/2$ and $S = 150$ splits. The influence functions account for the calibration of the row weights, and the covariance treats the predicted savings and the income model as known.

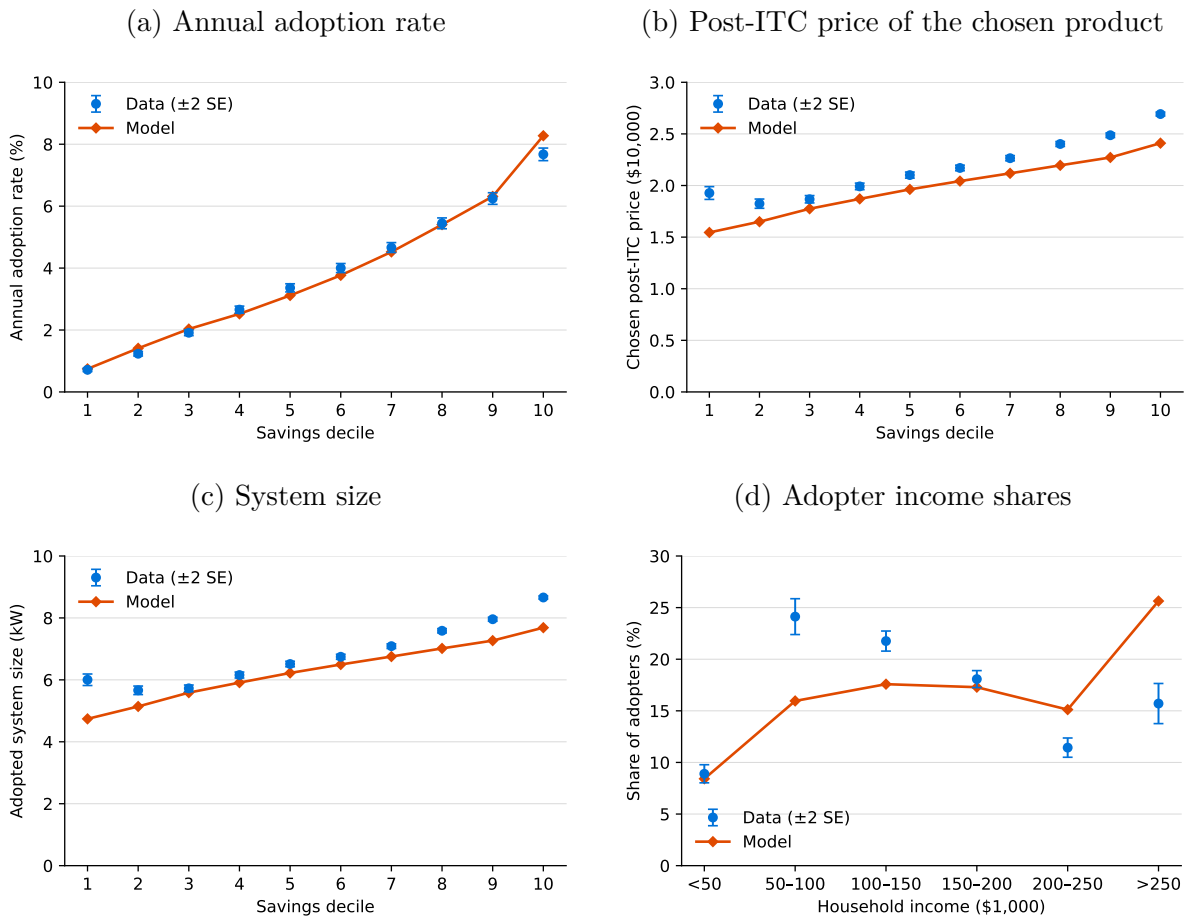
Table C7: Billing micromoments

Difference	Estimate	Std. err.
Panel A. Annual adoption rate (percentage points)		
Savings decile 2 minus 1	0.52	(0.05)
Savings decile 3 minus 1	1.20	(0.06)
Savings decile 4 minus 1	1.94	(0.06)
Savings decile 5 minus 1	2.65	(0.07)
Savings decile 6 minus 1	3.29	(0.08)
Savings decile 7 minus 1	3.95	(0.09)
Savings decile 8 minus 1	4.73	(0.09)
Savings decile 9 minus 1	5.53	(0.10)
Savings decile 10 minus 1	6.96	(0.11)
Panel B. Chosen-firm agents per 100,000 potential adopters		
Income bin 2 minus 1	−0.948	(0.158)
Income bin 3 minus 1	−2.227	(0.360)
Income bin 4 minus 1	−2.438	(0.390)
Income bin 5 minus 1	−2.636	(0.397)
Income bin 6 minus 1	−2.637	(0.400)
Income bin 7 minus 1	−2.790	(0.404)
Income bin 8 minus 1	−3.007	(0.403)
Income bin 9 minus 1	−3.128	(0.409)
Income bin 10 minus 1	−3.231	(0.410)
Income bin 11 minus 1	−3.300	(0.403)
Income bin 12 minus 1	−3.841	(0.404)
Savings decile 2 minus 1	0.884	(1.387)
Savings decile 3 minus 1	1.461	(1.279)
Savings decile 4 minus 1	1.762	(1.234)
Savings decile 5 minus 1	2.400	(1.217)
Savings decile 6 minus 1	2.901	(1.208)
Savings decile 7 minus 1	3.513	(1.200)
Savings decile 8 minus 1	3.976	(1.194)
Savings decile 9 minus 1	3.833	(1.183)
Savings decile 10 minus 1	3.476	(1.184)
Panel C. Chosen post-ITC system price ($10,000)		
Savings decile 2 minus 1	−0.103	(0.038)
Savings decile 3 minus 1	−0.060	(0.036)
Savings decile 4 minus 1	0.064	(0.035)
Savings decile 5 minus 1	0.174	(0.034)
Savings decile 6 minus 1	0.243	(0.034)
Savings decile 7 minus 1	0.337	(0.033)
Savings decile 8 minus 1	0.475	(0.033)
Savings decile 9 minus 1	0.560	(0.033)
Savings decile 10 minus 1	0.765	(0.032)

Notes: Differences relative to income bin 1 or savings decile 1, estimated on 2021–2022 billing households. Estimates are means over 150 sample splits; standard errors in parentheses are the square roots of the diagonal of the repeated-split covariance used in demand estimation. Panels B and C use adopters of large firms. Panel B reports the income and savings effects of the projection with market and size-bin fixed effects, over the twelve income bins of the demand cells. Panel C is the post-ITC price of the chosen product in real 2023 dollars.

Fit. Figure C8 compares the model at the demand estimates with the data in levels. Estimation targets the differences relative to the lowest decile or income bin, not the levels, and it does not use system size.

Figure C8: Micromoment fit



Notes: Data are 2021–2022 billing households by savings decile in panels (a) to (c) and LBNL adopters pooled over 2017–2022 by the six LBNL income bins in panel (d); bars are ±2 standard errors. Model levels are evaluated at the demand estimates in Table 3. Estimation targets each series' differences relative to its lowest decile or bin, not its level. Panel (b) uses adopters of large firms. Panel (c) is the mean capacity of adopted systems in kW and is held out of estimation.

D Estimation Appendix

D.1 Demand Estimation

Instruments. The cost shifters are electrician and sales occupation wages from the OEWS, per-watt module and inverter spot prices interacted with the characteristics of

the modules and inverters used by each firm, and an indicator for whether the market is different from the firm's headquarters. The rival-characteristics instruments are the sums of characteristics of rival products in the market and of rival products in the same size bin. The quadratic differentiation instruments sum the squared differences between each characteristic of a product and that of every other product in the market. Both sets are built from the three non-size characteristics included in demand: efficiency, log inverter loading ratio, and microinverter share. To identify α_y, I also use a price differentiation instrument. I predict prices from a regression of price on the exogenous characteristics, the price cost shifters, and year and firm fixed effects, and the instrument sums the squared differences between a product's predicted price and those of rival firms' products. I interact this instrument with a constant, the market mean of standardized log income, and the market mean of its square, so that it varies with the income composition of the market, which identifies how price sensitivity varies with income (Miravete, Seim and Thurk, 2023).

Weighting matrix. The weighting matrix is block diagonal, with one block for each set of moments in Section 3.3. Each block is the inverse of an estimate of the covariance of that set of moments:

- *Instrument moments.* In the first step, I use the residuals of an instrumented homogeneous logit regression of observed log inside-to-outside share ratios on net price, agent density, the product characteristics, mean income and its interaction with size, and firm and year effects, instrumenting price and agent density with the excluded instruments. The covariance sums the products of the instruments and these residuals within each market. In the second step, I update this covariance using the model's residuals at the first-step estimates.

- *Local adoption counts.* In the first step, the variance of each local area's count treats each potential adopter's choice to adopt as an independent binary outcome, evaluated at the observed counts of adopters and non-adopters. In the second step, I update this variance using the adoption probabilities that the model predicts at the first-step estimates.

- *Income micromoments.* I resample local areas from the LBNL data to bootstrap the covariance of the income shares.

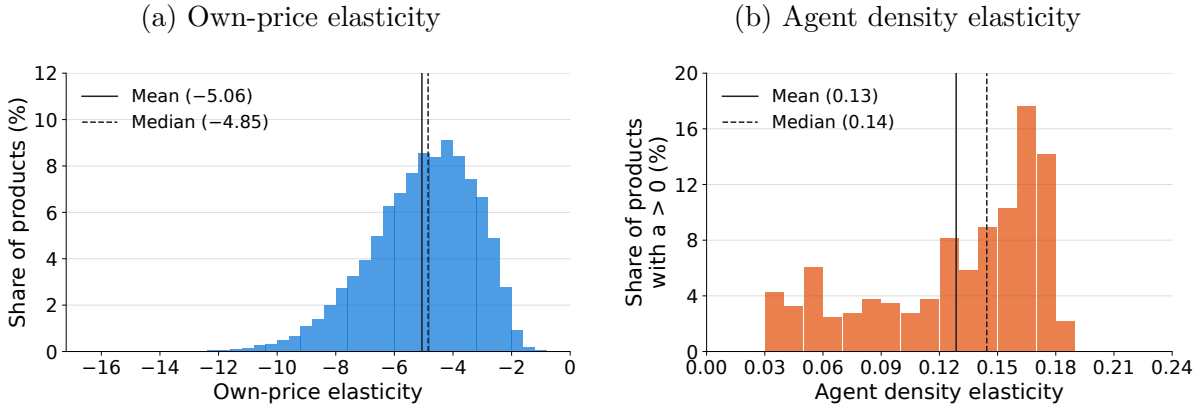
- *Billing micromoments.* Following the repeated-split inference procedure in Fava (2025), I compute the joint covariance of the billing differences across the 150 repeated sample splits of Section 4.

- *SEEDS moment.* I use the sampling variance of the survey share, which Appendix D.2 derives.

The billing, income, and SEEDS moments come from samples separate from the demand data, so their covariances are the same in both steps. In the second step, I re-estimate with the updated weighting matrix held fixed.

Standard errors. The standard errors are sandwich standard errors at the second-step estimate. The covariance of the instrument and count moments is clustered by market, which keeps the covariance between the two sets within each of the 180 markets. The billing, income, and SEEDS covariances enter as estimated above, since their samples are independent of the demand data and of one another. The demand cell weights, the savings estimates, and the predicted income distribution of billing households are treated as known.

Figure D1: Own-price and agent elasticities of demand



Notes: Distributions across products (firm, size bin, and market) at the demand estimates in Table 3, with each product weighted equally. Panel (a) is the elasticity of a product's demand with respect to its own price, over all 50,032 products. Panel (b) is the elasticity of a product's demand with respect to its firm's agent density, holding prices and rivals' salesforces fixed, over the 15,538 products of firms with agents in the market.

D.2 Agent Reach Moment

Survey share. The reach sample consists of the 954 California adopters in the SEEDS survey who installed between 2013 and 2015, bought or leased their system, and reported how they first connected with the installer they chose. The agent channel is direct sales: the installer knocked on the door or left a flyer. Because respondents may check several responses, I assume that each checked response is equally likely to be the one that delivered

the accepted offer, so respondent i contributes k_i^A/k_i, where k_i is the number of responses checked and k_i^A indicates direct sales.

The agent share is much higher among leased systems, $\bar{p}_L = 0.640$, than among purchased ones, $\bar{p}_O = 0.175$. I assume that, within each ownership type, respondents are a random sample of 2013–2015 adopters, and I weight the two means by the third-party-owned share of 2013–2015 installations in the three IOU territories, $s^{TPO} = 0.553$, among installations with a recorded ownership type:

$$\hat{p} = s^{TPO}\bar{p}_L + (1 - s^{TPO})\bar{p}_O = 0.432.$$

Treating s^{TPO} as known, the sampling variance of the survey share is

$$\mathrm{Var}(\hat{p}) = (s^{TPO})^2 \frac{S_L^2}{n_L} + (1 - s^{TPO})^2 \frac{S_O^2}{n_O},$$

where n_g is the number of respondents with ownership type g and S_g^2 is the sample variance of k_i^A/k_i among them. The standard error is 0.014.

Model analog. By equation (28), an adopter of firm f met the firm through an agent with probability $\phi a_f/(1 + \phi a_f)$, whatever the product and the household. The model analog of $\hat{p}$ averages this probability over 2013–2015 adopters:

$$p(\phi) = \sum_{f,m,t} w_{fmt} \frac{\phi a_{fmt}}{1 + \phi a_{fmt}}, \tag{30}$$

where the sum runs over the firm–markets of 2013–2015, w_{fmt} is the firm–market's share of the 254,094 installations in the three IOU territories with an application in those years, and $a_{fmt} = A_{fmt}/N_{mt}$ is its agent density, constructed from salesperson registrations and potential adopters as in the demand data. Because the survey years precede the estimation sample, the weights are observed rather than predicted installation shares. Installations by firms without agents, or whose installer I cannot identify, have $a_{fmt} = 0$ and contribute nothing, so $p(\phi)$ rises strictly in ϕ from zero toward 0.585, the share of installations by firms with agents.

Moment. The reach moment is $g_S(\phi) = p(\phi) - \hat{p}$, with variance $\mathrm{Var}(\hat{p})$. Since $p(\phi)$ depends on ϕ alone, while utility depends on ϕ and γ_c through $\gamma_c \log(1 + \phi a_{fmt})$, this moment separates agent reach from effectiveness. On its own, equation (30) gives $\phi = 2.57$ at $\hat{p}$, with density measured per 100,000 potential adopters. This is close to the joint estimate of 2.50 in Table 3, at which $p(\phi) = 0.430$.

D.3 Agent Cost Likelihood

Define the inverse Mills ratio $h(\iota) = \varphi(\iota)/(1 - \Phi(\iota))$. The first-order conditions for maximizing the log-likelihood in equation (25) with respect to $\boldsymbol{\beta}_A$ and the scale σ_k of each group $k \in \{\text{HQ}, \text{non-HQ}\}$ are

$$\begin{aligned}
\frac{\partial \log \mathcal{L}}{\partial \boldsymbol{\beta}_A} &= \sum_{f,m,t} \frac{\boldsymbol{W}_{fmt}}{\sigma_{fmt}} \Big[1\{a_{fmt} > 0\}\iota_{fmt} + 1\{a_{fmt} = 0\}h(\bar{\iota}_{fmt}) \Big] &&= 0, \\
\frac{\partial \log \mathcal{L}}{\partial \log \sigma_k} &= \sum_{(f,m,t)\in k} \Big[1\{a_{fmt} > 0\}(\iota_{fmt}^2 - 1) + 1\{a_{fmt} = 0\}\bar{\iota}_{fmt}h(\bar{\iota}_{fmt}) \Big] &&= 0,
\end{aligned}$$

where the second sum is over the firm–markets in group k. For a non-agent firm, $h(\bar{\iota}_{fmt})$ and $\bar{\iota}_{fmt}h(\bar{\iota}_{fmt})$ are the expectations of ι_{fmt} and $\iota_{fmt}^2 - 1$ given $\iota_{fmt} > \bar{\iota}_{fmt}$, the generalized residuals of Gouriéroux et al. (1987).

D.4 Selling Costs and Agent Costs per Watt

If firms paid agents per sale, those commissions would enter the marginal cost of an installation and the pricing first-order conditions would load them into recovered installation costs. Table D1 regresses recovered marginal installation costs on whether, and how intensively, the firm uses agents in the market. Column 1 shows that, across firms, agent-using firms have higher installation costs, but columns 2 and 3 show that within a firm, installation costs do not rise in the markets where it deploys agents, so per-sale selling costs are not absorbed into installation costs.

Figure D2, panel (a), displays agent costs per installed watt among firm–markets with agents. These costs are low because they are spread over many buyers who would have installed solar without agents. Panel (b) instead divides each market's agent spending by the installed watts that agents add, from the counterfactual without agents in Section 7. Per additional watt, agents cost \$0.75 in aggregate, about 7.5 times the \$0.10 per installed watt.

D.5 Computing Equilibria

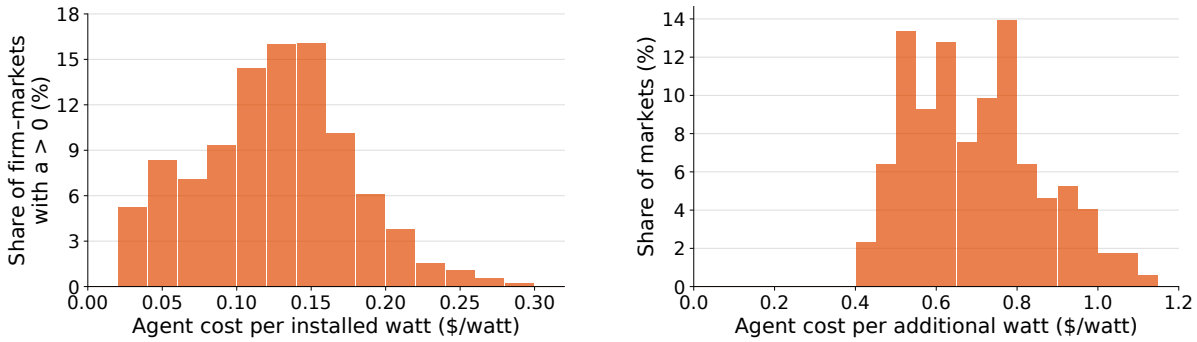
I solve for prices and salesforces jointly. For prices, I write the first-order conditions as the fixed point of Morrow and Skerlos (2011). Write the price derivatives of the market shares as $\partial s_{kmt}/\partial P_{jmt} = \Gamma_{jk} - 1\{j = k\}\Lambda_{jj}$, where Λ_{jj} integrates each household's price sensitivity times its choice probability of j and Γ_{jk} integrates its price sensitivity times

Table D1: Recovered installation costs and agent use

	(1)	(2)	(3)
Panel A. Any agents			
Any agents	0.280	0.036	−0.022
	(0.051)	(0.029)	(0.016)
Panel B. Agent density terciles			
Density tercile 1	0.224	−0.041	−0.044
	(0.054)	(0.025)	(0.021)
Density tercile 2	0.247	0.026	−0.003
	(0.068)	(0.026)	(0.019)
Density tercile 3	0.367	0.170	−0.016
	(0.107)	(0.116)	(0.034)
Market + size FE	Yes	Yes	Yes
Firm FE	No	Yes	No
Firm-year FE	No	No	Yes
Observations	49,151	49,148	49,050

Notes: The outcome is the recovered marginal installation cost in dollars per watt. Each column reports two regressions with the same fixed effects, one on an indicator for any agents in Panel A and one on indicators for terciles of agent density in Panel B; the omitted group in both is products of firms without agents. Density terciles are terciles of agent density among products of firms with agents. Standard errors in parentheses are clustered by firm.

Figure D2: Agent cost per installed and per additional watt



(a) Per installed watt, firm–markets

(b) Per additional watt, markets

Notes: Real 2023 dollars. Agent spending is the marginal agent cost times the number of agents. Panel (a) divides each firm's agent spending by its installed watts in the market, for the 3,923 firm–markets with agents. Panel (b) divides each market's total agent spending by the difference between its installed watts in the observed equilibrium and in the counterfactual equilibrium without agents, for the 172 markets with agents.

its choice probabilities of j and k, over the products k of the same firm. The pricing first-order conditions are then the fixed point

$$\boldsymbol{P}_{mt} = \boldsymbol{C}^{I}_{mt} + \boldsymbol{\Lambda}^{-1}_{mt}\left[\boldsymbol{s}_{mt} + \boldsymbol{\Gamma}'_{mt}\left(\boldsymbol{P}_{mt} - \boldsymbol{C}^{I}_{mt}\right)\right]. \tag{31}$$

Salesforces update by each firm's best response. Because agents enter demand through $\log(1+\phi a_{fmt})$, marginal gross profit is $\pi^{*\prime}_{fmt}(a) = \phi B_{fmt}/(1+\phi a)$, where B_{fmt} is the derivative of gross profit with respect to $\log(1+\phi a_{fmt})$, and, holding B_{fmt} at its current value, stationarity gives $a_{fmt} = B_{fmt}/C^{A}_{fmt} - 1/\phi$. A firm whose marginal gross profit at zero agents falls below its agent cost hires no agents, and a firm at zero resumes hiring when its marginal gross profit at zero exceeds its agent cost. In counterfactuals in which salesforces respond, firms with agents keep the agent costs they reveal in the data, and for each firm without agents I draw its agent cost from the estimated distribution conditional on exceeding the return to its first agent, averaging outcomes over 100 draws. When salesforces are set by the counterfactual, as when agents are banned, only prices update. In the equilibrium in which firms maximize joint industry profits, B_{fmt} is instead the derivative of industry gross profit with respect to $\log(1+\phi a_{fmt})$, at given prices.

For the effect of one more agent in Section 7.1, I differentiate each firm's installations q with respect to firm f's agents at the observed equilibrium, letting prices respond:

$$\frac{dq}{dA_f} = \frac{\partial q}{\partial A_f} + \sum_k \frac{\partial q}{\partial P_k}\frac{dP^*_k}{dA_f}. \tag{32}$$

The price response $d\boldsymbol{P}^*/dA_f$ follows from the implicit function theorem on the pricing first-order conditions: it solves a linear system whose matrix is the derivative of those conditions with respect to prices and whose right-hand side is minus their derivative with respect to A_f. I compute the same derivatives for each firm's gross profit, installed capacity, and avoided environmental damages, and weight firms by their agents when I aggregate.

E Welfare Appendix

This appendix describes how I compute each component of the welfare change in Section 8.2. Welfare is

$$W = CS + PS + G + E + F,$$

where G is minus the ITC outlay, E is the environmental benefit of solar generation, and F is the utility account. I evaluate each component at the observed equilibrium and at each counterfactual equilibrium, using each equilibrium's prices, salesforces, and cell shares s_{cjmt}, and holding N_{mt}, ω_{cmt}, product menus, and installation costs fixed. The changes in Table 5, panel (b), are the differences between each counterfactual and the observed equilibrium. All values are in real 2023 dollars. Flows over the life of a system are discounted to its installation year and then summed over the 180 markets.

Lifetime flows have the same timing as bill savings in the demand model. A system lasts $H = 25$ years, its output degrades by $d = 0.75$ percent a year, and the flow in year h is discounted by $(1 + r)^h$. A flow that starts at one dollar and grows at real rate g' before degradation has present value

$$\mathrm{PV}(r, g') = \sum_{h=1}^{H} \frac{[(1 + g')(1 - d)]^{h-1}}{(1 + r)^h}. \tag{33}$$

Three factors appear below: $\mathrm{PV}(0.03, 0) = 16.12$ for environmental benefits, $\mathrm{PV}(0.05, 0) = 13.14$ for the costs the utility avoids, and $\mathrm{PV}(0.05, g) = 19.60$ for the bill revenue it loses, where $g = 4.03$ percent is the real growth of retail electricity prices from Section 3.2. Every per-kW input to the environmental and utility accounts is constant within a cell, so these accounts depend on the equilibrium only through the capacity that each cell installs,

$$\bar{G}_{cmt} = N_{mt}\, \omega_{cmt} \sum_{j \in \mathcal{J}_{mt}} s_{cjmt}\, G_q, \tag{34}$$

where G_q is the DC capacity in kW of product $j = (f, q)$.

E.1 Generation

The bill savings in demand come from billing data and do not measure how much electricity a system generates, which both the utility's avoided costs and the environmental benefits require. I take generation from Sexton et al. (2021), who simulate the annual output of a south-facing rooftop system in every U.S. ZIP code with the National Renewable Energy Laboratory's System Advisor Model. Their simulated system has 4.32 kW of DC capacity, so I divide each ZIP code's output by 4.32 to obtain kWh per DC kW. They report that the model overpredicts the output of 500 installed California systems by about 6 percent, so I divide these yields by 1.06. I assign ZIP codes to 2010 ZCTAs, weighting by population where several ZIP codes share a ZCTA, and give the 113 of the 1,761 California

ZCTAs in my crosswalk that have no estimate the value of the nearest ZCTA that does. I then average over the ZCTAs in each local area ℓ, weighting by housing units, to obtain first-year generation y_ℓ in kWh per DC kW, which every cell in the local area shares. Across baseline installations, first-year generation averages 1,671 kWh per DC kW, a capacity factor of 19 percent, and totals 7.88 TWh. Output then declines by 0.75 percent a year over 25 years. I assume no rebound, so each kWh of solar generation displaces one kWh of grid supply.

E.2 Environmental Benefits

Sexton et al. (2021) estimate how the hourly emissions of each fossil plant respond to electricity load in each region of its grid interconnection, by hour of day and month, separately for carbon dioxide, sulfur dioxide, nitrogen oxides, and fine particulate matter. Combining these responses with the simulated hourly output of the system in each ZIP code gives the emissions that solar capacity there avoids. They value the three local pollutants with the AP3 integrated assessment model in 2014 dollars and carbon dioxide at $41 per short ton. I use these avoided damages as the environmental benefit of solar. I use the estimates from 2012–2016, the most recent period they report, and leave out their energy value, which the utility account covers.

I convert their damages for each ZIP code into factors per kWh of simulated output. Dividing their carbon damages by $41 recovers short tons of avoided carbon dioxide, which I convert to metric tons, giving e_ℓ metric tons per kWh. I add the damages from the three local pollutants and inflate them to 2023 dollars with the CPI-U, giving damages h_ℓ per kWh. Because both factors are per kWh of simulated output, the 6 percent output adjustment enters once, through y_ℓ. I aggregate the factors to local areas in the same way as generation and hold them fixed across cohorts and over each system's life. Environmental benefits in market mt are

$$E_{mt} = \mathrm{PV}(0.03, 0) \sum_c \bar{G}_{cmt}\, y_{\ell(c)} \left(\mathrm{SCC} \cdot e_{\ell(c)} + h_{\ell(c)} \right). \tag{35}$$

The social cost of carbon is SCC = $62.40 per metric ton, the California Public Utilities Commission's 2020 base value of $53 converted to 2023 dollars and held constant in real terms, and the discount rate is the commission's 3 percent real social rate (California Public Utilities Commission, 2024).

In their first year, baseline installations avoid 0.50 metric tons of carbon dioxide per MWh, worth $31.18 per MWh, and $6.24 per MWh in damages from local pollutants.

Over their lives they avoid 90.1 million metric tons of carbon dioxide, and their discounted environmental benefits are $3.96 billion from carbon and $0.79 billion from local pollutants, or $1.01 per watt. Without agents, first-year generation falls by 1.05 TWh and lifetime avoided carbon dioxide by 12.0 million metric tons, so E falls by $0.63 billion: $0.53 billion from carbon and $0.11 billion from local pollutants.

E.3 Utility Account

Under net metering, a household's bill falls by the retail value of all the electricity its system generates, both what it consumes and what it exports. The utility loses this bill revenue, B, and avoids the cost of supplying the same electricity, K. California's investor-owned utilities recover their authorized revenue through rates, so the difference $B - K$ falls on other ratepayers, the cost shift of Borenstein (2024). As in Gillingham, Kirkpatrick and Bollinger (2025), I count the utility account $F = K - B$ in welfare.

Lost bill revenue uses the estimated savings directly. A household in savings decile r that adopts in year t saves $(1+g)^{-(2024-t)}\rho_r$ per kW in its first year, the same savings that enter its effective price, and the savings grow with retail prices and decline with output:

$$B_{mt} = \mathrm{PV}(0.05, g)\,(1+g)^{-(2024-t)} \sum_{c} \bar{G}_{cmt}\,\rho_{r(c)}. \tag{36}$$

Baseline installations save households $1.21 billion in their first year, $258 per kW or 15.4 cents per kWh generated.

Avoided costs come from the California Public Utilities Commission's 2022 Avoided Cost Calculator, which gives hourly marginal avoided costs by utility and climate zone through 2052 (California Public Utilities Commission, 2022). I keep the components that measure resources the utility no longer uses: energy, generation capacity, transmission capacity, distribution capacity, ancillary services, and losses. I exclude the calculator's climate components, which are the cap-and-trade allowance price, the GHG adder and its rebalancing term, and methane leakage, so that climate damages enter welfare only through E. I weight each climate zone's hourly avoided costs by a typical-year solar production profile at the calculator's weather station for that zone (National Laboratory of the Rockies, 2026), convert them to 2023 dollars at the calculator's 2 percent inflation rate, and levelize them over a 25-year life starting in 2023 at the 5 percent utility rate, with output declining by 0.75 percent a year. Averaged across each utility's climate zones, the avoided cost of solar is $41.75 per MWh for Pacific Gas & Electric, $54.50 for Southern California Edison, and $48.64 for San Diego Gas & Electric, well below the

all-hour averages of \$67 to \$81 because solar produces when energy is cheap and before the evening peak that drives capacity costs. Each local area's avoided cost k_ℓ averages the three utility values with the shares of the local area's 2017–2022 installed capacity in each utility's territory:

$$K_{mt} = \mathrm{PV}(0.05, 0) \sum_c \bar{G}_{cmt}\, y_{\ell(c)}\, k_{\ell(c)}. \tag{37}$$

Baseline installations avoid \$0.37 billion in costs in their first year, 4.8 cents per kWh generated.

I discount both flows at 5 percent real, the utility weighted average cost of capital in the commission's staff report on the societal cost test (California Public Utilities Commission Energy Division and Energy and Environmental Economics, 2022). The two present-value factors differ because bill savings grow with retail prices while the avoided cost does not: a dollar of first-year bill revenue is worth \$19.60 over the life of the system, and a dollar of first-year avoided cost \$13.14. Together with the gap between the 15.4-cent retail value and the 4.8-cent avoided cost, this makes lifetime lost revenue 4.8 times lifetime avoided cost. For baseline installations, B is \$23.81 billion and K is \$4.92 billion, so F is −\$18.89 billion, or −\$4.01 per watt. Without agents, B falls by \$3.16 billion and K by \$0.65 billion, so F rises by \$2.51 billion.

E.4 Consumer Surplus, Producer Surplus, and Government Spending

Consumer surplus is the expected maximum utility of Small and Rosen (1981), converted to dollars with each cell's price sensitivity:

$$CS_{mt} = N_{mt} \sum_c \frac{\omega_{cmt}}{\alpha_c} \int \log\left(1 + \sum_{j \in \mathcal{J}_{mt}} \exp\left(\bar{u}_{cjmt} + \gamma_c \log(1 + \phi a_{f(j)mt}) + \sigma_\nu \nu\right)\right) d\Phi(\nu). \tag{38}$$

The agent term is inside the log-sum, so consumer surplus counts it as value to households, and savings are discounted at the households' own rates δ_y. The utility of the outside option is normalized to zero, so the level of consumer surplus is not identified, but its change between two equilibria with the same normalization is.

Producer surplus is gross profit net of agent expense,

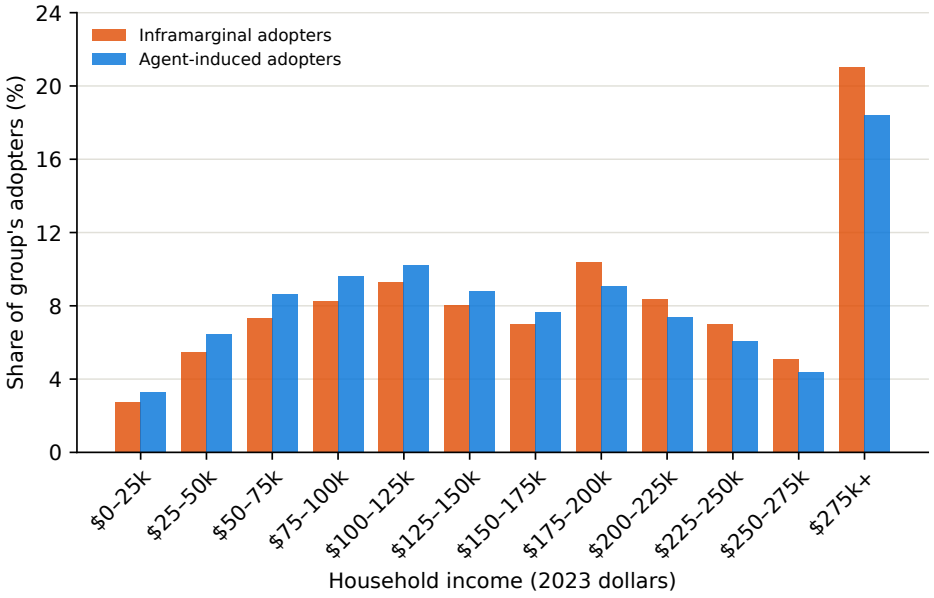
$$PS_{mt} = \sum_{j \in \mathcal{J}_{mt}} \left(P_{jmt} - C^I_{jmt}\right) N_{mt} s_{jmt} - \sum_f C^A_{fmt} A_{fmt}, \tag{39}$$

which includes the fringe with zero agents. For firms with agents, C^A_{fmt} is the agent cost recovered from the salesforce first-order condition, and agent expense is zero when agents are banned. Government spending is the ITC outlay, $G_{mt} = -\tau_t \sum_j P_{jmt} N_{mt} s_{jmt}$, with τ_t equal to 30 percent in 2017–2019 and 2022 and 26 percent in 2020–2021, valued dollar for dollar in the installation year.

E.5 Income of Agent-Induced Adopters

Figure E1 compares the income distributions of inframarginal adopters and the agent-induced adopters that agents bring into the market. The average agent-induced adopter has an income of $199,000, against $214,000 among adopters without agents.

Figure E1: Income of inframarginal and agent-induced adopters

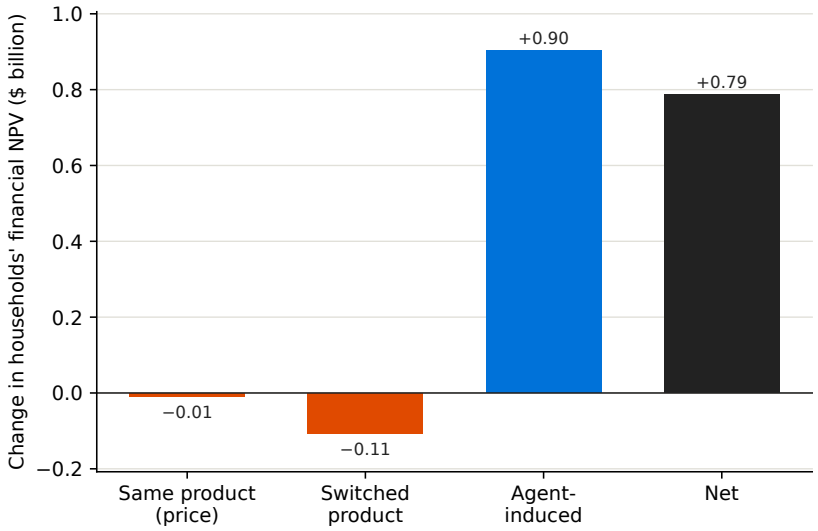


Notes: Share of each group's adopters in each household income bin, in 2023 dollars, in the observed equilibrium, 2017–2022. Inframarginal adopters also adopt in the counterfactual equilibrium without agents; agent-induced adopters adopt only with agents.

E.6 Households' Financial Gains

Figure E2 reports the change in households' financial returns from agents at the 5.5 percent real rate of a typical solar loan, by group of households.

Figure E2: Change in households' financial returns from agents



Notes: Change in the net present value of households' bill savings net of the price they pay after the ITC, the observed equilibrium relative to the counterfactual equilibrium without agents, discounted at a 5.5 percent real rate, 2017–2022, in billions of 2023 dollars. Households who buy the same product in both equilibria gain or lose only through its price; households who switch buy from a different installer with agents; agent-induced adopters adopt only with agents.

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